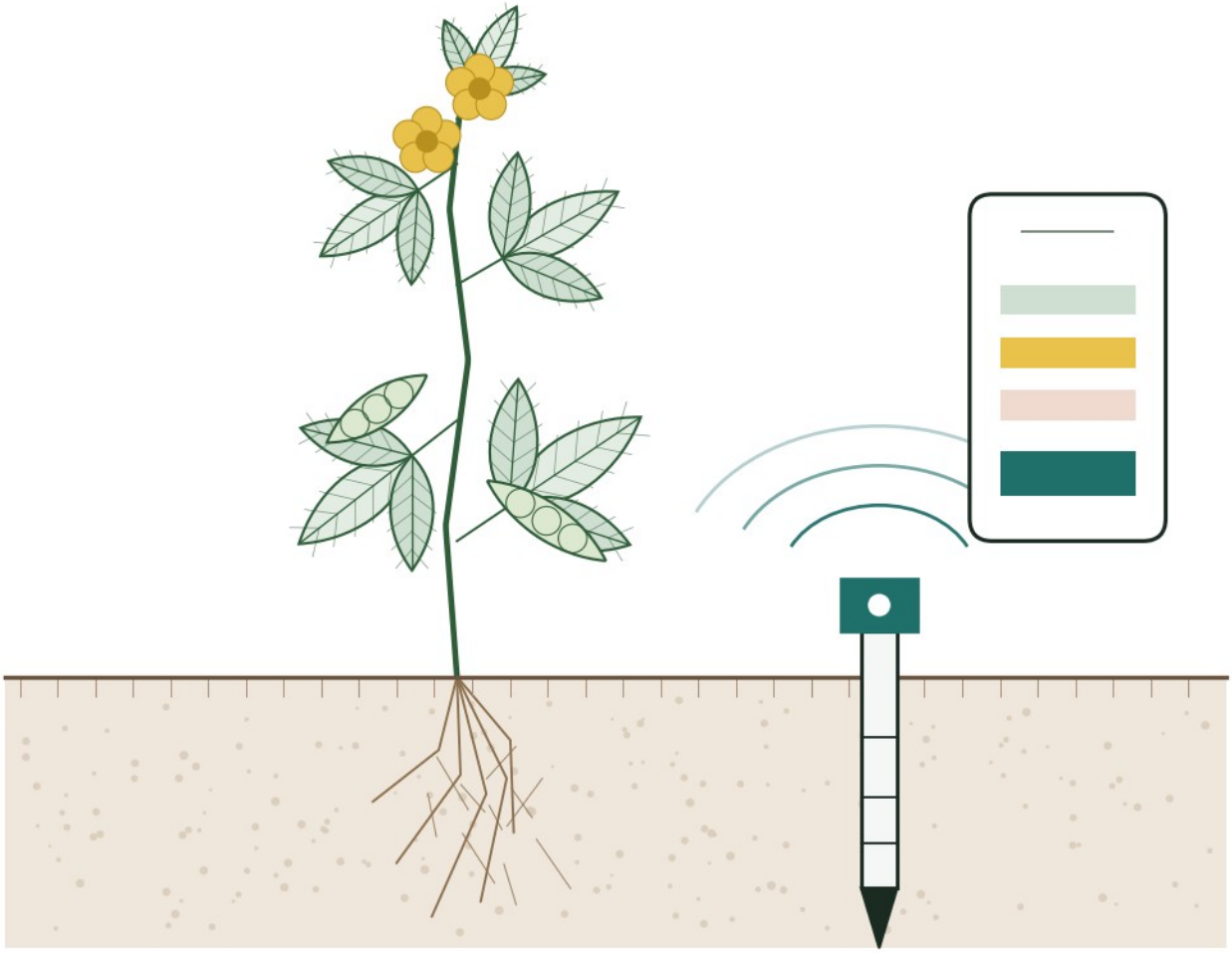
GROUP O

Software Requirements Specification

Multi-Site Smart Soil and Environmental Monitoring System



TEMPERATURE · MOISTURE · EC · pH · NITROGEN · PHOSPHORUS · POTASSIUM

DELIVERY TEAM

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Version	Date	Status
5.0	2 November 2026	Final, as built

Document Control

Item	Detail
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Prepared for	Department of Agriculture, Plant Sciences, Faculty of Health and Environmental Sciences, Central University of Technology, Free State
Stakeholder and end user	Ms Goitsemang Dikane
Prepared by	Group O project delivery team
Source repository	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Shared backend	Firebase (Authentication, Realtime Database). Cloud Storage and Cloud Functions are not used (see C27)
Hosting of the web console	Render (Docker), with UptimeRobot keeping it awake

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Revision History

Version	Date	Summary of change
1.0	25 August 2026	Initial specification prepared from the stakeholder requirements document and the consultation meeting of 19 August 2026

Version	Date	Summary of change
2.0	30 August 2026	Revised following the stakeholder follow-up responses. Data provisioning changed to representative sample data; user roles fixed at two; threshold basis established; feasibility, risk register, change log and refined scope incorporated
3.0	4 September 2026	Findings and requirements confirmed by the stakeholder on 3 September 2026. Repository and backend details added, and Appendices A to E incorporated. Issued as the approved baseline for design and development.
4.0	13 October 2026	Adds Section 22, System Architecture and Design: architecture, use case and sequence diagrams, database design, interface designs, design alignment, design sources, project plan and repository. The shared backend changes from Supabase to Firebase; changes C15 to C25 are recorded in Section 22.7. Sections 1 to 21 and Appendices A to E are kept as approved.
5.0	2 November 2026	Final as-built specification. Adds Section 23, As-Built System and Final Status: the delivered architecture and data, the final status of every requirement and acceptance criterion, changes C26 to C47, new requirements FR86 to FR92, resolved open decisions, updated risks, limitations and test evidence. Sections 1 to 22 and Appendices A to E are kept as approved.

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1. Introduction

1.1 Purpose of this Document

This specification defines what the Multi-Site Smart Soil and Environmental Monitoring System must do, the conditions under which it must operate, and how its components work together. It is the agreed reference against which the system is designed, built and evaluated.

The document is written for two audiences. The stakeholder uses it to confirm that the system being built addresses the problem described. The delivery team uses it to guide design, divide development work and prepare test cases.

1.2 Scope of this Document

The specification covers the complete prototype: a mobile application, a web application, a shared backend, and the data that flows between them. It records the requirements agreed with the stakeholder, the assessment of whether those requirements can be delivered within the available time and resources, the risks identified, and the scope decisions that follow from both.

It does not cover the sensor hardware, its installation, calibration or maintenance. That equipment is owned and operated by a separate party.

1.3 Document Structure

The document follows the order in which the work was carried out. Sections 2 to 4 describe the stakeholder, the current situation and what the system is intended to achieve. Sections 5 to 9 define the boundaries of the system and how it is put together. Sections 10 to 16 state the requirements. Sections 17 to 20 record the feasibility assessment, the risks, and the resulting changes to scope. Section 21 lists the supporting evidence.

1.4 Basis of this Specification

Every requirement in this document traces to a statement made by the stakeholder, either in the written requirements document supplied by the organisation or in the written responses to the delivery team's follow-up questions. Both are held as supporting evidence and are listed in Section 21. The stakeholder confirmed these findings and approved the requirements on 3 September 2026.

1.5 How to Read This Document

A specification uses a small number of terms in a particular way. They are explained here so that the rest of the document can be read without a technical background. A fuller list of terms appears in Section 6.

1.5.1 The Four Types of Requirement

Requirements are separated into four kinds. Each answers a different question about the system, and each is numbered with its own prefix so that any requirement can be referred to later by its identifier.

Type	Question it answers	Example
Functional FR	What must the system do?	A user must be able to see the battery level of a sensor.
Non-functional NFR	How well, or under what conditions, must it do it?	Only an administrator may change an acceptable range.
System SR	What must the system be built	The mobile application must run on Android.

Type	Question it answers	Example
	with, or run on?	
Integration IR	How must the separate parts work together?	A user account created on the website must also work in the mobile application.

The distinction between the first two matters in practice. “A user must be able to view alerts” describes a function the system performs. “Only an administrator may change an acceptable range” describes a rule that governs how that function behaves. Keeping them apart makes it clear what will be built, and separately what it will be judged against.

1.5.2 Priority

Not every requirement carries the same weight. Each is given one of three priorities, so that if time becomes short it is already agreed what is set aside first. The priorities follow the order the stakeholder gave when asked which functions matter most, recorded in Section 14.2.

Priority	Meaning
Must Have	Essential. The system does not address the problem without it, and it will be built first.
Should Have	Important, and intended, but the system would still be usable without it in this first version.
Could Have	Included only if time and resources allow once everything above is complete.

1.5.3 Reading a Requirements Table

Each requirement is listed on one row with four columns.

Column	What it means
ID	A permanent reference, such as FR39. Requirement numbers do not change, even if a requirement is later altered or set aside, so that any discussion can point to a specific line.
Requirement	A single statement of one thing the system must do or satisfy. Each is written to be testable, so that at the end it can be shown to be either met or not met.
Component	Which part of the system it applies to. Mobile is the phone application; Web is the website; Both means it applies to each; Backend is the shared part behind them, which stores the information and applies the rules, and which users do not see directly.
Priority	Must Have, Should Have or Could Have, as described above.

1.5.4 What the Numbering Means

Requirement identifiers are permanent. If a requirement is simplified or deferred, it keeps its number and the change is written into the change log in Section 19 rather than the requirement being deleted. This means the record always shows what was originally asked for, what was agreed instead, and why. Nothing disappears quietly.

2. Stakeholder and Current Process

2.1 Stakeholder Profile

Item	Detail
Organisation	Department of Agriculture, Plant Sciences, Faculty of Health and Environmental Sciences, Central University of Technology, Free State.
Activities	Plant science research and the monitoring of soil and environmental conditions for agricultural benefit.
Stakeholder and end user	Ms Goitsemang Dikane, Plant Sciences (nGAP), Department of Agriculture. Confirms the requirements and accepts the system on behalf of the intended users.
Engagement channel	Ms Mpho Mbele, Departmental Manager: Information Technology, through whom the written requirements document was supplied and the follow-up questions were routed.
Deployment	Fifteen 7-in-1 RS485 soil sensors installed across three agricultural locations, five sensors at each location.
Crop	Pigeon pea (Cajanus cajan) at all three locations.
Reason for engagement	The organisation has sensing equipment in place and measurements being produced, but no single system through which those measurements can be viewed, compared and acted upon.

2.2 Current Process

Soil condition at the three locations is currently assessed by physical sampling. Soil samples are taken once, at the start of the planting cycle, and sent to a laboratory. The laboratory result is used to determine which fertiliser is suitable for the season.

Between that single test and the end of the season, the organisation has no continuous view of soil condition. Fifteen sensors are deployed and producing measurements, but there is no system that brings those readings together, compares conditions across the three locations, or indicates when a site requires attention. Assessment between sampling points therefore depends on physical inspection of each location.

2.2.1 Participants in the Current Process

Participant	Involvement
Site or farm manager	Responsible for the condition of the land and for decisions about intervention
Agricultural staff	Carry out work at the locations based on those decisions
Field workers	Present at the locations and first to observe visible changes
Researchers	Interpret conditions and assess results across the three sites
Laboratory	Analyses the soil samples taken at the start of the planting cycle
Sensor operator	Owns and operates the sensing equipment and publishes the measurements

2.2.2 Current Process Flow

Figure 1 shows the same process in diagram form.

Step	Action	Frequency
1	Soil samples are taken at each location	Once, at the start of the planting cycle
2	Samples are sent to a laboratory for analysis	Once per cycle
3	Laboratory results determine the fertiliser to be applied	Once per cycle
4	Sensors measure soil and environmental conditions continuously	Every 30 minutes
5	Measurements are published by the sensor operator	Continuous
6	Conditions between sampling points are assessed by physical inspection	Irregular
7	Changes in condition are identified when noticed on site	Irregular, and often late

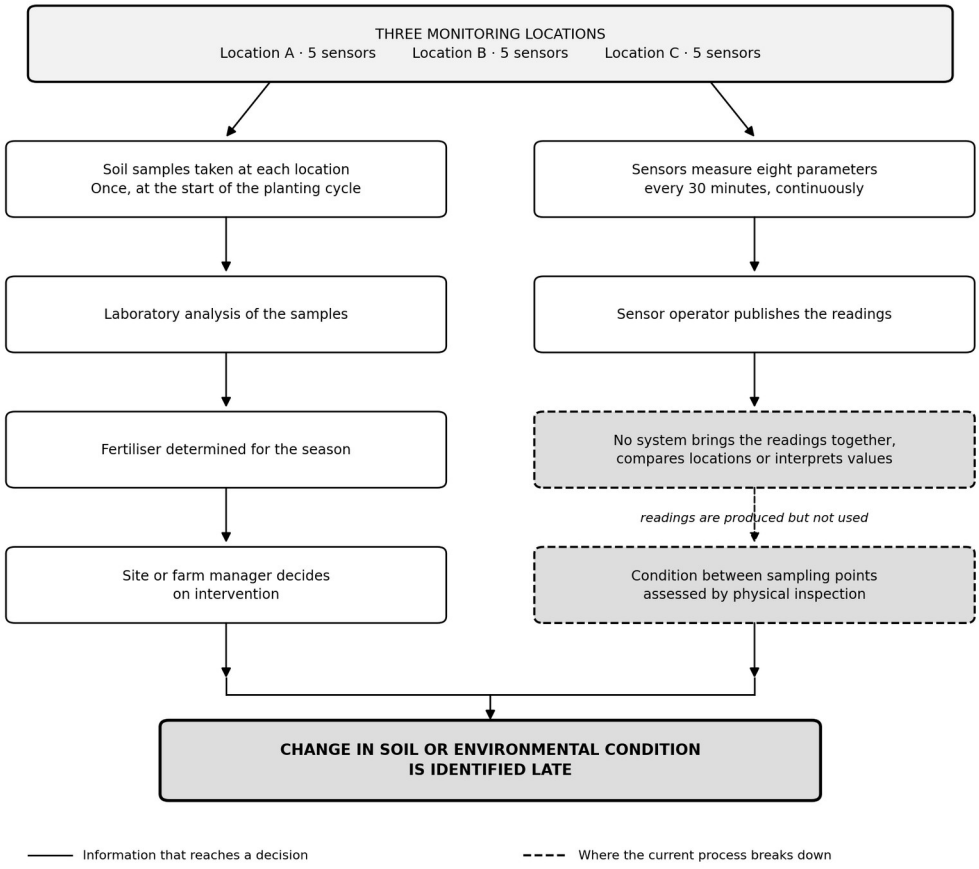


Figure 1 — Current process at the three monitoring locations

2.3 Difficulties in the Current Process

Difficulty	Effect on the organisation
Soil condition is tested only once per cycle	Conditions change during the season with no corresponding measurement being reviewed

Difficulty	Effect on the organisation
Readings are produced but not brought together	Fifteen sensors generate data continuously that no one is in a position to read
No comparison across the three locations	A site performing differently from the others is not identified
Change in condition is noticed late	Responses to low soil moisture, unusual nutrient levels or other environmental change are delayed
Assessment depends on physical presence	Three separate locations cannot be inspected frequently enough to be useful
No visibility of equipment condition	A sensor that has stopped reporting is indistinguishable from a sensor reporting normal conditions

3. Problem Statement

The organisation currently uses single-point laboratory soil sampling and physical inspection to assess soil condition across three agricultural locations. This results in significant changes in soil and environmental conditions being identified late, while measurements capable of revealing those changes are produced continuously by fifteen deployed sensors and are not accessible in any usable form.

The problem affects site and farm managers, agricultural staff, field workers and researchers, because decisions about intervention depend on information they cannot see. Conditions such as low soil moisture or unusual nutrient levels are noticed only when they become visible on site or when the next sampling cycle occurs, by which time a response is late.

An improved solution is required to bring the readings from all three locations into one place, interpret them against acceptable ranges for the crop grown, indicate which location or sensor requires attention, report whether the sensing equipment is functioning, and make that information available to the responsible people without requiring a visit to the site.

3.1 Elements of the Problem

Element	Statement
Current situation	Soil is tested once per planting cycle. Sensor measurements are produced continuously but are not presented in a usable form.
Problem	Significant change in soil and environmental condition is identified late.
Affected users	Site or farm managers, agricultural staff, field workers and researchers.
Consequences	Delayed response to low soil moisture, unusual nutrient levels and other environmental change. Equipment failure goes unnoticed.
Need	A central system that organises the readings, shows the condition of each site, and identifies changes requiring attention.

4. Project Aim and Objectives

4.1 Aim

To deliver an integrated prototype that brings soil and environmental measurements from fifteen sensors across three agricultural locations into a single system, interprets those measurements against acceptable ranges for the crop, reports the operational condition of the sensing equipment, and presents the result to authorised users through a mobile and a web application.

4.2 Objectives

1. Store soil and environmental readings for all fifteen sensors in a single shared database, without duplication.
2. Present the current condition of each location and each individual sensor.
3. Compare measured values against thresholds configured for each location and raise an alert when a value falls outside its acceptable range.
4. Report the operational condition of each sensor, including battery level, sensor status and time of last communication.
5. Raise an alert when a sensor has stopped communicating.
6. Retain readings over time and present them as graphs for a user-selected period.
7. Restrict access to information and configuration according to the user's assigned role.
8. Allow an administrator to manage locations, sensors, thresholds and user accounts.
9. Allow conditions at the three locations to be compared side by side.
10. Notify responsible users when a location requires attention.

5. Project Scope

Scope records what the project includes and what it does not. Agreeing this early prevents a common difficulty, where each side assumes something different was included and only discovers the difference at the end.

5.1 Deliverables

Deliverable	Description
Mobile application	Flutter application for viewing conditions, sensor detail, equipment status, history and alerts
Web application	ASP.NET MVC application for administration, configuration, comparison and reporting
Shared backend	Supabase project providing the database, authentication, access rules and storage
Data provisioning component	Loads representative sample readings into the database at a defined interval
Software Requirements Specification	This document, maintained as the agreed reference
Project schedule	Baselined schedule showing tasks, responsibilities, dependencies and milestones
Source repository	Version-controlled repository containing documentation and source code
Integrated demonstration	Working demonstration of the prototype against the acceptance criteria

5.2 Acceptance Criteria

These are the conditions the finished system must demonstrate before it is accepted. Each is written so that it can be shown to be either met or not met, rather than being a matter of judgement. Together they are the practical test of whether the problem described in Section 3 has been addressed.

Ref	Criterion
AC1	An authorised user can sign in, and an unauthorised user cannot.
AC2	A dashboard displays counts of locations, registered sensors, active sensors, inactive sensors and open alerts.
AC3	All three locations are listed, and the sensors at each location can be reached from the location.
AC4	All fifteen sensors are registered and individually identifiable.
AC5	All seven measured parameters are displayed for a selected sensor, together with the time of the reading.
AC6	Battery level, sensor status and time of last communication are displayed for each sensor.
AC7	A reading falling outside its configured threshold generates an alert visible in both applications.
AC8	A sensor that has not communicated within the configured period is reported as inactive and generates an alert.
AC9	Readings for a selected sensor and period are displayed as a graph.

Ref	Criterion
AC10	An administrator can create, edit and remove locations, sensors, thresholds and user accounts.
AC11	A standard user cannot change any configuration.
AC12	Conditions at the three locations can be compared in a single view.
AC13	Data recorded through the web application is visible in the mobile application, and the reverse where applicable.
AC14	Readings can be exported from the web application.

5.3 Exclusions

- Installation, configuration, calibration or maintenance of the sensor hardware
- Direct communication between the applications and the sensors or gateway equipment
- Live integration with the sensor operator's published data store, which is treated as a later integration step
- Agronomic advice, fertiliser recommendation or crop treatment guidance beyond comparison against configured thresholds
- Automated irrigation or any control of physical equipment
- Predictive analysis or forecasting of future soil condition
- Payment processing
- Public access to readings without an authenticated account

5.4 Constraints

- The prototype must be delivered within one academic semester, with approximately three weeks available for development
- The technology set is fixed: Flutter for mobile, ASP.NET MVC for web, Supabase for the shared backend
- Live sensor readings are produced by a separate party and are not available to the delivery team for this version
- Services must operate within free service tiers
- Eight team members with differing availability and levels of development experience
- Only representative sample data may be used in the prototype and in the demonstration

5.5 Assumptions

- The three locations, the fifteen sensors and the seven measured parameters remain as described
- Pigeon pea remains the crop at all three locations
- Two user roles are sufficient for this version
- The stakeholder remains available to confirm findings and to evaluate the prototype
- Sample data structured as described in Section 8.3 is an acceptable substitute for live readings
- The free service tier is sufficient for the volume of readings the prototype generates

6. Definitions and Abbreviations

6.1 Terms Used About the System

Term	Meaning
Alert	A record raised automatically when a reading falls outside its configured threshold, or when a sensor stops communicating
ASP.NET MVC	The framework used to build the web application
Dashboard	The summary view presenting counts and overall condition across all locations
EC	Electrical conductivity, a measure of dissolved salts in the soil, expressed in decisiemens per metre
Flutter	The framework used to build the mobile application
Inactive sensor	A sensor that has not produced a reading within the configured period
Location	One of the three sites at which sensors are deployed
NPK	Nitrogen, phosphorus and potassium
Parameter	One of the eight quantities measured by a sensor
Reading	One complete set of measurements captured by one sensor at one point in time
Role	The permission level assigned to a user account, either Administrator or Standard User
RS485	The serial communication standard used by the sensor equipment
Sensor	One of the fifteen deployed measuring devices
Sensor status	The reported operational state of a sensor
Supabase	The shared backend providing the database, authentication, access rules and storage
Threshold	A configured acceptable range for one parameter at one location

6.2 Terms Used About the Project

These terms describe how the work is planned and recorded rather than what the system does.

Term	Meaning
Acceptance criterion	A condition the finished system must demonstrate before it is accepted. The full list is in Section 5.2. Each is written so that it can be shown to be either met or not met, rather than judged as a matter of opinion.
Assumption	Something taken to be true for planning purposes, which has not been separately verified. If an assumption turns out to be wrong, the plan may need to change.
Constraint	A limit the project must work within, such as available time or the technologies that must be used. A constraint is not a choice.
Contingency	What will be done if a risk actually happens.

Term	Meaning
Deliverable	Something the delivery team will produce and hand over.
Entity	A kind of information the system stores, such as a location or a reading. Section 16 lists them.
Exclusion	Something deliberately not included, recorded so that expectations are clear on both sides.
Feasibility	Whether the project can realistically be completed with the time, skills and resources available. Section 17 assesses this in six areas, each reaching one of three conclusions: feasible, feasible with changes, or not currently feasible.
Impact	How seriously a risk would affect the project if it happened, rated from 1 to 3.
Issue	A problem that has already happened. This differs from a risk, which has not.
Likelihood	How likely a risk is to happen, rated from 1 to 3.
Mitigation	What will be done in advance to make a risk less likely, or less damaging if it occurs.
Prototype	A working version built to demonstrate the proposed solution. It shows that the approach works, and is not a finished product for daily operational use.
Risk	Something that may happen and would affect the project if it did. A risk that has already happened is no longer a risk; it is an issue.
Risk score	Likelihood multiplied by impact, giving a number from 1 to 9. Scores of 1 to 3 are low, 4 to 6 are medium, and 7 to 9 are high.
Scope	What the project includes and, equally importantly, what it does not.
Specification	This document. It records what has been agreed, and is the reference the finished system is measured against.

7. Stakeholders and User Groups

7.1 Stakeholders

Stakeholder	Interest in the system
Department of Agriculture, Plant Sciences	Commissions the system. Represented by Ms Goitsemang Dikane, who confirms the requirements and accepts the prototype as the end user.
Sensor operator	Owns and operates the sensing equipment and publishes the measurements. Not a user of the system.
Delivery team	Designs, builds and delivers the prototype

7.2 User Roles

The stakeholder confirmed that two roles are required for this version, and that additional roles may be introduced later if needed.

Role	Permissions
Administrator	Full access. Manages user accounts and roles, locations, sensor registration and threshold configuration. Views all readings, history, alerts and reports.
Standard User	Read access. Views the dashboard, locations, sensors, current readings, equipment status, history and alerts. Cannot change any configuration.

7.3 User Groups by Function

Group	Typical role	Application
Site or farm manager	Administrator or Standard User	Both
Agricultural staff	Standard User	Mobile
Field workers	Standard User	Mobile
Researchers	Standard User	Both
System administrator	Administrator	Web

The exact number of users is not fixed. The system does not impose a limit.

8. Product Perspective

8.1 System Composition

The prototype operates as a single integrated system built on one shared backend. The mobile application and the web application both connect to the same Supabase project and read from the same tables. Neither application holds its own copy of the data.

Component	Technology	Function
Mobile application	Flutter	Viewing conditions, sensor detail, equipment status, history and alerts in the field
Web application	ASP.NET MVC	Administration, configuration, comparison, reporting and export
Shared backend	Supabase	Database, authentication, role-based access rules, storage
Data provisioning	Scheduled backend function	Loads representative sample readings at a defined interval

8.2 Data Ownership Boundaries

Each table has exactly one writer. This prevents two sources of truth for the same information.

Data	Written by	Read by
Readings	Data provisioning component only	Both applications
Sensor status	Data provisioning component only	Both applications
Alerts	Backend only, generated automatically	Both applications
Locations	Web application only	Both applications
Sensors	Web application only	Both applications
Thresholds	Web application only	Both applications and the backend
Users and roles	Web application only	Backend access rules

The mobile application does not create or modify readings, configuration or user accounts. Alerts are generated in the backend so that both applications present the same alert state.

8.3 Data Provisioning

For this version the system operates on representative sample data rather than live sensor readings. The stakeholder confirmed in writing that this is acceptable, that access to the operator's published data store is not required at this stage, and that live integration may be treated as a later step once access and the final data format are available.

Sample readings are generated to match the structure and behaviour of real readings: fifteen sensors across three locations, a complete reading every thirty minutes, values within and occasionally outside the configured thresholds, and sensors that occasionally stop reporting so that inactivity detection can be demonstrated.

8.3.1 Reading Record Structure

Field	Type	Description
Sensor identifier	Reference	The sensor that produced the reading
Location identifier	Reference	The location at which the sensor is installed
Timestamp	Date and time	When the reading was captured
Soil moisture	Numeric	Percentage
Soil temperature	Numeric	Degrees Celsius
Electrical conductivity	Numeric	Decisiemens per metre
pH	Numeric	pH units
Nitrogen	Numeric	Milligrams per kilogram
Phosphorus	Numeric	Milligrams per kilogram
Potassium	Numeric	Milligrams per kilogram

8.3.2 Sensor Status Record Structure

Field	Type	Description
Sensor identifier	Reference	The sensor being reported on
Battery level	Numeric	Percentage of remaining charge
Power status	Text	Reported power or charging state
Sensor status	Text	Active or inactive
Last communication	Date and time	When the sensor last produced a reading
Recorded at	Date and time	When the status record was written

8.3.3 Later Integration

The data provisioning component is the only part of the system that writes readings. When live integration is required, that component is replaced with one that retrieves readings from the operator's published data store and maps them to the same structure. No other part of the system changes. This isolation is the reason the component exists as a separate element rather than as sample data loaded directly into the database.

9. Main Product Functions

Component	Functions
Mobile application	Sign in and out; view the dashboard summary; view the three locations; view the sensors at a location; view all seven parameters for a selected sensor; view battery level, power status and sensor status; view time of last communication; view readings over time as graphs; view open alerts and alert detail; compare conditions across locations.
Web application	Sign in and out; view the dashboard summary; create, edit and remove locations; register, edit and remove sensors; create and manage user accounts and roles; configure thresholds per location and parameter; view and filter historical readings; compare locations; view and manage alerts; export readings and alerts.
Shared backend	Store users, roles, locations, sensors, readings, sensor status, thresholds and alerts; authenticate users; enforce role-based access; evaluate readings against thresholds; detect inactive sensors; generate alerts; retain historical readings.
Data provisioning	Generate and store representative readings for all fifteen sensors at a thirty-minute interval; record sensor status; avoid duplicate records.

10. Functional Requirements

Functional requirements state what the system must do. Each is written as a single action that can be shown to work or not work, and each is numbered so that it can be referred to later. They are grouped by area of the system, and the grouping follows the order of the stakeholder priority list in Section 14.2.

The four columns are explained in Section 1.5.3, and the three priority levels in Section 1.5.2.

10.1 Authentication and Access

ID	Requirement	Component	Priority
FR01	A user must be able to sign in using an email address and password.	Both	Must
FR02	The system must reject sign-in attempts with invalid credentials.	Both	Must
FR03	A user must be able to sign out.	Both	Must
FR04	The system must assign each user account one of two roles: Administrator or Standard User.	Backend	Must
FR05	The system must restrict every function according to the signed-in user's role.	Backend	Must
FR06	The system must prevent access to any reading, alert or configuration by an unauthenticated request.	Backend	Must
FR07	A user must be able to change their own password.	Both	Should

10.2 Dashboard

ID	Requirement	Component	Priority
FR08	The system must display a dashboard on sign-in showing the overall state of the deployment.	Both	Must
FR09	The dashboard must display the number of monitoring locations.	Both	Must
FR10	The dashboard must display the number of registered sensors.	Both	Must
FR11	The dashboard must display the number of active sensors.	Both	Must
FR12	The dashboard must display the number of inactive sensors.	Both	Must
FR13	The dashboard must display the number of open alerts.	Both	Must
FR14	The dashboard must indicate the overall condition of each location.	Both	Must
FR15	The dashboard must display the time at which the information was last updated.	Both	Should

10.3 Locations

ID	Requirement	Component	Priority
FR16	The system must list all monitoring locations.	Both	Must
FR17	A user must be able to open a location and view the sensors installed at it.	Both	Must
FR18	The system must display the current condition of a selected location.	Both	Must
FR19	The system must display the number of open alerts at a selected location.	Both	Must
FR20	An administrator must be able to create a location.	Web	Must
FR21	An administrator must be able to edit the details of a location.	Web	Must
FR22	An administrator must be able to remove a location that has no sensors assigned to it.	Web	Should

10.4 Sensors

ID	Requirement	Component	Priority
FR23	The system must list all sensors installed at a selected location.	Both	Must
FR24	Each sensor must be individually identifiable.	Both	Must
FR25	A user must be able to open a sensor and view its detail.	Both	Must
FR26	The system must display which sensor produced any given reading.	Both	Must
FR27	An administrator must be able to register a sensor and assign it to a location.	Web	Must
FR28	An administrator must be able to edit the details of a registered sensor.	Web	Must
FR29	An administrator must be able to deactivate a sensor without deleting its readings.	Web	Should
FR30	An administrator must be able to move a sensor to a different location.	Web	Could

10.5 Readings

ID	Requirement	Component	Priority
FR31	The system must display the most recent reading for a selected sensor.	Both	Must
FR32	The system must display soil moisture for a selected sensor.	Both	Must
FR33	The system must display soil temperature for a selected sensor.	Both	Must
FR34	The system must display electrical conductivity for a selected sensor.	Both	Must
FR35	The system must display pH for a selected sensor.	Both	Must
FR36	The system must display nitrogen, phosphorus and potassium for a selected sensor.	Both	Must
FR37	The system must display the date and time at which each reading was captured.	Both	Must
FR38	The system must indicate, for each displayed parameter, whether the value falls inside or outside its configured threshold.	Both	Must

10.6 Equipment Condition

ID	Requirement	Component	Priority
FR39	The system must display the battery level of each sensor.	Both	Must
FR40	The system must display the power status of each sensor.	Both	Must
FR41	The system must display whether each sensor is active or inactive.	Both	Must
FR42	The system must display the time at which each sensor last communicated.	Both	Must
FR43	The system must mark a sensor as inactive when it has not produced a reading within the configured period.	Backend	Must
FR44	An administrator must be able to configure the period after which a sensor is treated as inactive.	Web	Should

10.7 Thresholds

ID	Requirement	Component	Priority
FR45	The system must hold a configurable acceptable range for each parameter at each location.	Backend	Must
FR46	An administrator must be able to set the minimum and maximum value of a threshold.	Web	Must
FR47	An administrator must be able to edit an existing threshold.	Web	Must
FR48	The system must record the source or basis on which a threshold was set.	Web	Should
FR49	A standard user must be able to view thresholds but not change them.	Both	Must
FR50	An administrator must be able to apply one location's thresholds to another location.	Web	Could

10.8 Alerts

ID	Requirement	Component	Priority
FR51	The system must generate an alert when a reading falls outside its configured threshold.	Backend	Must
FR52	The system must generate an alert when a sensor becomes inactive.	Backend	Must
FR53	An alert must record the sensor, the location, the parameter, the measured value and the time raised.	Backend	Must
FR54	The system must list all open alerts.	Both	Must
FR55	A user must be able to open an alert and view its detail.	Both	Must
FR56	The system must present the same alert state in both applications.	Backend	Must
FR57	A user must be able to acknowledge an alert so that it is removed from the open list.	Both	Should
FR58	The system must notify users when a new alert is raised.	Both	Should
FR59	The system must filter alerts by location, by sensor and by severity.	Web	Should

10.9 History and Comparison

ID	Requirement	Component	Priority
FR60	The system must retain readings so that previous values can be viewed.	Backend	Must

ID	Requirement	Component	Priority
FR61	A user must be able to view previous readings for a selected sensor.	Both	Must
FR62	The system must display readings over time as a graph.	Both	Must
FR63	A user must be able to select the period shown in a graph.	Both	Must
FR64	A user must be able to select which parameter is graphed.	Both	Must
FR65	The system must allow conditions at the three locations to be compared in a single view.	Both	Must
FR66	The system must allow historical readings to be filtered by location, sensor, parameter and date range.	Web	Should

10.10 Reporting

ID	Requirement	Component	Priority
FR67	An administrator must be able to export readings for a selected location and period.	Web	Must
FR68	An administrator must be able to export the alert history for a selected period.	Web	Should
FR69	The system must produce a summary report of conditions across the three locations.	Web	Should

10.11 User Administration

ID	Requirement	Component	Priority
FR70	An administrator must be able to create a user account.	Web	Must
FR71	An administrator must be able to assign a role to a user account.	Web	Must
FR72	An administrator must be able to change a user's role.	Web	Must
FR73	An administrator must be able to deactivate a user account.	Web	Must
FR74	The system must list all user accounts and their assigned roles.	Web	Must

10.12 Data Provisioning

ID	Requirement	Component	Priority
FR75	The system must store readings for all fifteen sensors at a thirty-minute interval.	Backend	Must
FR76	The system must store a sensor status record for each sensor.	Backend	Must
FR77	The system must not store duplicate readings for the same sensor and timestamp.	Backend	Must
FR78	The system must record the outcome of each provisioning run so that failures are visible.	Backend	Should

11. Non-Functional Requirements

Non-functional requirements describe qualities, rules and conditions rather than functions. They answer how well the system must work and under what rules, rather than what it does. A requirement such as “only an administrator may change an acceptable range” does not add a feature; it governs how an existing feature behaves.

These are grouped below by the kind of condition they describe.

11.1 Security and Access

ID	Requirement	Priority
NFR01	Passwords must be stored in a form from which the original cannot be recovered.	Must
NFR02	Access rules must be enforced in the backend and not only in the user interface.	Must
NFR03	Backend keys and service credentials must be held outside the source repository.	Must
NFR04	Credentials must not be embedded in the mobile application.	Must
NFR05	Only representative sample data and test accounts may be used in the prototype and in the demonstration.	Must
NFR06	The system must record which account performed each configuration change.	Should

11.2 Usability

ID	Requirement	Priority
NFR07	Every screen must be reachable within three interactions from the dashboard.	Must
NFR08	A parameter outside its threshold must be visually distinguishable from one inside it.	Must
NFR09	Error messages must state what went wrong and what the user can do about it.	Must
NFR10	The mobile interface must display correctly on an Android smartphone.	Must
NFR11	The system must confirm to the user when an action has succeeded or failed.	Must
NFR12	Screen labels and terminology must be consistent across both applications.	Should

11.3 Performance and Reliability

ID	Requirement	Priority
NFR13	The dashboard must display without unnecessary delay for the current data volume.	Must
NFR14	A graph covering thirty days for one sensor must render without	Must

ID	Requirement	Priority
	unnecessary delay.	
NFR15	The failure of a provisioning run must not prevent existing readings from being viewed.	Must
NFR16	Readings must be retained for at least twelve months.	Should

11.4 Correctness and Boundaries

ID	Requirement	Priority
NFR17	The system must compare measured values against configured thresholds and must not present the result as agronomic or treatment advice.	Must
NFR18	Every displayed reading must show the time at which it was captured, so that the age of the information is visible.	Must
NFR19	A threshold must record the basis on which it was set, so that its origin can be established.	Should
NFR20	Functions not completed in the prototype must be documented rather than presented as working.	Must

12. System Requirements

System requirements state what the system must be built with and what it must run on. They are recorded because they constrain the design, and because they were fixed before the project began rather than chosen by the delivery team.

ID	Requirement
SR01	The mobile application must be developed using Flutter.
SR02	The web application must be developed using ASP.NET MVC.
SR03	The shared backend must be provided by Supabase.
SR04	The database must be relational and must enforce referential integrity between locations, sensors, readings, thresholds and alerts.
SR05	Authentication must be provided by the shared backend rather than implemented separately in each application.
SR06	Role-based access must be enforced by database access rules.
SR07	Source code and project documentation must be held in a version-controlled repository.
SR08	The mobile application must run on Android.

13. Integration Requirements

The system is made of separate parts: a phone application, a website and a shared store of information behind both. Integration requirements state how those parts must work together, so that they behave as one system rather than three that happen to look similar. These are the requirements most easily overlooked, and the ones most likely to cause difficulty if they are.

ID	Requirement
IR01	Both applications must connect to the same Supabase project and read from the same tables.
IR02	Configuration recorded through the web application must be visible in the mobile application.
IR03	Alerts generated in the backend must be visible in both applications with the same state.
IR04	Readings written by the data provisioning component must be available to both applications.
IR05	A user account created through the web application must be able to sign in to the mobile application.
IR06	A threshold changed through the web application must be applied to subsequent alert evaluation without further action.
IR07	The data provisioning component must be the only writer of readings and sensor status, so that it can later be replaced by a live integration without changing any other component.

14. Requirements Prioritisation

14.1 Summary

Type	Prefix	Total	Must	Should	Could
Functional	FR	92	67	21	4
Non-functional	NFR	20	16	4	0
System	SR	8	8	0	0
Integration	IR	8	8	0	0
Total	—	**128**	**99**	**25**	**4**

Counts at version 5.0, taken from the requirement tables of Sections 10 to 13 and 22.7.2 and the new requirements FR86 to FR92 (Section 23.5). The final status of every requirement is in Section 23.6.

14.2 Stakeholder Priority Order

The stakeholder was asked which functions matter most if development time becomes limited. The order below is the answer given, and it governs the sequence in which work is completed.

Order	Function	Requirements
1	User login and access control	FR01–FR06
2	The three monitoring locations	FR16–FR19
3	Display of all fifteen sensors	FR23–FR26
4	Current sensor readings	FR31–FR37
5	Battery and sensor status	FR38–FR42
6	Historical readings	FR59–FR64
7	Alerts for abnormal conditions and inactive sensors	FR51–FR56
8	A clear dashboard for each location	FR08–FR14

The stakeholder identified advanced analytics, prediction and complex reporting as suitable for a later version. Those are not included in this specification.

15. Agricultural Threshold Basis

Thresholds are configurable per location and per parameter. The values below are the initial settings, established from the South African Department of Agriculture production guideline for pigeon pea and from standard soil salinity classification. They are a starting point for configuration, not fixed limits, and the stakeholder may adjust any of them.

Parameter	Minimum	Maximum	Basis
pH	5.0	7.0	Pigeon pea production guideline. Excessive acidity inhibits nodulation and causes chlorosis and die-back.
Soil temperature	18 °C	35 °C	Pigeon pea production guideline growing range. See note below.
Soil moisture	20 %	60 %	Growth-stage dependent. See note below.
Electrical conductivity	0	2 dS/m	Standard salinity classification. Above 2 dS/m is slightly saline; above 4 dS/m is saline.
Nitrogen	Configurable	Configurable	Pigeon pea fixes atmospheric nitrogen and shows little response to nitrogen fertiliser. Configured against laboratory soil analysis.
Phosphorus	Configurable	Configurable	Configured against laboratory soil analysis.
Potassium	Configurable	Configurable	The guideline notes that adequate potassium is required. Configured against laboratory soil analysis.

15.1 Stated Limitations of the Initial Values

11. The production guideline gives air temperature, not soil temperature. The soil temperature range above is an approximation applied from that figure and should be reviewed against local observation.
12. Soil moisture requirements vary by growth stage. Pigeon pea prefers moist conditions during its first two growing months and drier conditions during flowering and harvesting. A single fixed range will produce alerts during flowering that do not indicate a problem. Stage-aware thresholds are recorded as a later enhancement.
13. Nitrogen, phosphorus and potassium have no single acceptable range independent of the soil. These are configured from the laboratory analysis the organisation already obtains at the start of each planting cycle.
14. The system reports whether a measured value falls inside or outside a configured range. It does not recommend fertiliser, irrigation or any other intervention.

15.2 Agreed Position on Calibration

The stakeholder reviewed these values and confirmed the approach in writing on 3 September 2026, noting that the calibration and setting of range measurements can be amended once the platform has been established, and that the specification meets the entry requirements of the end user as it stands.

This settles a question the delivery team could not answer alone. The initial values above are a working starting point rather than final limits. Because every range is stored in the backend and configured per location, refining

them later requires no change to the applications and no new development. The pipeline will be revisited with the stakeholder once the system is running and real readings can be compared against the ranges in use.

16. Minimum Database Entities

An entity is a kind of information the system stores. A location is one entity, a reading is another. Listing them here establishes what the system must keep a record of. How they relate to one another, and the exact structure of each, is settled during design.

The final structure and relationships are produced during system design. The entities below are the minimum required to satisfy the requirements in this document.

Entity	Purpose	Key attributes
Users	System users and assigned roles	Identifier, email, role, account status, created date
Locations	The three monitoring sites	Identifier, name, description, created date
Sensors	The fifteen registered devices	Identifier, name, location, model, installation date, active status
Readings	Individual measurements with timestamps	Identifier, sensor, timestamp, moisture, temperature, conductivity, pH, nitrogen, phosphorus, potassium
Sensor status	Operational condition of each device	Identifier, sensor, battery level, power status, sensor status, last communication, recorded at
Thresholds	Acceptable ranges per parameter and location	Identifier, location, parameter, minimum, maximum, basis, updated by, updated at
Alerts	Automatically generated warnings	Identifier, sensor, location, alert type, parameter, measured value, severity, state, raised at, acknowledged by, acknowledged at

17. Feasibility Study

The assessment below covers whether the system as specified can be delivered within the available time, skills and resources.

17.1 Technical

The application layer uses frameworks the delivery team has worked with previously. The required functions are conventional application development: authenticated access, list and detail screens, form-driven management, filtered queries, charting and export.

The uncertainty that existed at the start of the project concerned retrieving data from a system operated by a separate party. The stakeholder has since confirmed that live integration is not required for this version and that representative sample data is acceptable. That removes the dependency the delivery team did not control.

Skills still to be developed are database access rules, scheduled backend functions, and charting in both frameworks. Time for this is allocated before development begins.

Decision	Feasible
Reason	The required technologies are within the team's demonstrated capability, and the external dependency that carried the technical risk has been removed from this version.

17.2 Operational

The system requires no change to how measurements are produced and no manual data entry by users. Users read information rather than supplying it, which keeps the burden of adoption low.

Two roles were confirmed by the stakeholder as sufficient. Configuration functions are restricted to administrators, because a threshold changed by one user alters the alerts every other user sees.

Guidance required is minimal for the mobile application, which is read-only. Brief guidance is needed for threshold configuration and user management on the web application.

Decision	Feasible
Reason	The solution fits the organisation's existing activities, the user roles are confirmed, and the functions map directly to the difficulties described in Section 2.3.

17.3 Economic

This assessment covers whether the delivery team has access to the resources required, rather than a business financial analysis.

Resource	Position
Development tools	Flutter, the .NET SDK, the development environments and version control are available at no cost.
Shared backend	Supabase operates within its free tier at the volume this prototype generates.

Resource	Position
Hardware procurement	None. The sensing equipment is owned and operated by a separate party.
Development machines	Confirmed available across the team.
Test devices	At least one Android device is required for the demonstration.
Internet access	Required by both applications and by the shared backend.
Presentation resources	Venue, display and a reliable connection for the demonstration.

Decision	Feasible
Reason	All required software operates within free tiers, no hardware procurement falls to the delivery team, and the resources needed are available.

17.4 Schedule

Approximately three weeks are available for development across a mobile application, a web application and a shared backend. The delivery team has eight members, which permits parallel work, but components developed separately require time to combine and that time is planned rather than assumed.

Removing the live integration from this version reduces the work by one component and removes the item that could not be designed until an external data format was known.

Required changes

15. Development is planned backwards from the acceptance criteria. Work not contributing to one of them is deferred.
16. The database and access rules are completed before parallel application development begins, because both applications depend on them.
17. Time for combining components is reserved within the development window rather than added at the end.
18. Each of the mobile, web, backend and provisioning components has a named owner.

Decision	Feasible with changes
Reason	The available time is sufficient only if scope is restricted to the acceptance criteria and the shared backend is completed before parallel development begins.

17.5 Legal

Area	Position
Personal information	The system stores user accounts only: email address, assigned role and account status. Measurements are not personal information.
Data protection	Handling of personal information is subject to the Protection of Personal Information Act. Restricting the prototype to test accounts keeps the project outside the handling of real personal information.
Data ownership	Measurements produced by the sensing equipment belong to the operating party. This version does not access them.
Credentials	Backend keys and passwords are held outside the source repository and are not distributed in client applications.
Confidentiality	Stakeholder correspondence is held as evidence. Contact details and identifiers are excluded from the source repository.
Attribution	External sources used in this document are cited where used.

Decision	Feasible
Reason	The prototype handles no real personal information, does not access third-party data in this version, and the confidentiality request has been applied.

17.6 Ethical

Area	Position
Purpose understood	The stakeholder understands that this is a prototype with a defined delivery period, not a production system.
Information collected	Only the account details required for authentication and role assignment.
Risk of over-reliance	If the system presented threshold comparison as agronomic advice, a user might act on a recommendation the system is not qualified to make. The system reports measured values against configured ranges and offers no crop or treatment recommendation. This is recorded as NFR17.
Use of information	Measurements are used to display conditions, raise alerts and produce reports for the organisation, and for no other purpose.
Fair treatment	Access is assigned by role rather than by individual. Incomplete functions are documented rather than presented as working.

Decision	Feasible
Reason	No unnecessary information is collected, information is used only for its stated purpose, and an explicit boundary is recorded against advice the system is not qualified to give.

17.7 Summary

Area	Decision	Main reason	Change required
Technical	Feasible	Required technologies within the team's capability; external dependency removed from this version	None
Operational	Feasible	Users read information rather than entering it; roles confirmed	Configuration restricted to administrators
Economic	Feasible	Software free or already held; no hardware procurement	None
Schedule	Feasible with changes	Approximately three development weeks across three components	Restrict scope to the acceptance criteria; complete the backend before parallel work
Legal	Feasible	No real personal information; no third-party data accessed in this version	None
Ethical	Feasible	Explicit boundary against agronomic advice	None

17.8 Conclusion

The proposed project is feasible with changes.

The main strengths of the project are that the required technologies are within the delivery team's demonstrated capability, that the software and services needed are available at no cost, that no hardware procurement or maintenance falls to the delivery team, and that the system requires no change to how the organisation currently produces its measurements. Users read information rather than supplying it, which keeps the burden of adoption low.

The main limitation identified is time. Approximately three weeks are available for development across a mobile application, a web application and a shared backend, and eight people working in parallel carry an integration cost that must be planned rather than assumed.

To ensure that the project can be completed within the available period, the delivery team will plan development backwards from the acceptance criteria and defer work that does not contribute to one of them, complete the database and access rules before any parallel application development begins, reserve time for combining components within the development window, assign a named owner to each component, and defer the additional features listed in Section 20 until the required functions are complete.

18. Project Risk Register

A risk is something that may happen and would affect the project if it did. It is recorded before it happens, so that action can be taken in advance rather than in response. Something that has already happened is not a risk but an issue, and is handled separately.

Each risk below is rated for how likely it is and how serious it would be. Multiplying the two gives a score that allows risks to be compared and ranked. Each also carries a mitigation, meaning what will be done in advance to reduce it, and a contingency, meaning what will be done if it happens anyway. Every risk has one named person responsible for watching it.

18.1 Rating Scale

Rating	Likelihood	Impact
1	Unlikely to occur	Minor delay or inconvenience
2	Possible	Noticeable effect requiring action
3	Likely	Serious effect on delivery or quality

Risk score = likelihood × impact. Scores of 1 to 3 are Low, 4 to 6 are Medium, and 7 to 9 are High.

18.2 Register

ID	Risk	Category	L	I	Score	Level
R01	Development time may be insufficient for the full scope, leaving required functions incomplete	Schedule	3	3	9	High
R02	Components developed in parallel may fail to integrate, so that parts work individually but not together	Technical	2	3	6	Medium
R03	Credentials may be committed to the repository, exposing keys that deletion alone cannot withdraw	Security	2	3	6	Medium
R04	Threshold values may prove unsuitable for the sites, producing alerts that do not indicate a real problem	Quality	2	3	6	Medium
R05	Contribution may be uneven across an eight-member team, concentrating work on a few people	Team	2	2	4	Medium
R06	Scope may expand during development as features are added mid-build	Scope	2	2	4	Medium
R07	A team member may become unavailable, leaving a component without an owner	Team	2	2	4	Medium
R08	The stakeholder may be unavailable when confirmation or evaluation is required	Stakeholder	2	2	4	Medium
R09	Sample data may not resemble real readings closely enough to demonstrate	Quality	2	2	4	Medium

ID	Risk	Category	L	I	Score	Level
	the system convincingly					
R10	Learning time for access rules and scheduled functions may exceed the allowance made	Technical	2	2	4	Medium
R11	Free service limits may be reached as readings accumulate	Resource	1	2	2	Low
R12	Later live integration may reveal a data structure the current design cannot accommodate	Technical	2	1	2	Low

18.3 Response and Ownership

ID	Mitigation	Contingency	Owner
R01	Plan development backwards from the acceptance criteria; complete the backend before parallel work; reserve integration time explicitly	Deliver Must Have requirements only; document incomplete functions rather than presenting partial features as complete	Project manager
R02	Finalise the database structure before application development begins; agree field names and types in writing; integrate early rather than at the end	Reserve time in the final week for correction; prioritise the integrated demonstration over individual features	Technical lead
R03	Commit an ignore file before any configuration exists; hold keys in environment variables; brief every member before their first commit	Rotate the exposed key immediately. The file must be removed and the credential replaced, as deletion does not remove it from history	Technical lead
R04	Record the basis of every threshold; make all values configurable; state the known limitations in Section 15.1	Adjust values with the stakeholder and record the change in the change log	Specification team
R05	Assign named responsibility for every task; record contributions with evidence; review progress at a fixed weekly meeting	Reallocate outstanding work and record the reallocation	Project manager
R06	Record every requirement with a priority; treat Should Have and Could Have as excluded from the initial build; log any change with a reason	Return added work to the backlog and document the decision	Specification team
R07	Ensure at least two members understand each component; keep all work in the shared repository rather than on individual machines	Redistribute the affected work and record the change	Project manager
R08	Agree confirmation and evaluation dates in advance; put requests in writing	Proceed on previously confirmed findings and request written feedback	Project manager
R09	Generate values within realistic ranges, including occasional threshold breaches and periods of sensor inactivity	Adjust the generator; demonstrate against a prepared data set	Backend developer
R10	Allocate explicit learning time before development begins; build a small working example of each first	Simplify the affected function and document the simplification	Technical lead
R11	Monitor database size; store readings at the defined interval rather than at maximum frequency; use summarised data for long-range charts	Archive older readings; reduce the provisioning interval	Backend developer
R12	Isolate all provisioning in one component so that a replacement affects nothing else	Adapt the replacement component only; no other component changes	Technical lead

18.4 Highest Risks

R01 is the only High risk. R02, R03 and R04 share a score of 6 and are treated as the next priority.

Rank	Risk	Why it matters	Action now
1	R01 — Insufficient development time	Approximately three weeks are available for fourteen acceptance criteria across three components. Scope control is the only available response and it must be applied before development starts, not during it.	Treat the refined scope in Section 20 as closed. Sequence the backend first. Assign a named owner to each component.
2	R02 — Integration failure	The prototype is evaluated as one integrated system. Components that work individually but do not combine would not satisfy the integration requirements, and integration problems surface late unless deliberately tested early.	Agree the database structure and field names in writing before any application work begins. Build one end-to-end path early.
3	R04 — Unsuitable thresholds	Alerts are the point of the system. Thresholds that fire when nothing is wrong, or stay silent when something is, would undermine the whole prototype in front of the stakeholder.	Confirm the initial values in Section 15 with the stakeholder. Keep every value configurable and record its basis.

18.5 Monitoring

The register is reviewed at each weekly progress meeting. For each open risk the delivery team records whether likelihood or impact has changed, whether the mitigation has been carried out, and whether the risk can be closed. New risks are added with the same fields. A risk that materialises is recorded as an issue, together with the action taken.

19. Requirements Change Log

A specification changes as understanding improves. Rather than quietly editing the document, every change is recorded here with the reason for it, so that the history of what was agreed remains visible to both sides.

A requirement that has been set aside is never simply deleted. It is recorded below, with the reason, so that it can be revisited in a later version without having to be rediscovered.

Ref	Original position	Change	Reason
C01	Readings retrieved from the sensor operator's published data store and written to the shared backend	Replaced by a data provisioning component using representative sample data	The stakeholder confirmed in writing that live integration is not required for this version and that sample data is acceptable
C02	Live data retrieval listed as a project deliverable	Recorded as a later integration step	Follows from C01
C03	Battery and device status dependent on fields being published by the operator	Included as a confirmed requirement	The stakeholder confirmed that battery level, sensor status and last communication time must be present in the sample data
C04	An eighth parameter, humidity, had been recorded in an earlier draft	Removed. The specification records seven measured parameters	The stakeholder requirements document, section 3, defines seven soil parameters, and the sample reading fields supplied in the follow-up responses list the same seven. Humidity was not a stakeholder requirement
C05	User roles undefined	Fixed at two roles: Administrator and Standard User	The stakeholder confirmed two roles are sufficient for this version, with more added later if required
C06	Thresholds required but no values available	Initial values established from the pigeon pea production guideline and salinity classification, all configurable	The stakeholder confirmed the crop as pigeon pea and did not specify values
C07	Sensor inactivity undefined	A sensor is inactive after two hours without a reading, configurable	Reading interval confirmed at thirty minutes; two hours represents four consecutive missed readings
C08	Reading frequency unspecified	Set at thirty minutes	Confirmed by the stakeholder
C09	Notification method unspecified	Push notification recorded as the requirement, delivered as in-application alerts in the first version	Confirmed as the preferred channel. Simplified for the first version to reduce integration work
C10	Complex reporting included	Reduced to export of readings and alerts	The stakeholder identified advanced analytics, prediction and complex reporting as suitable for a later version
C11	Prediction and analytics	Removed	Identified by the stakeholder as a later

Ref	Original position	Change	Reason
	under consideration		version
C12	System to interpret readings and recommend action	Boundary recorded as NFR17	The system compares values against configured ranges and does not offer agronomic advice
C13	Historical retention unspecified	Set at twelve months	Proposed by the delivery team; recorded as NFR16
C14	Threshold values researched by the delivery team and not yet agreed	Confirmed as a working starting point, to be refined once the system is running	The stakeholder confirmed on 3 September 2026 that calibration and range settings can be amended after the platform is established

Changes made after version 4.0

The changes made while the system was built, C26 to C47, continue this log. They are recorded in Section 23.4 (As-Built System and Final Status). A requirement that was changed or not built keeps its ID, and Section 23.6 gives its final status.

20. Refined Project Scope

Following the feasibility assessment in Section 17 and the risks in Section 18, every proposed function was reviewed against the time available. Each is marked with one of four decisions.

Decision	Meaning
Keep	Included as originally planned
Simplify	Included, but in a reduced form for this version, with the fuller version to follow
Postpone	Not built in this version, but recorded as a later enhancement rather than abandoned
Remove	Excluded, because it does not belong in this system

20.1 Required Functions

Function	Decision	Reason
Authentication and role-based access	Keep	First in the stakeholder's priority order and a dependency of every other function
Dashboard summary	Keep	Acceptance criterion AC2
Locations and sensor navigation	Keep	Acceptance criteria AC3 and AC4
Display of all seven parameters	Keep	The core purpose of the system
Battery, power and sensor status	Keep	Confirmed by the stakeholder as required
Inactive sensor detection	Keep	Named by the stakeholder in the priority order
Threshold configuration	Keep	Without it alerts have no basis
Alert generation and viewing	Keep	Acceptance criteria AC7 and AC8
Historical readings and graphs	Keep	Acceptance criterion AC9
Location comparison	Keep	Acceptance criterion AC12
User administration	Keep	Required for the organisation to operate the system independently
Export of readings	Keep	Acceptance criterion AC14
Data provisioning	Keep	Without it the system holds no data

20.2 Additional Functions

Function	Decision	Reason
Push notification	Simplify	Delivered as in-application alerts in the first version. Background push added once the required functions are complete.
Live dashboard updates	Simplify	Refresh on view rather than a continuous

Function	Decision	Reason
		subscription, which reduces integration work
Alert acknowledgement	Keep	Low cost, and without it the alert list becomes unusable as alerts accumulate
Alert filtering	Keep	Required once alert volume grows across fifteen sensors
Complex reporting and analytics	Remove	Identified by the stakeholder as suitable for a later version
Prediction and forecasting	Remove	Identified by the stakeholder as suitable for a later version
Offline mode	Postpone	The most time-intensive item proposed. Reconsidered if connectivity at the sites proves poor.
Map view of locations	Postpone	Adds a mapping dependency without contributing to an acceptance criterion
Sensor selection by code scanning	Postpone	Selection from a list achieves the same outcome
Biometric sign-in	Postpone	Standard authentication satisfies the access requirement
Site visit logging with photographs	Postpone	Introduces file storage and a write path from an application that currently reads only
Weather overlay	Postpone	Adds an external dependency without contributing to an acceptance criterion
Dark mode and animated transitions	Postpone	Presentation only
Home screen widget	Postpone	Platform-specific work outside the required scope
Interface language selection	Postpone	Retained as a later enhancement. Translation requires verification by a first-language speaker and is not undertaken until the required functions are complete.

21. Supporting Evidence

Appendix	Evidence	Status
A	Follow-up questions issued to the stakeholder	Held
B	Stakeholder written responses to the follow-up questions	Held
C	Stakeholder requirements document supplied by the organisation	Held
D	Stakeholder written confirmation of the findings, Ms Goitsemang Dikane, 3 September 2026	Held
E	Engagement record: meeting dates, attendees and correspondence	Held

Contact details, identity numbers, credentials and confidential records are excluded from the appendices and from the source repository.

21.1 Confirmation

A written summary of the findings was issued to the stakeholder for review. Ms Goitsemang Dikane confirmed the findings and the requirements recorded in this document on 3 September 2026, stating that the document meets the entry requirements of the end user and can be used as it stands.

She further noted that minor corrections can be amended as the work proceeds, and that the calibration and setting of range measurements can be revisited once the platform has been established. Those positions are recorded in Section 15.2 and in change log entry C14. The confirmation is held as Appendix D, and this version is issued as the approved baseline for design and development.

Confirmation	Detail
Reviewed by	Ms Goitsemang Dikane
Position	Plant Sciences (nGAP), Department of Agriculture — stakeholder and end user
Date of confirmation	3 September 2026
Findings confirmed as accurate	Yes
Requirements approved	Yes — the document meets the entry requirements of the end user
Corrections required	None blocking. Minor corrections to be amended as the work proceeds.
Outstanding for a later stage	Calibration and range settings to be revisited once the platform is established
Evidence	Written confirmation held as Appendix D

21.2 Sources

Reference	Source
S1	Department of Agriculture, Forestry and Fisheries, Republic of South Africa. Pigeon peas production guideline. Directorate: Plant Production.
S2	Food and Agriculture Organization of the United Nations. Crop salt tolerance data, salinity classification by electrical conductivity.
S3	Project requirements document, Multi-Site Smart Soil Monitoring System, supplied by the stakeholder organisation.
S4	Stakeholder written responses to the follow-up questions issued by the delivery team.

21.3 Project Repository and Backend

Project documentation and source code are held in a single version-controlled repository. The shared backend is a single Supabase project used by both applications.

Item	Detail
Source repository	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Repository name	soil-sensor-monitoring
Repository owner	Nido-Maphosa86
Repository contents	README, documentation folder holding this specification and its appendices, the project schedule, and the source code for the mobile application, the web application and the backend
Access	All eight members of the delivery team have write access
Supabase project	https://supabase.com/dashboard/project/dsejhpcdshrkpmaewbcq
Supabase project reference	dsejhpcdshrkpmaewbcq
Backend contents	PostgreSQL database, authentication, access rules and storage for readings, sensor status, thresholds and alerts

No password, API key, service role key, telephone number, identity number or confidential organisational record has been committed to the repository at any point. A .gitignore file was committed before any configuration file was created.

22. System Architecture and Design

Section 22 is new in version 4.0 of this specification. It records the system architecture and design that the delivery team will build from: the architecture, the UML diagrams, the database design, the interface designs, and the way each of them traces back to the approved requirements.

Sections 1 to 21 are the approved baseline (version 3.0) and are not rewritten. Where the design changes a baseline requirement, Section 22.7 records the change with its reason, so the history of what was agreed stays visible. Appendices A to E remain the stakeholder evidence.

The design at a glance

- One Firebase project is the shared backend. The Flutter and ASP.NET applications read and write the same Realtime Database, so a reading, a threshold or an alert exists once and both applications show it.
- Each application reaches Firebase in the way that suits it. The phone uses the Firebase SDK under Security Rules. The web server calls the REST API with a service-account token and checks the user's role itself before every write.
- Four rules are written once and enforced everywhere: a value outside its band raises an alert, a sensor silent for two hours raises an Offline alert, Acknowledge and Resolve both need a note, and a Standard User sees only assigned locations.
- Readings and sensor status have one writer: the sensor gateway, which writes the actual readings of the 15 sensors every 30 minutes.
- Thirteen entities, 26 use cases, seven sequence diagrams and 32 screens and dialogs all trace back to the approved requirements. Section 22.7 records the changes this design makes to the baseline.

Sections of this part

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22.1 System Architecture

22.1.1 Overview

SensorGrid is one system made of two client applications and one shared backend. The Flutter mobile application serves field monitoring on Android. The ASP.NET MVC web application serves administration and analysis in a browser. Both connect to the same Firebase project. A third-party sensor gateway writes the readings.

Platform decision.
The shared backend is Firebase: Firebase Authentication, the Firebase Realtime Database and Cloud Storage for Firebase. This replaces Supabase in SR03, SR05, SR06 and IR01 and is recorded as change C15 in Section 22.7. The structure the requirements describe is unchanged: one shared backend, a Flutter client and an ASP.NET client.

22.1.2 Users and Roles

User or system	Application used	Main purpose
Administrator	ASP.NET web application. An Administrator account can also sign in on the phone and then has Standard User functions only.	Oversees all three locations. Sets thresholds, manages user accounts and location access, reviews readings, alerts and reports, and exports readings.
Standard User (field worker, farm manager, researcher)	Flutter mobile application. The web application is available as a fallback.	Monitors the locations assigned to them: current readings, sensor status, trends and alerts. Acknowledges or resolves alerts with a note and downloads reports.
Sensor gateway (external system)	None. Writes straight to Firebase.	A third party operates the ESP32 gateway and the 15 RS485 soil sensors. It writes one reading and one status record per sensor every 30 minutes.

22.1.3 Components and Responsibilities

Component	Main responsibility
Flutter mobile application	Field monitoring on an Android phone: dashboard, locations, readings, trends, sensor status, alerts (acknowledge and resolve with a note), report download, profile and notification preferences. Talks to Firebase directly through the Firebase SDK.
ASP.NET MVC web application	Administration and analysis: thresholds, user accounts and location access, locations and sensors, readings log with CSV export, alerts and reports. Talks to Firebase only from the server, so who may do what is checked on the server before anything is written.
Firebase Authentication	Verifies email and password, creates accounts, sends reset and setup emails. It does not hold roles; the role and the assigned locations are stored in the Realtime Database, under the user's node.
Firebase Realtime Database	The single database for users, locations, sensors, sensor status, readings, thresholds, alerts, notifications, reports and data-load records. It stores everything as one JSON tree that both applications read and write, and it can push a change to an app that is listening. Protected by Firebase Security Rules.
Cloud Storage for Firebase	Holds the generated report files (PDF and CSV). The web application writes them; both applications download them. Needs the paid Blaze plan (22.7.4).

Component	Main responsibility
Cloud Functions	Alert evaluator: compares each new reading with its location band, checks for silent sensors and writes alerts and notifications. Needs the paid Blaze plan, so where it runs is open (22.7.4). The rule is specified once, in 22.1.7.
Sensor gateway (external)	Reads the sensors and writes readings and sensor status to the Realtime Database every 30 minutes. Operated by a third party, outside the delivery team's scope.

22.1.4 Architecture Diagram

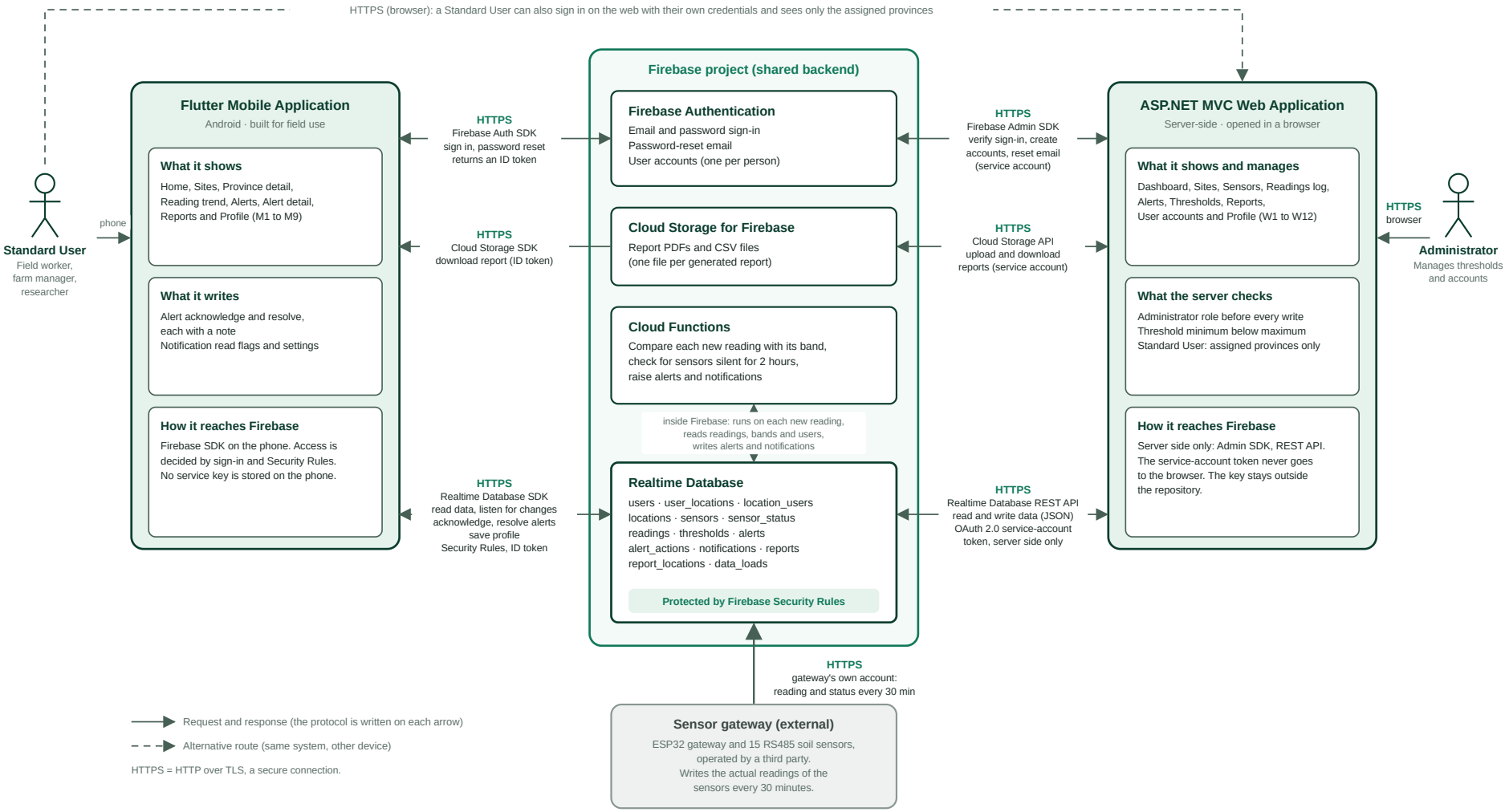


Figure 22.1 — System architecture. Solid arrows are requests and responses over HTTPS. The dashed route is the alternative web access for a Standard User.

22.1.5 Main Data Flows

From	To	Information exchanged	How
Standard User	Flutter app	Views readings, status, trends and alerts. Acknowledges or resolves alerts with a note. Downloads reports.	Phone
Standard User	ASP.NET app	Signs in from a browser when there is no phone. Sees only the assigned locations.	Browser (alternative route)
Administrator	ASP.NET app	Thresholds, accounts, locations, sensors, readings log, exports, alerts and reports.	Browser
Flutter app	Firebase Authentication	Sign in, sign out, password-reset email.	Firebase SDK over HTTPS
Flutter app	Realtime Database	Reads the user's own record and assigned locations, then the sensors, status, readings, thresholds, alerts and notifications of those locations. Writes alert acknowledge and resolve, notification read flags and notification preferences.	Firebase SDK over HTTPS, under Security Rules
Flutter app	Cloud Storage	Downloads report files.	Firebase SDK over HTTPS
ASP.NET app	Firebase Authentication	Verifies the sign-in token, creates accounts, sends password-setup emails, changes passwords.	Admin SDK, server side
ASP.NET app	Realtime Database	Reads and writes users, locations, sensors, thresholds, alerts and report records. Reads readings and status.	REST API with a service-account token, server side
ASP.NET app	Cloud Storage	Uploads generated reports and serves downloads.	Server side, service account
Sensor gateway	Realtime Database	One reading and one status record per sensor every 30 minutes.	HTTPS, the gateway's own account
Alert evaluator	Realtime Database	Reads each new reading, the location band and the location's users. Writes alerts and notifications.	Inside Firebase

22.1.6 One Integrated System

Who uses what. Standard Users use the Flutter application. Administrators use the ASP.NET web application. A Standard User can also sign in on the web with their own credentials, for example when they do not have their phone, and sees only the assigned locations.

What Firebase does. Firebase Authentication checks who is signing in. The Realtime Database stores all project data in one place and can push a change to an app that is listening. Cloud Storage holds the report files. The sensor gateway writes readings into the same database, so a reading is stored once and both applications show it.

How each application reaches Firebase. The Flutter application uses the Firebase SDK on the phone, under Security Rules. The ASP.NET application talks to Firebase only from the server. It calls the Realtime Database REST API with a service-account access token that is never sent to the browser.

One writer per kind of data. Neither application keeps its own copy of the data, and each kind of data has one writer, so there is never a second source of truth.

Data	Written by	Read by
Readings, sensor status	Sensor gateway only	Both applications

Data	Written by	Read by
Data-load records	Sensor gateway, one record per load	Administrator on the web
Alerts (creation)	Alert evaluator only	Both applications
Alerts (state, notes, who and when)	Standard User or Administrator, in either application	Both applications
Notifications	Alert evaluator creates them. The owner marks them read.	The owner, in both applications
Thresholds	Administrator, on the web only	Both applications
Users, role, assigned locations	Administrator, on the web only. A user edits only their own notification preferences.	Both applications
Locations and sensors	Administrator, on the web only	Both applications
Reports	Web application	Both applications

Configuration flows across. A band saved on the web is used by the phone and by the next alert check without further action. When a user resolves an alert with a note on the phone, the Administrator sees the alert as Resolved and can read the note on the web.

22.1.7 Business Rules Enforced by the Design

Rule	Definition	Source
One reading, one record	A reading is stored under its sensor and its capture time to the minute (for example readings/FS/FS-01/202609081430). The same sensor and time can only ever be one record, and the rules let the gateway create a reading but not overwrite one, so duplicates cannot occur.	FR77
Threshold breach	For every new reading, each of the seven values is compared with the minimum and maximum of its location band. A value outside the band raises a Breach alert for that parameter. The system compares values and never gives agronomic advice.	FR38, FR51, NFR17
Offline sensor	A sensor with no reading for 2 hours (four missed 30-minute readings) is marked Offline and raises one Offline alert with no parameter. The period is two hours for every sensor.	FR43, FR52, C07
Low battery	A sensor reporting a battery level below 25 percent raises a Warning alert for the parameter Battery. The level is 25 percent for every sensor.	FR39, FR52
Severity and state	Severity (Critical or Warning) says how serious the alert is. State (Open, Acknowledged, Resolved) says where it is in handling. They are separate fields. The rule that picks Critical or Warning is a setting of the evaluator, to be agreed with the stakeholder.	FR53, FR57
Note required	Acknowledge and Resolve both need a note. Each has its own note, user and time, stored as an Alert_action record of the alert.	FR57, FR80, FR81
Valid band	A threshold minimum must be below its maximum. Every band records its basis, who set it and when.	FR46, FR48, NFR06, NFR19
Location access	A Standard User sees only the locations assigned to them, in both applications.	FR82
Notifications	Each new alert creates one notification for every user assigned to its location.	FR58
Sensor identity	A sensor code (for example FS-06) is unique everywhere and cannot be changed. A Modbus slave ID only has to be unique within one location.	FR27–FR30

Rule	Definition	Source
Humidity	Humidity is not stored. The seven soil parameters are the only reading values.	C04, C17

22.1.8 Firebase Services and Security

Firebase service	Used for	Used by	Notes
Authentication	Email and password sign-in, account creation, reset and setup emails, password changes	Both	Roles are not stored here. One account per person.
Realtime Database	All structured data (13 entities, see 22.4)	Both, plus gateway and evaluator	One JSON tree. The phone can listen for changes; the web server reads on request.
Security Rules	Access control on the database and on Storage	Flutter app (direct access) and the gateway	The web server's token bypasses the rules, so the web server checks the role itself.
Cloud Storage	Report files	Both	One file per generated report. Path stored on the report record. Needs the Blaze plan.
Cloud Functions	Alert evaluator and offline check	Evaluator only	Needs the Blaze plan (22.7.4).

- **Identity.** Firebase Authentication proves who the user is. The user's node in the database (users/{user_id}) holds the role and the account status, and the node user_locations/{user_id} lists the assigned location IDs (FR04, FR05, FR82).
- **Direct access from the phone.** Security Rules refuse any request that is not signed in (FR06). The data of each location is stored under its location_id, so one rule checks the path against the user's own user_locations node and limits a Standard User to assigned locations. A Standard User may only add the acknowledge or resolve record of an alert and move its state forward, and only with a non-empty note (FR81).
- **Access from the web server.** The ASP.NET application talks to Firebase only from the server. It calls the Realtime Database REST API with a service-account access token. That token bypasses Security Rules, so the web server checks the caller's role before every write (shown in Sequence Diagrams 2 to 4).
- **Secrets.** The Firebase service-account key stays outside the source repository and is loaded from the server's environment (NFR03). The mobile application contains only the public Firebase project identifiers, never a service credential (NFR04); access is decided by sign-in and Security Rules.
- **Passwords.** Passwords are held and hashed by Firebase Authentication; neither application stores them (NFR01).

Cost and limits. The no-cost Spark plan covers the Realtime Database for this load. Cloud Storage and Cloud Functions are the two items that need the paid Blaze plan (see 22.7.4).

Spark plan limit	Allowance	How the design stays inside it
Simultaneous connections	100	Each open app that is listening for changes counts as one connection. A small group of users and the web server stay far below it.
Data stored	1 GB	About 0.1 GB a year of readings and status records (estimate: 720 of each a day, about 350 bytes per pair). Twelve months of retention (NFR16) fits.
Data downloaded	10 GB a month	A listener downloads only what changes. Trend and log queries read a time range under one sensor and never the whole readings branch.

Spark plan limit	Allowance	How the design stays inside it
Cloud Storage and Cloud Functions	Not on Spark	Both need the paid Blaze plan. Report files and the alert evaluator depend on it (22.7.4).

22.1.9 Non-Functional Requirements Met by the Design

Requirement	How the design meets it
NFR02	Access rules are enforced in Firebase Security Rules and again by the web server before every write, never only in the screens.
NFR03, NFR04	The service-account key stays on the server, outside the repository. The phone holds only public project identifiers.
NFR06, NFR19	Every threshold stores basis, updated_by and updated_at.
NFR08	Each parameter is shown in a traffic-light state (in range, warning, breach) on M4 and W4.
NFR15	A failed data load leaves existing readings in place. Both applications keep reading them, and the failure is recorded in Data_load.
NFR16	Readings are never deleted automatically, so at least twelve months are retained.
NFR17	The evaluator only compares a value with its band. Screens state the comparison and give no treatment advice.
NFR18	Every reading carries captured_at and every screen that shows a reading shows that time.

22.2 Use Case Diagram

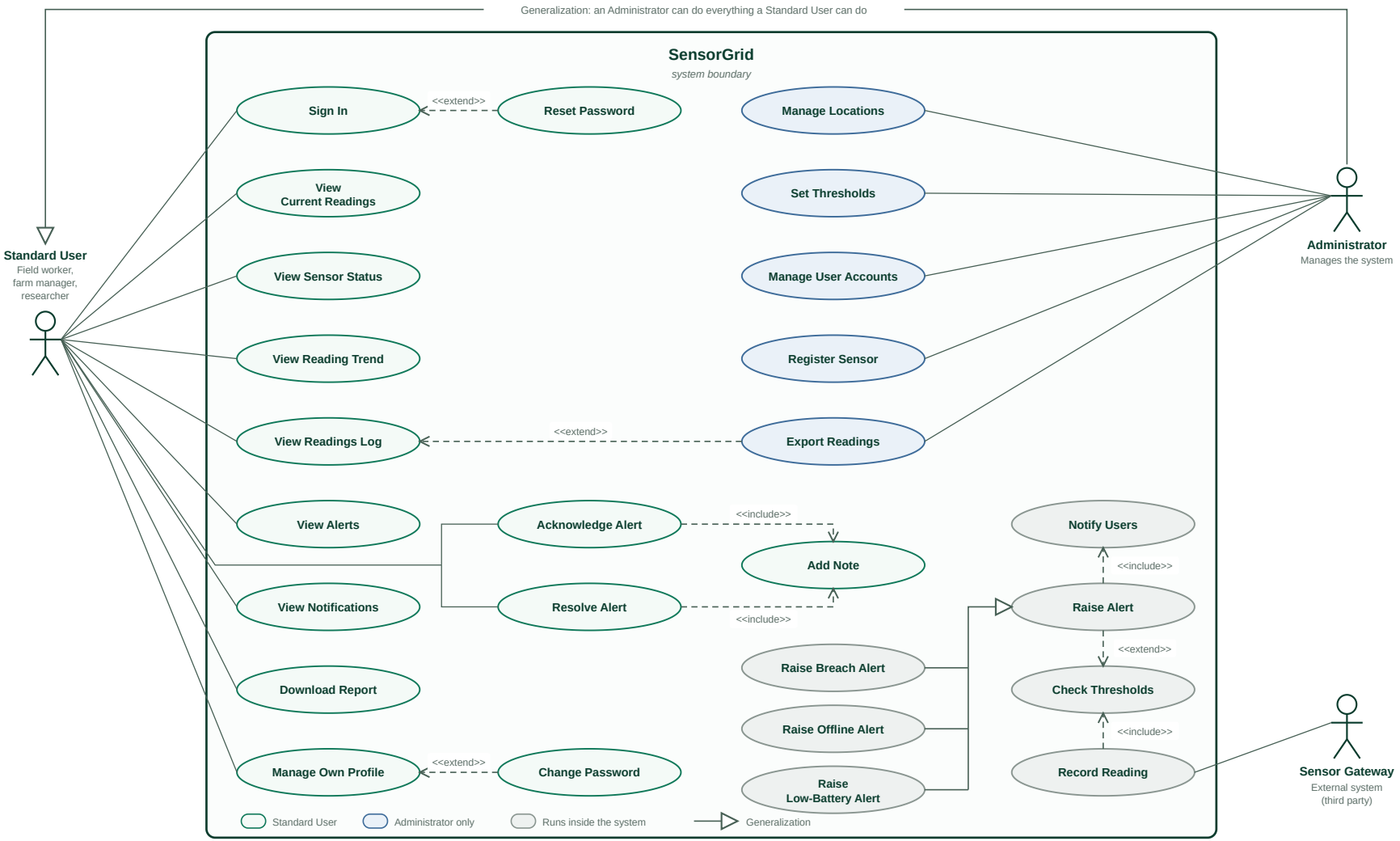


Figure 22.2 — Use case diagram. Green: Standard User. Blue: Administrator only. Grey: performed by the system when a reading arrives. A hollow triangle arrow shows generalization.

The diagram above shows the SensorGrid system boundary, the three actors and 26 use cases. The tables describe each actor and use case and the requirements they serve.

22.2.1 Actors

Actor	Type	Role in the system
Standard User	Primary	Monitors assigned locations, handles alerts, downloads reports and manages their own profile.
Administrator	Primary. A specialisation of Standard User.	Inherits every Standard User use case, and also manages locations, sensors, thresholds and user accounts, and exports readings.
Sensor Gateway	External system	Writes readings and sensor status. Starts the automatic use cases that check thresholds, raise alerts and notify users.

22.2.2 UML Relationships Used

- **System boundary.** The rectangle labelled SensorGrid contains every use case. The three actors sit outside it.
- **Associations.** A line joins each actor to the use cases that actor starts. Standard User use cases are green; Administrator-only use cases are blue; use cases the system performs by itself are grey.
- **Generalization of actors.** The Administrator is a specialisation of the Standard User. A hollow triangle arrow runs from the Administrator to the Standard User. An Administrator can do every Standard User use case (the green ones) as well as the Administrator-only use cases (the blue ones), so the lines to the green use cases are drawn once, from the Standard User.
- **Generalization of use cases.** Raise Alert is the general use case. Raise Breach Alert, Raise Offline Alert and Raise Low-Battery Alert are its specialisations, and a hollow triangle arrow points from them to Raise Alert. Each specialisation does everything Raise Alert does, including Notify Users, and adds its own condition and alert type.
- **«include».** The base use case always runs the included one. Acknowledge Alert and Resolve Alert always include Add Note, because a note is required. Record Reading always includes Check Thresholds. Raise Alert always includes Notify Users.
- **«extend».** The extension runs only when its condition occurs. Reset Password extends Sign In when the password is forgotten. Export Readings extends View Readings Log when the Administrator chooses to export. Change Password extends Manage Own Profile when the user asks for it. Raise Alert extends Check Thresholds only when a value is outside its band, a sensor has been silent for two hours, or a battery is low.

22.2.3 Use Case Descriptions

ID	Use case	Actor	Description	Relationship	Requirements
UC01	Sign In	Both	Authenticate with email and password.	Extended by UC02	FR01–FR06
UC02	Reset Password	Both	Request a reset email when the password is forgotten.	Extends UC01	FR79
UC03	View Current Readings	Both	See the latest seven soil values per location and sensor, on the dashboard and site pages.		FR08–FR10, FR14–FR19, FR31–FR38
UC04	View Sensor Status	Both	See battery, power, signal, state and last communication per sensor.		FR11, FR12, FR23–FR26, FR39–FR42
UC05	View Reading Trend	Both	See one parameter over 24 hours, 7 days or 30 days, and compare locations.		FR60–FR65
UC06	View Readings Log	Both	Browse and filter the timestamped readings.	Extended by UC07	FR61, FR66

ID	Use case	Actor	Description	Relationship	Requirements
UC07	Export Readings	Administrator	Download the filtered readings or the alert history as CSV.	Extends UC06	FR67, FR68
UC08	View Alerts	Both	List breach, offline and low-battery alerts by severity and state.		FR13, FR19, FR54–FR56, FR59
UC09	Acknowledge Alert	Both	Mark an open alert as being worked on.	Includes UC11	FR57, FR81
UC10	Resolve Alert	Both	Close an alert once it has been dealt with.	Includes UC11	FR80, FR81
UC11	Add Note	Both	Record what was done. Always part of UC09 and UC10.	Included by UC09, UC10	FR81
UC12	View Notifications	Both	See unread alert notifications (tab badge, bell panel) and mark them read.		FR58, FR84
UC13	Download Report	Both	Generate and download a nutrient and in-range summary, a raw readings export or a threshold breach log.		FR69
UC14	Manage Own Profile	Both	See account details, assigned locations and notification preferences. Sign out.	Extended by UC15	FR03, FR85
UC15	Change Password	Both	Set a new password while signed in.	Extends UC14	FR07
UC16	Set Thresholds	Administrator	Set the minimum, maximum and basis of a band per parameter per location.		FR45–FR48, FR50
UC17	Manage User Accounts	Administrator	Invite users, set roles and status, assign locations.		FR70–FR74, FR83
UC18	Register Sensor	Administrator	Add or edit a sensor and assign it to a location.		FR27–FR30
UC19	Record Reading	Sensor Gateway	Store a reading and a status record for a sensor.	Includes UC20	FR75–FR78
UC20	Check Thresholds	System	Compare a stored reading with the location bands and check silent sensors and battery.	Included by UC19. Extended by UC21	FR38, FR51, FR43
UC21	Raise Alert	System	Create an alert when a check fails. The general use case: UC24 to UC26 are its specialisations.	Generalises UC24, UC25, UC26. Extends UC20. Includes UC22	FR51–FR53
UC22	Notify Users	System	Create a notification for each user assigned to the location.	Included by UC21	FR58
UC23	Manage Locations	Administrator	Create, edit or remove a location.		FR20–FR22
UC24	Raise Breach Alert	System	Create a Breach alert for the parameter whose value is outside its band.	Specialises UC21	FR38, FR51, FR53
UC25	Raise Offline Alert	System	Create an Offline alert, with no parameter, for a sensor that has sent nothing for two hours.	Specialises UC21	FR43, FR52, FR53
UC26	Raise Low-Battery Alert	System	Create a Warning alert for the parameter Battery when the battery level is below 25 percent.	Specialises UC21	FR39, FR52

22.3 Sequence Diagrams

Seven sequence diagrams cover the main processes. Diagrams 1 and 6 are Flutter and Firebase processes. Diagrams 2 to 5 are ASP.NET and Firebase processes. Diagram 7 shows how the two applications are tied together when a reading arrives. Each one shows only the people, the application they use, Firebase Authentication and the Realtime Database. A check that an application makes on its own is drawn as an arrow from the application to itself. Each diagram relates to approved use cases and requirements.

Diagram	Participant	Role
1	Standard User	Starts the process from the alert.
	Flutter app	Shows the alert and the note sheet, checks the note is not empty and writes the result.
	Realtime Database	Stores the Resolved action at alert_actions and the new state at alerts, both under the location. Security Rules check the user, the location and the note.
2, 3	Administrator	Starts the management action.
	Web application	Checks the Administrator role, checks the details and writes to the database. Its token bypasses Security Rules, so these checks are made here.
	Realtime Database	Stores the band (thresholds) or the sensor (sensors), which both applications read.
4	Administrator	Invites a user.
	Web application	Checks the role and the fields, then coordinates the account and the database records.
	Firebase Authentication	Creates the sign-in account and sends the password-setup email.
	Realtime Database	Stores the user, user_locations and location_users records in one update.
5	User	Any signed-in user, Administrator or Standard User.
	Web application	Checks the new password and asks Firebase to change it.
	Firebase Authentication	Re-checks the current password and updates the password.
6	Standard User	Signs in on the phone.
	Flutter app	Sends the credentials, reads the user's record and locations, and shows the dashboard.
	Firebase Authentication	Checks the credentials and returns an ID token.
	Realtime Database	Returns the user's record, locations and the data of those locations only, under Security Rules.
7	Sensor Gateway	Writes the reading and the sensor status.
	Realtime Database	Stores readings, thresholds, alerts and notifications, and pushes changes to a listening phone.
	Alert evaluator	Compares each reading with its band and writes the alert and notifications (Cloud Function).
	Flutter app and Standard User	Receive the alert and show the badge and list.

22.3.1 Sequence Diagram 1: Standard User resolves an alert with a note

Platform Flutter app and Firebase Realtime Database

Use cases UC10 Resolve Alert, which includes UC11 Add Note

Requirements FR56, FR57, FR80, FR81, IR03, NFR11

The user taps Resolve on the alert detail screen (M7). The note sheet (M7b) opens and the user types a note and confirms. The app refuses an empty note and asks for one. With a note, the app adds the Resolved action (note, user and time) and sets the alert's state to Resolved in one update. Security Rules confirm that the user is signed in, is assigned to the alert's location and sent a note. The alert then shows as Resolved with a confirmation, and the Administrator sees the same state, and the note, on W9. If the rules refuse the write, the screen says the alert cannot be changed.

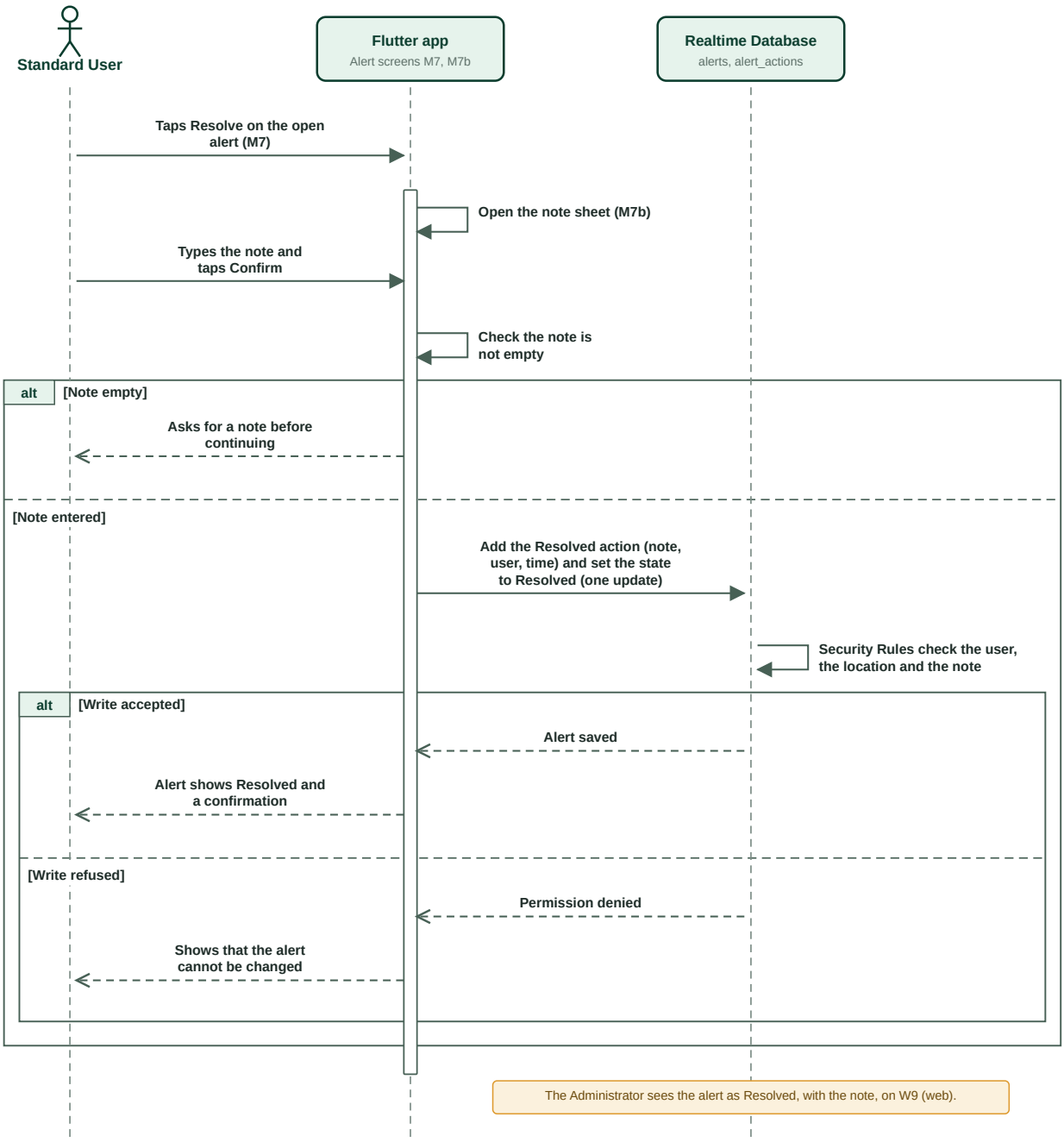


Figure 22.3 — Sequence Diagram 1: Standard User resolves an alert with a note.

22.3.2 Sequence Diagram 2: Administrator edits a threshold band

Platform ASP.NET MVC web application and Firebase Realtime Database

Use cases UC16 Set Thresholds

Requirements FR05, FR45–FR48, IR06, NFR06, NFR19

The Administrator opens Thresholds (W8). The web application checks the Administrator role and reads the bands for the page. The Administrator chooses a location and a parameter, enters the minimum, maximum and basis, and saves. The web application checks the role again, then that the numbers are valid, the minimum is below the maximum and a basis is given. Only then does it write the band with updated_by and updated_at. The next check of readings uses the new band, and the phone shows it on M4. An invalid band returns an error beside the field and nothing is written.

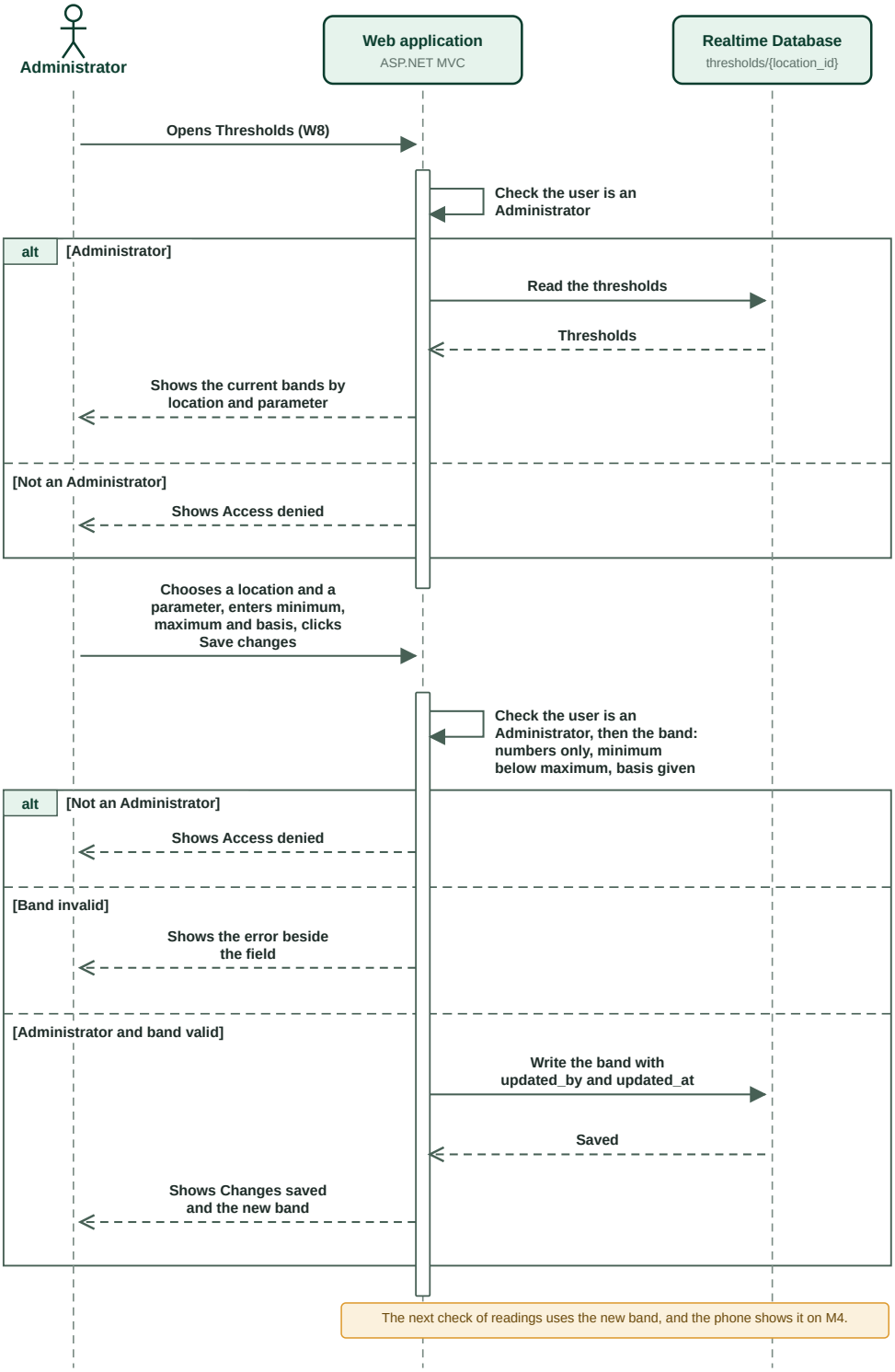


Figure 22.4 — Sequence Diagram 2: Administrator edits a threshold band.

22.3.3 Sequence Diagram 3: Administrator registers a sensor

Platform ASP.NET MVC web application and Firebase Realtime Database

Use cases UC18 Register Sensor

Requirements FR27–FR30, FR05

The Administrator fills in the sensor form (W6) and clicks Save sensor. The web application checks the role and the fields (code format, depth, slave ID is a number). The Realtime Database cannot refuse a duplicate by itself, so the web application reads the sensors of the location and compares the sensor code and the Modbus slave ID with them. The code begins with the location's prefix, so checking it against that location is enough to keep it unique everywhere; the slave ID only has to be unique within one location. A free code and ID are written as a new sensor. A duplicate returns an error beside the field, and for a slave ID the next free ID is offered.

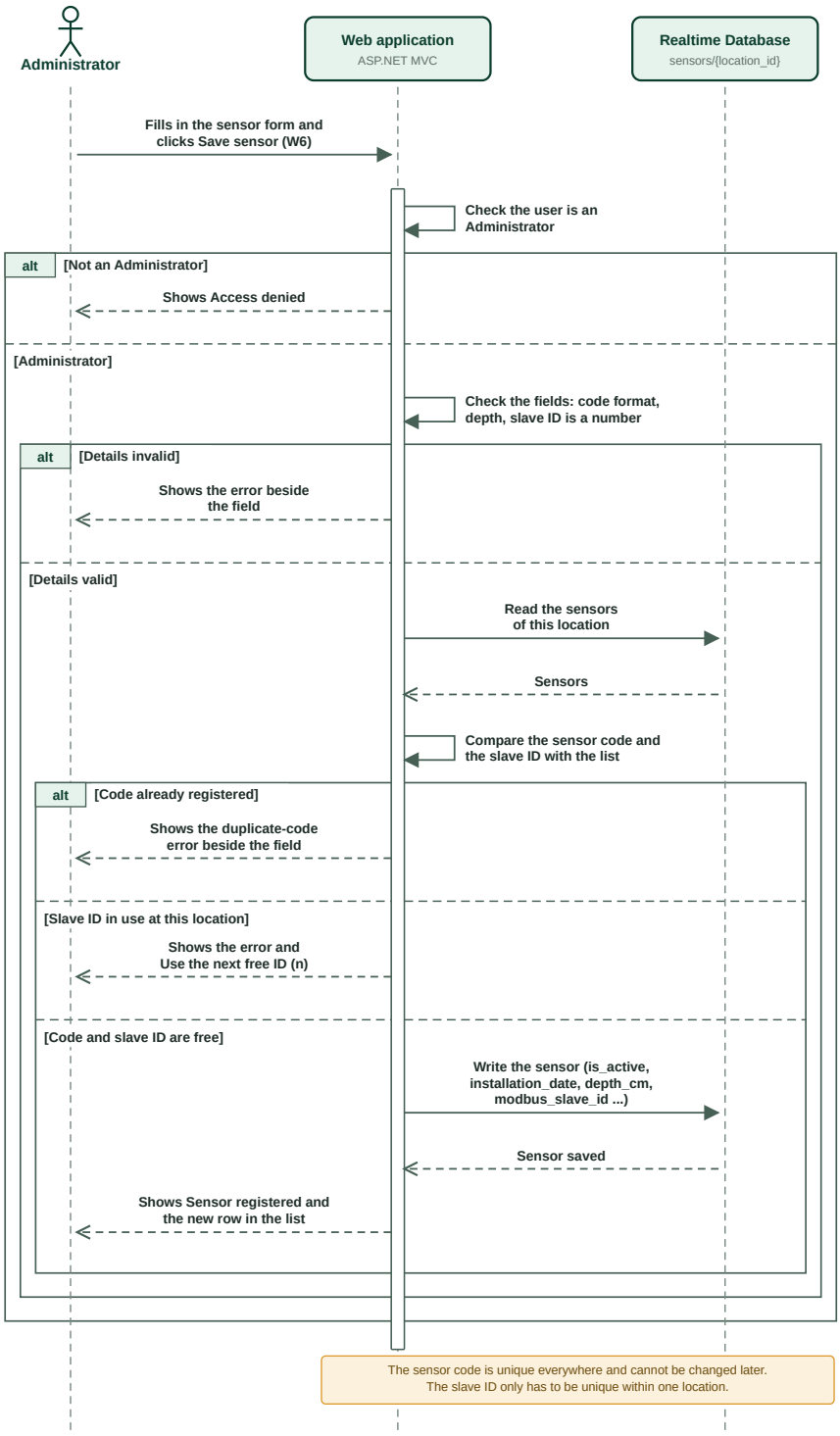


Figure 22.5 — Sequence Diagram 3: Administrator registers a sensor.

22.3.4 Sequence Diagram 4: Administrator invites a user

Platform ASP.NET MVC web application, Firebase Authentication and Firebase Realtime Database

Use cases UC17 Manage User Accounts

Requirements FR05, FR70, FR71, FR82, FR83, IR05

The Administrator clicks Invite user on User accounts (W11), fills in the invite form (W11b) with name, email, role and assigned locations, and sends it. The web application checks the Administrator role and the fields, then asks Firebase Authentication to create the sign-in account. If the email is new, it writes the user (account_status Invited), user_locations and location_users in one multi-path update and asks Firebase to send the password-setup email. The accounts list (W11c) shows the new row with the status Invited, and the new user can then sign in on the phone (IR05). A duplicate email returns a clear error and creates nothing.

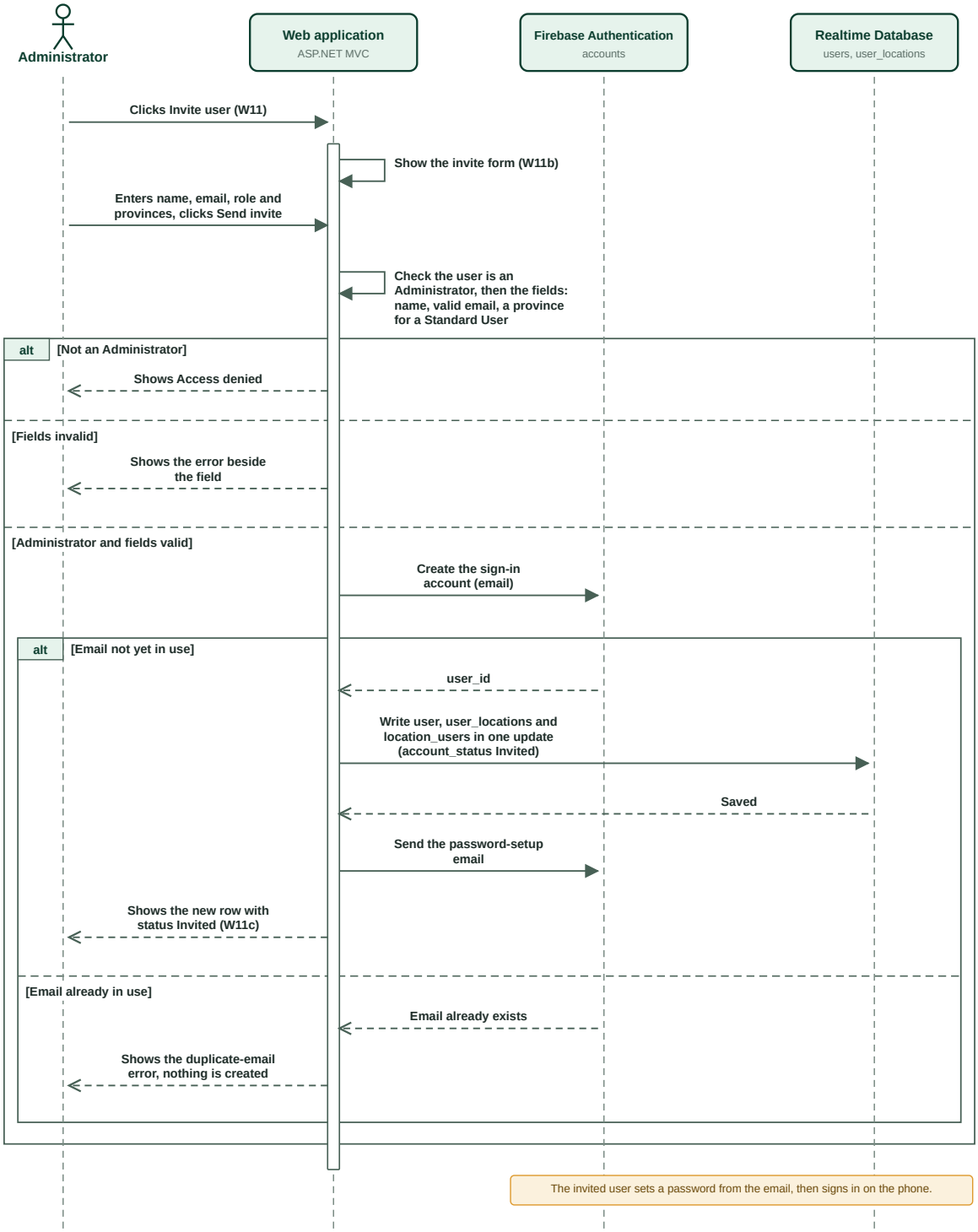


Figure 22.6 — Sequence Diagram 4: Administrator invites a user.

22.3.5 Sequence Diagram 5: User changes their password

Platform ASP.NET MVC web application and Firebase Authentication

Use cases UC15 Change Password, which extends UC14 Manage Own Profile

Requirements FR07, NFR01

On the change password dialog (W12b) the user enters the current, new and confirm passwords. The web application finds the signed-in user and checks the new password (8 or more characters, letters and numbers, the same as the confirmation). It then signs in again with the current password, so a wrong current password is refused by Firebase. After a fresh sign-in it updates the password. A password change is a major account change for Firebase, so the user's other devices are asked to sign in again. W12 then shows Password updated and the Last changed date, which Firebase Authentication holds.

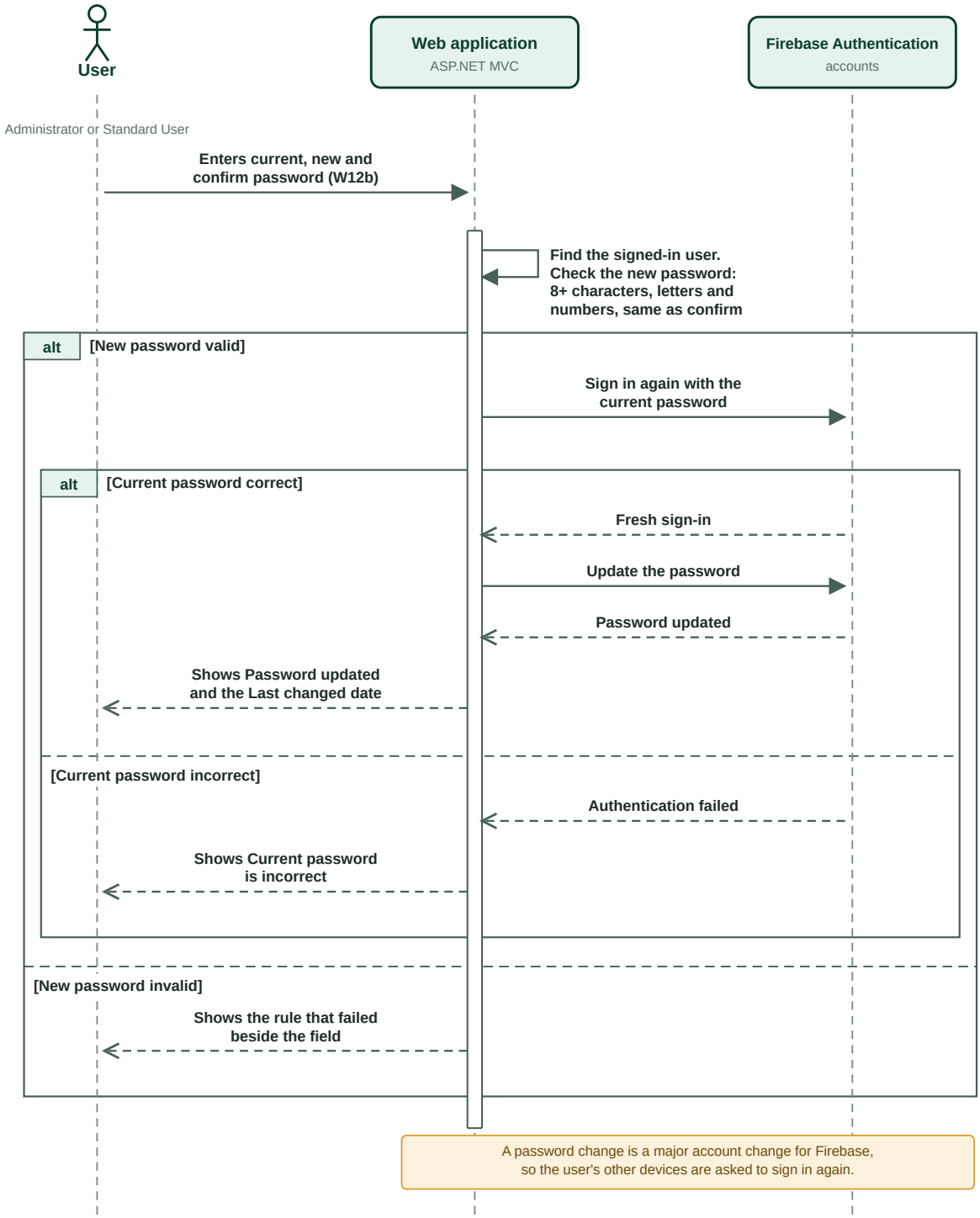


Figure 22.7 — Sequence Diagram 5: User changes their password.

22.3.6 Sequence Diagram 6: Standard User signs in on the phone

Platform Flutter app, Firebase Authentication and Firebase Realtime Database

Use cases UC01 Sign In, followed by UC03 View Current Readings

Requirements FR01–FR06, FR82, IR05

The user enters email and password on the sign-in screen (M1). Firebase Authentication checks them and returns an ID token and the user ID. The app reads the user's record (role and account_status) under Security Rules, which let a user read only their own record. A deactivated account is signed out and shown Access denied. An active account reads its assigned locations, then the readings, sensor status and open alerts of those locations only, and shows the dashboard (M2). An Administrator who signs in on the phone gets the Standard User functions only. Wrong credentials return one message that does not say which part was wrong.

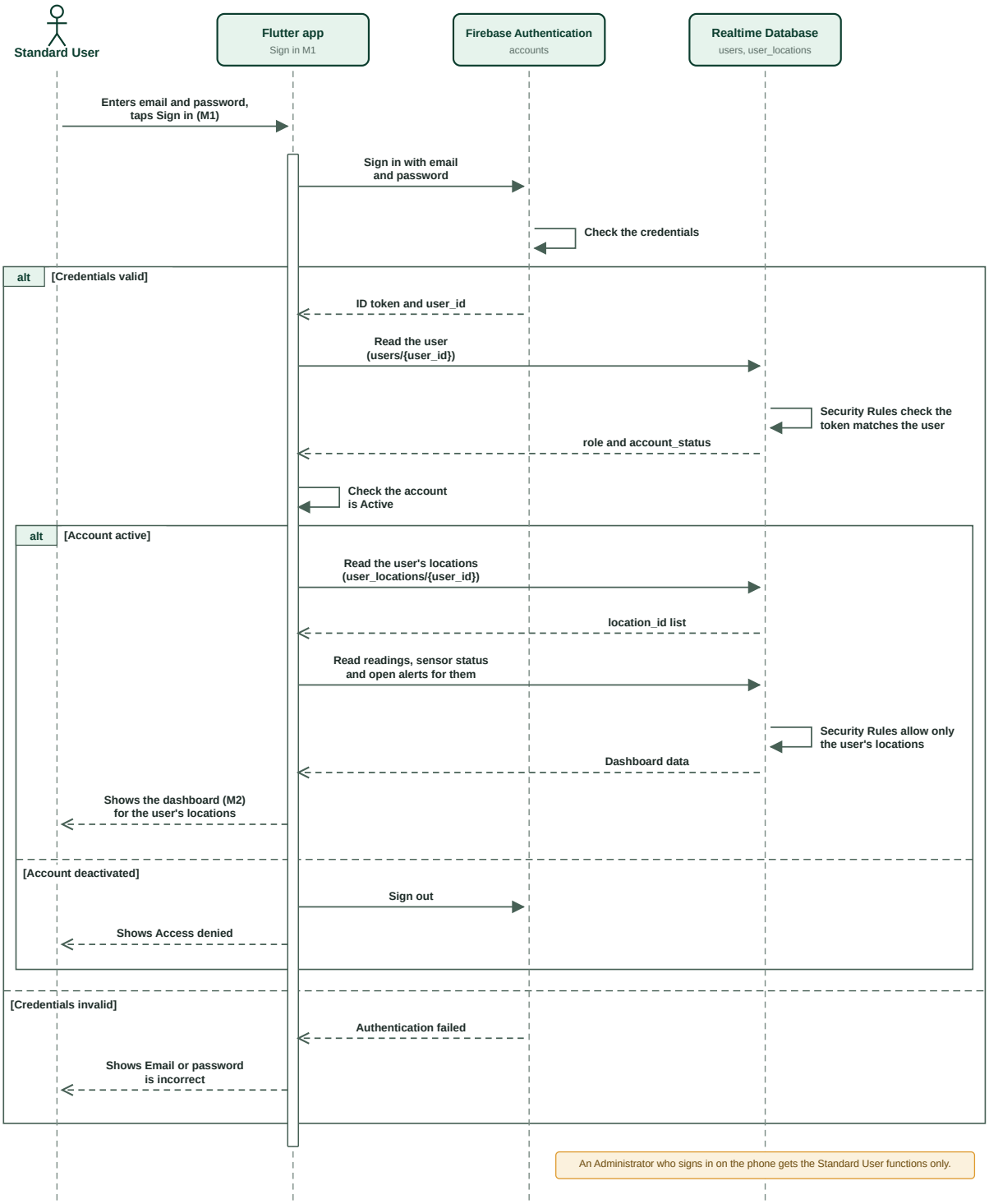


Figure 22.8 — Sequence Diagram 6: Standard User signs in on the phone.

22.3.7 Sequence Diagram 7: A reading arrives and an alert is raised

Platform Sensor gateway, Realtime Database, alert evaluator and Flutter app

Use cases UC19 Record Reading, UC20 Check Thresholds, UC21 Raise Alert (specialised by UC24 to UC26), UC22 Notify Users

Requirements FR51–FR53, FR58, FR75, FR77, IR03, IR04, IR06

Every 30 minutes the gateway writes a reading and a status record for each sensor. The reading is stored under its sensor and capture time, so a repeat cannot create a duplicate. A new reading triggers the evaluator, which reads the location's thresholds and assigned users, then compares each value. A value outside its band produces an alert (Open, with severity, measured value and threshold) and one notification for each assigned user, written together in one multi-path update. A phone that is listening receives the change at once and the Alerts badge and list update; otherwise it shows them the next time the screen opens. A separate scheduled check raises the Offline alert for sensors silent for two hours.

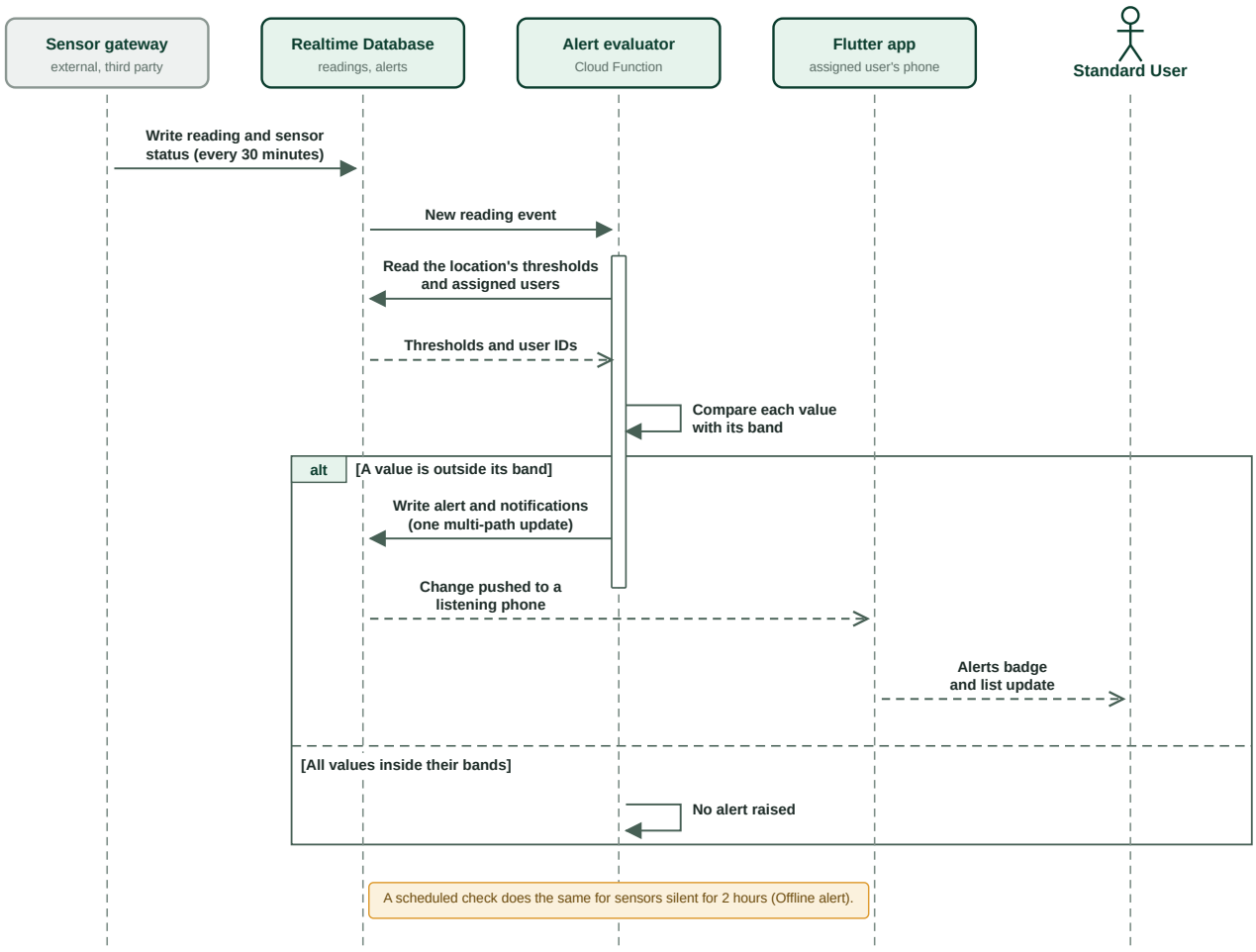


Figure 22.9 — Sequence Diagram 7: A reading arrives and an alert is raised.

22.4 Database Design

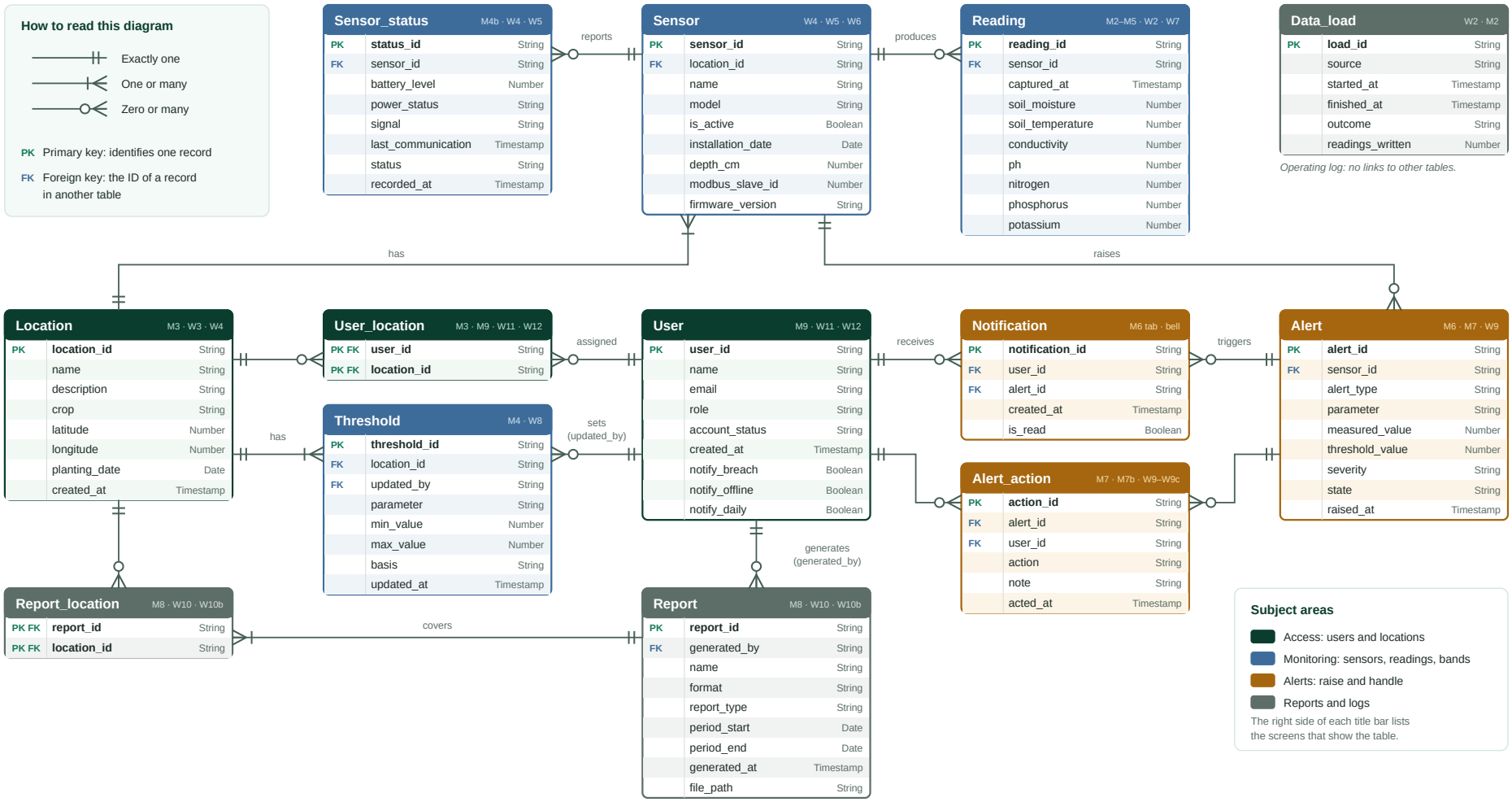


Figure 22.10 — Entity relationship diagram. PK is a primary key, FK a foreign key, M and W are mobile and web screens.

The ERD is the logical model of everything the system stores. It extends the minimum entities of Section 16 with what the designs need: the link between users and locations, the record of each acknowledge and resolve, notifications, reports, the locations a report covers, and a record of each data load. In Firebase the data is one JSON tree in the Realtime Database, and each entity is a branch of that tree (22.4.5). A foreign key is the key of the related record. The database has no foreign-key engine, so Security Rules, the web server's checks and fixed keys keep these references valid.

22.4.1 Entities and Keys

Entity	Realtime Database path	Purpose	Primary key	Foreign keys
User	users/{user_id}	A person who can sign in. Holds name, email, role, account status and notification preferences.	user_id (the Firebase Authentication ID)	None
User_location	user_locations/{user_id}/{location_id}, mirrored at location_users/{location_id}/{user_id}	Gives a user access to a location. Makes users and locations many-to-many.	user_id + location_id	user_id, location_id
Location	locations/{location_id}	One of the three monitoring locations, shown as a province in the screens. Holds name, description, crop, coordinates, planting date and creation time.	location_id	None
Sensor	sensors/{location_id}/{sensor_id}	A physical soil sensor installed at a location.	sensor_id (the sensor code, for example FS-06)	location_id
Sensor_status	sensor_status/{location_id}/{sensor_id}/{key}	Equipment condition reported with each cycle: battery, power, signal, state, last communication.	status_id (sensor_id + time)	sensor_id
Reading	readings/{location_id}/{sensor_id}/{key}	One timestamped set of seven soil values from one sensor.	reading_id (sensor_id + time)	sensor_id
Threshold	thresholds/{location_id}/{parameter}	The minimum and maximum band for one parameter at one location, with its basis.	threshold_id (location_id + parameter)	location_id, updated_by
Alert	alerts/{location_id}/{alert_id}	A breach, offline or low-battery event with its severity, state, and the value and threshold involved.	alert_id	sensor_id
Alert_action	alert_actions/{location_id}/{alert_id}/{action}	One acknowledge or resolve of an alert: its user, time and required note.	action_id (alert_id + action)	alert_id, user_id
Notification	notifications/{user_id}/{alert_id}	A per-user, read or unread, notice about an alert.	notification_id (user_id + alert_id)	user_id, alert_id
Report	reports/{report_id}, file in Cloud Storage	A generated report listed on the Reports pages: its type, format, period and file.	report_id	generated_by
Report_location	report_locations/{report_id}/{location_id}	Lists the locations a report covers. Makes reports and locations many-to-many.	report_id + location_id	report_id, location_id
Data_load	data_loads/{load_id}	One data load by the sensor gateway, with its outcome, so that failures are visible.	load_id	None (operating log)

22.4.2 Relationships and Cardinality

Relationship	Cardinality	Meaning
User to User_location	1 : 0..many	A user can be assigned to many locations.
Location to User_location	1 : 0..many	A location can be assigned to many users. With the row above, users and locations are many-to-many.
Location to Sensor	1 : many	A location has five sensors.
Location to Threshold	1 : many	A location has one band for each of the seven parameters.
User to Threshold (updated_by)	1 : 0..many	A user can set many bands. Each band records the user who last changed it.
Sensor to Reading	1 : 0..many	A sensor produces a reading every 30 minutes.
Sensor to Sensor_status	1 : 0..many	A sensor reports a status record every 30 minutes.
Sensor to Alert	1 : 0..many	A sensor can raise many alerts over time.
Alert to Alert_action	1 : 0..many	At most one Acknowledged and one Resolved action each.
User to Alert_action	1 : 0..many	A user can take many actions. Each keeps the signed-in user.
Alert to Notification	1 : 0..many	One alert notifies every user assigned to its location.
User to Notification	1 : 0..many	A user receives many notifications.
User to Report (generated_by)	1 : 0..many	A user can generate many reports.
Report to Report_location	1 : many	A report covers one or more locations.
Location to Report_location	1 : 0..many	A location can appear in many reports.

22.4.3 Allowed Values

Field	Allowed values or unit
User.role	Administrator, Standard User
User.account_status	Active, Invited, Deactivated
Threshold.parameter	Soil moisture, soil temperature, conductivity, pH, nitrogen, phosphorus, potassium (stored as soil_moisture, soil_temperature, conductivity, ph, nitrogen, phosphorus, potassium)
Threshold.basis	Free text naming the source of the band (for example the pigeon pea production guideline)
Reading values	Soil moisture %, soil temperature °C, conductivity µS/cm, pH, nitrogen, phosphorus and potassium in mg/kg
Sensor_status.status	Online, Battery low, Needs attention, Offline
Sensor_status.signal	Strong, Fair, Weak, None
Sensor_status.battery_level	Percent, 0 to 100
Sensor_status.power_status	Power source or state reported by the sensor. The exact values come from the gateway.
Alert.alert_type	Breach, Offline, Battery
Alert.severity	Critical, Warning
Alert.state	Open, Acknowledged, Resolved

Field	Allowed values or unit
Alert.parameter	One of the seven parameters, or Battery. Empty for an Offline alert.
Alert_action.action	Acknowledged, Resolved
Alert_action.user_id	The user_id of the signed-in person who took the action
Report.report_type	Monthly nutrient summary, Raw readings export, Threshold breach log
Report.format	PDF, CSV
Data_load.source / outcome	Gateway / Succeeded, Partial, Failed
Timestamp fields	UTC, stored as a number: milliseconds since 1 January 1970
Date fields	Text in the form yyyy-MM-dd
Reading and status key	Capture time to the minute, UTC, written yyyyMMddHHmm (for example 202609081430)
Empty values	Not stored while empty. The Realtime Database removes empty values, so parameter, measured_value and threshold_value do not exist on an alert that has no value for them.

22.4.4 Design Notes

- **What the design adds to Section 16.** User_location, Alert_action, Notification, Report, Report_location and Data_load are new. User_location exists because a user can be assigned to many locations (FR82, FR83). Alert_action exists because FR80 and FR81 need the user, time and note of each acknowledge and resolve. Notification exists because the bell and the Alerts tab badge must know what each user has and has not seen. Report and Report_location exist because the Reports pages list generated reports, who made them and which locations they cover. Data_load exists because FR78 requires each data load to be recorded so that failures are visible. Location also holds its coordinates, crop and planting date, which W3 and W4 show.
- **Acknowledge and resolve are separate.** Each is one Alert_action record with the signed-in user, the time and its own note. The key is the alert_id plus the action, so an alert has at most one of each. Alert keeps the measured value and the threshold that was crossed, so M7 and W9 can show the comparison.
- **Location in the path.** Sensors, status records, readings, thresholds, alerts and alert actions are stored under their location_id, so one Security Rule protects a whole branch. A reading or an alert reaches its location through its sensor, so the ERD does not repeat location_id on them. A notification is stored under its user and keeps a copy of its alert's location_id, so the app can open the alert.
- **User_location as its own node.** The link between users and locations is stored as user_locations/{user_id}/{location_id}, so a rule can test membership with one lookup. The same pairs are also stored as location_users/{location_id}/{user_id}, so the alert evaluator can list the users of a location without reading every user. Both are written together in one multi-path update.
- **Keys.** A reading or status key is the capture time under its sensor, which makes duplicates impossible (FR77). A threshold key is the parameter under its location, which gives one band per parameter per location. A notification key is the alert ID under its user, which gives one notification per user per alert (FR58). The sensor code is the sensor key and is unique everywhere. Alerts use Firebase push IDs, which sort by creation time.
- **Built for the questions the screens ask.** A query in the Realtime Database orders by one field, so the tree is laid out for the common questions: the latest reading (the last key under a sensor), a time range (a range of keys), the open alerts of a location (an index on state) and a user's notifications (their own node).

22.4.5 Realtime Database Structure

The tree below shows where each entity lives. Braces mark a key that is different for each record. FS stands for the Free State location and FS-01 for its first sensor; the values are only illustrations of the layout. The key of a record is its own ID: for example, the reading_id of a reading is its capture time under its sensor.

```
users
  {user_id}                  name, email, role, account_status, created_at,
                             notify_breach, notify_offline, notify_daily
user_locations
  {user_id}
    FS: true                 (one entry for each assigned location)
location_users
  FS
    {user_id}: true          (the same pairs, grouped by location)
locations
  FS                         name, description, crop, latitude, longitude, planting_date, created_at
sensors
  FS
    FS-01                    name, model, is_active, installation_date, depth_cm, modbus_slave_id,
                             firmware_version
sensor_status
  FS
    FS-01
      202609081430           battery_level, power_status, signal, status, last_communication, recorded_at
readings
  FS
    FS-01
      202609081430           captured_at, soil_moisture, soil_temperature, conductivity, ph, nitrogen,
                             phosphorus, potassium
thresholds
  FS
    soil_moisture            updated_by, min_value, max_value, basis, updated_at
alerts
  FS
    {alert_id}               sensor_id, alert_type, parameter, measured_value, threshold_value, severity,
                             state, raised_at
alert_actions
  FS
    {alert_id}
      acknowledged, resolved user_id, note, acted_at
notifications
  {user_id}
    {alert_id}               location_id, created_at, is_read
reports
  {report_id}                generated_by, name, format, report_type, period_start, period_end,
                             generated_at, file_path
report_locations
  {report_id}
    FS: true                 (the locations the report covers)
data_loads
  {load_id}                  started_at, finished_at, source, outcome, readings_written
```

22.4.6 Security Rules

A rule on a branch also covers everything below it, so the rules for a location are written once at the location_id level. Each rule reads the user's own user_locations node to learn whether the location is assigned to them.

Path	Read	Write
users/{user_id}	A user reads their own node. The Administrator reads all.	Administrator, through the web server. A user edits only their own notification preferences.
user_locations, location_users	A user reads their own user_locations node.	Administrator, through the web server, writing both nodes together.
locations, sensors, thresholds	Signed-in users, for assigned locations only. The Administrator reads all.	Administrator, through the web server.
readings, sensor_status	Same as locations.	The gateway's own account only. It can create a record but not overwrite one (FR77).
alerts	Same as locations. Indexed on state and raised_at for the alert lists.	A client may only move the state forward, in the same update that adds its action record. Creation is by the evaluator only.

Path	Read	Write
alert_actions	Same as locations.	A user assigned to the location may add each record once, with their own user_id and a note. Records cannot be changed.
notifications/{user_id}	The owner only.	The owner may change only is_read. Creation is by the evaluator only.
reports, report_locations and the reports/ storage folder	Signed-in users, for the locations they are assigned.	Web server only.
data_loads	Administrator.	The gateway only.

22.5 Interface Design

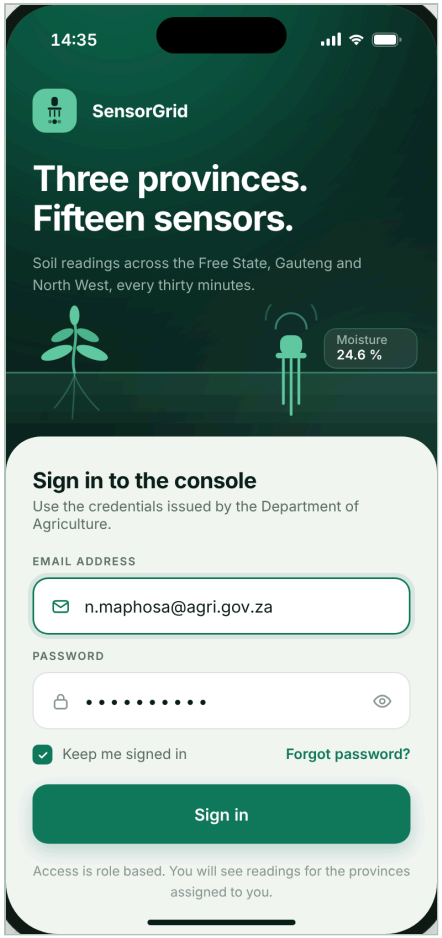
22.5.1 Approach

The interface designs show layout, navigation, data capture, information display and the actions each user must perform. They are drawn in Figma, with a clickable prototype for each application (22.8.1). Every screen is tied to a use case and to requirements, and every piece of information on a screen is supported by an entity in the ERD. The two applications use the same labels and terms (NFR12).

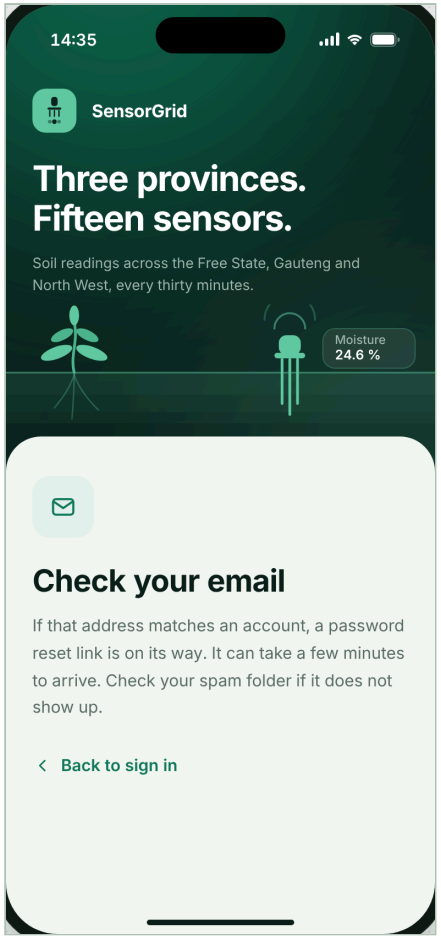
22.5.2 Flutter Mobile Screens

Screen	Name	Purpose	Use case	Requirements
M1	Sign in	Authenticate with email and password.	UC01	FR01, FR02
M1b	Forgot password, state 2	Confirms a reset email was sent. The same message shows whether or not the email has an account.	UC02	FR79
M2	Home	Network summary, alert preview and location cards.	UC03	FR08–FR15
M3	Sites	Assigned locations with the share of readings in range.	UC03	FR16, FR18, FR19, FR82
M4	Province detail	Seven parameters against their target bands. Readings, Trends and Sensors tabs.	UC03, UC04	FR17, FR31–FR38
M4b	Province detail, Sensors tab	Sensors of the location with battery, signal, state and last communication.	UC04	FR23–FR25, FR39–FR42
M5	Reading trend	One parameter over 24 hours, 7 days or 30 days.	UC05	FR62–FR64
M6	Alerts	Alert list with filters and an open-alert badge on the tab.	UC08, UC12	FR54, FR58, FR84
M7	Alert detail	Reading against threshold, activity, Acknowledge and Resolve.	UC09, UC10	FR55, FR57, FR80
M7b	Note sheet	A note is required before an alert is acknowledged or resolved.	UC11	FR81
M8	Reports	Nutrient and in-range summaries with report download.	UC13	FR69
M9	Profile	Account details, assigned locations, notification preferences, sign out.	UC14	FR03, FR85

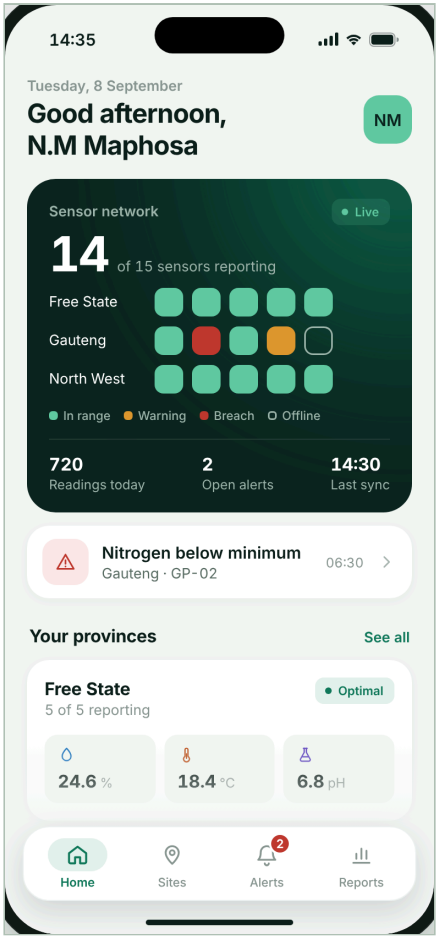
Figure 22.11 — Flutter mobile screens (1 of 3)



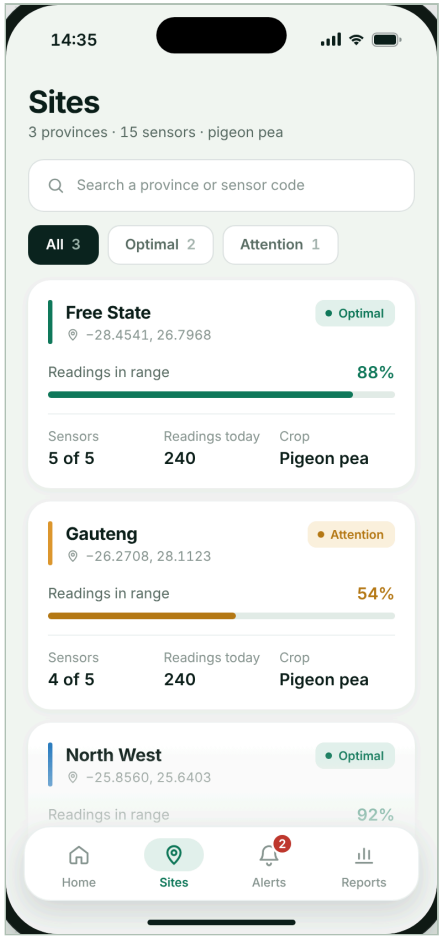
M1 Sign in



M1b Forgot password, state 2

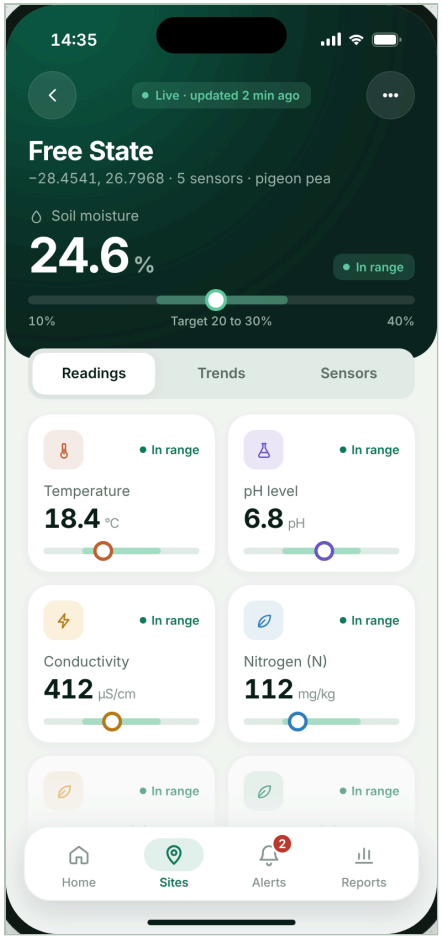


M2 Home

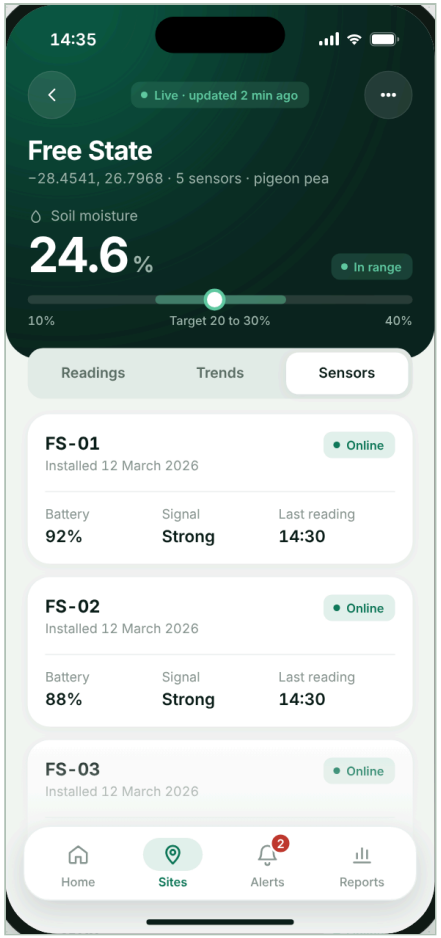


M3 Sites

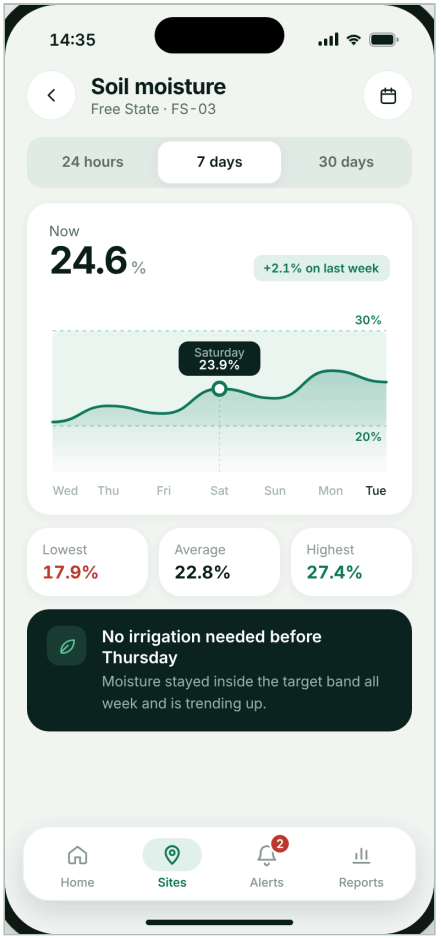
Figure 22.12 — Flutter mobile screens (2 of 3)



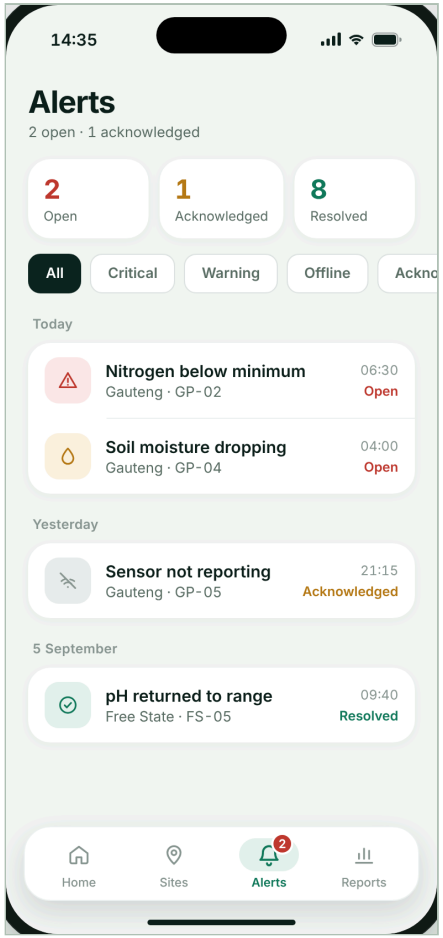
M4 Province detail



M4b Province detail, Sensors tab

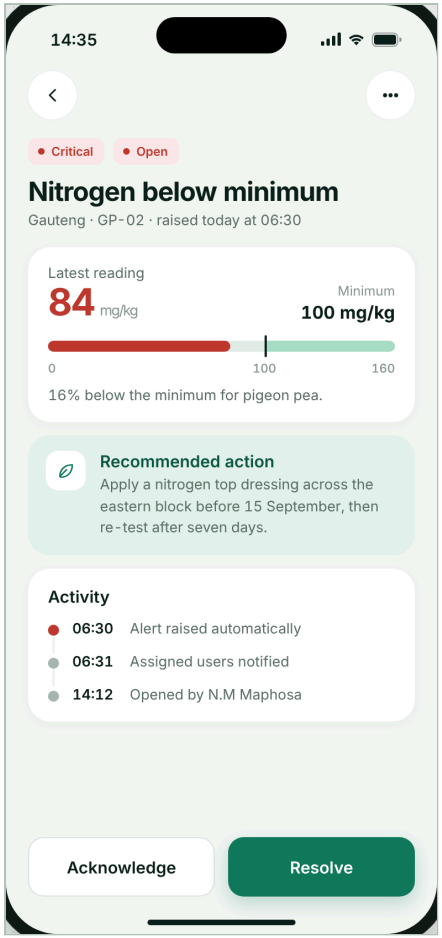


M5 Reading trend

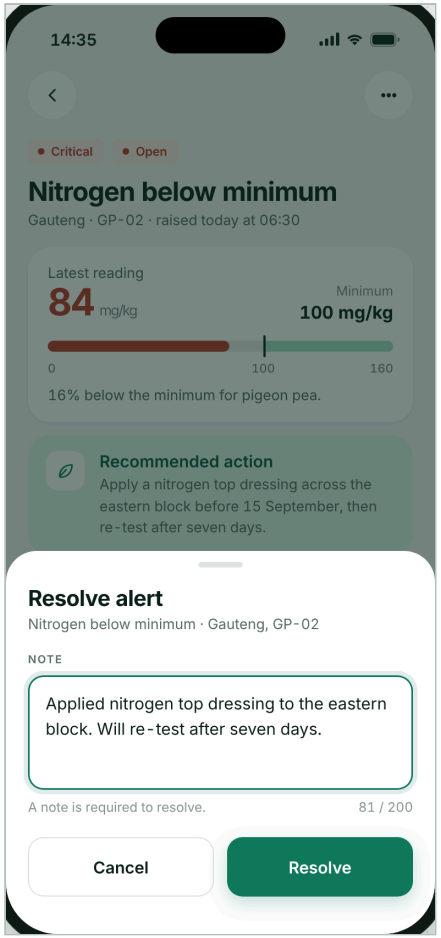


M6 Alerts

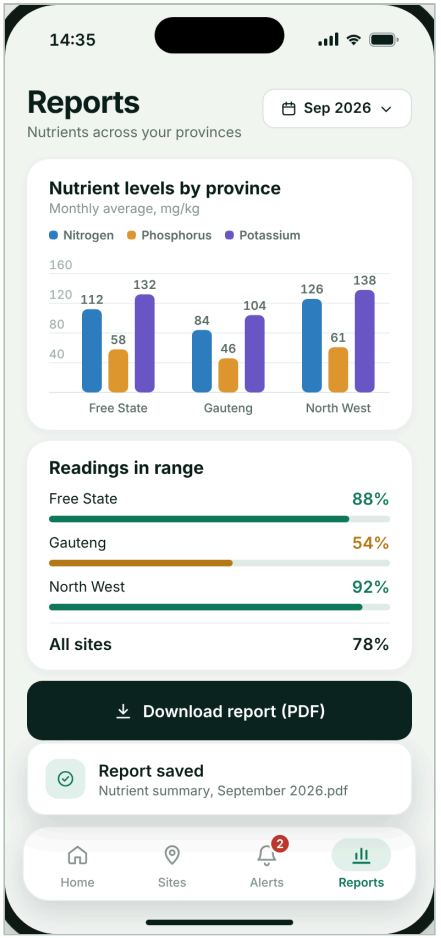
Figure 22.13 — Flutter mobile screens (3 of 3)



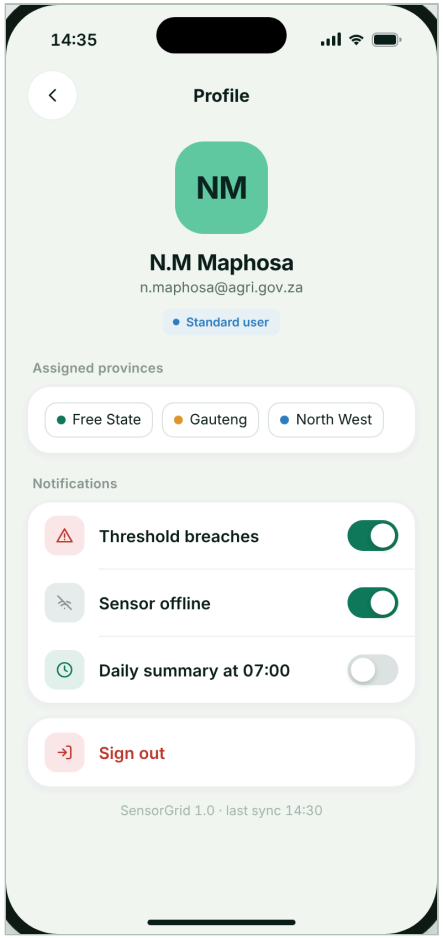
M7 Alert detail



M7b Note sheet



M8 Reports



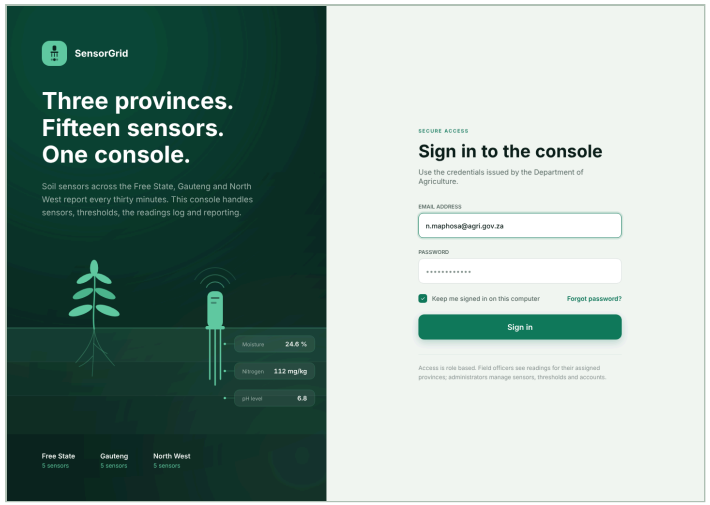
M9 Profile

22.5.3 ASP.NET Web Pages

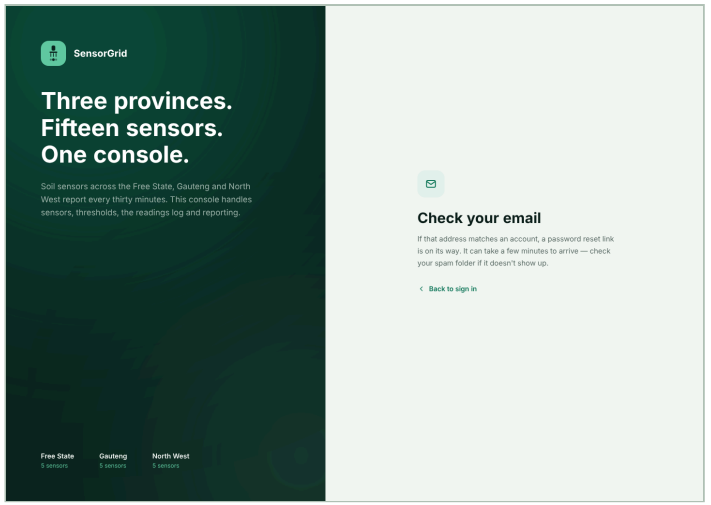
Both means Administrator and Standard User. A Standard User does not see Thresholds, User accounts or Register sensor.

Page	Name	User	Purpose	Use case	Requirements
W1	Sign in	Both	Authenticate.	UC01	FR01, FR02
W1b	Forgot password, state 2	Both	Reset email confirmation.	UC02	FR79
W2	Dashboard	Both	Sensors reporting, open alerts, moisture chart, location status, latest readings.	UC03, UC05	FR08–FR15, FR62
W3	Sites	Both	Location cards with status. The Administrator adds and edits locations.	UC03, UC23	FR16–FR22
W4	Site detail	Both	One location: parameters, sensors, thresholds.	UC03, UC04	FR17, FR23, FR31–FR38
W5	Sensors	Both	Sensor inventory with battery, signal, firmware and status.	UC04	FR23–FR26, FR39–FR42
W6	Register sensor	Administrator	Add or edit a sensor and assign it to a location.	UC18	FR27–FR30
W7	Readings log	Both	Timestamped readings, filters. CSV export for the Administrator.	UC06, UC07	FR61, FR66, FR67
W8	Thresholds	Administrator	Minimum, maximum and basis per parameter per location.	UC16	FR45–FR48
W9	Alerts	Both	Alert history with Acknowledge and Resolve.	UC08	FR54–FR56, FR59
W9b	Resolve alert dialog	Both	A note is required to resolve (same pattern as M7b).	UC10, UC11	FR80, FR81
W9c	Acknowledge alert dialog	Both	A note is required to acknowledge (same pattern as M7b).	UC09, UC11	FR57, FR81
W10	Reports	Both	Nutrient chart, in-range summary, monthly summary, generated reports.	UC13	FR69
W10b	Generate report dialog	Both	Choose report type, period and locations.	UC13	FR69
W11	User accounts	Administrator	Accounts, roles and location access.	UC17	FR70–FR74, FR83
W11b	Invite user dialog	Administrator	Name, email, role and assigned locations. Firebase sends the setup email.	UC17	FR70, FR71, FR83
W11c	User accounts, invite sent	Administrator	The list after an invite: the new account shows the status Invited.	UC17	FR70–FR72
W12	Profile	Both	Account details, preferences, change password, sign out.	UC14	FR03, FR85
W12b	Change password dialog	Both	Current, new and confirm password.	UC15	FR07
Bell	Notification panel	Both	Latest alert notifications from the top bar, with Mark all as read and an empty state.	UC12	FR58, FR84

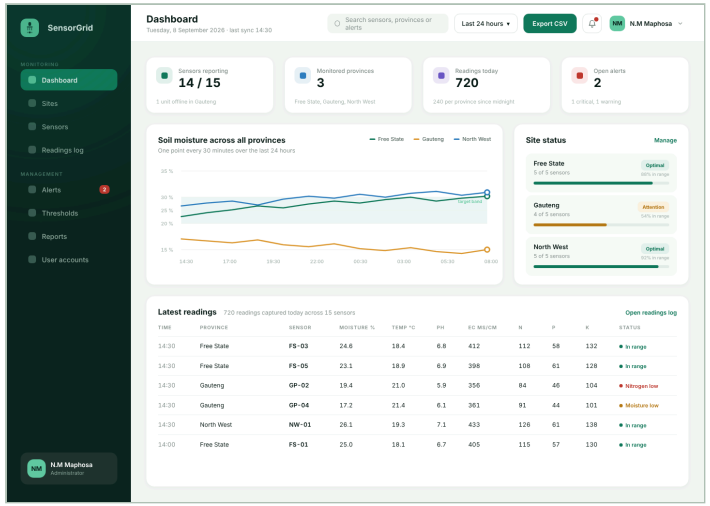
Figure 22.14 — ASP.NET web pages (1 of 6)



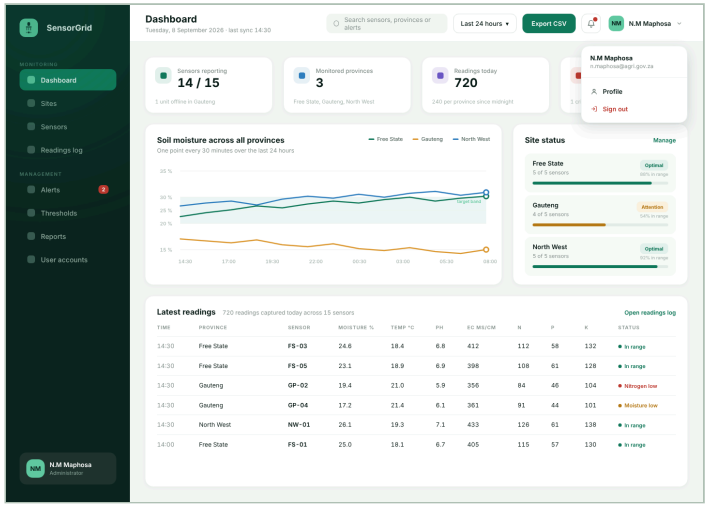
W1 Sign in



W1b Forgot password, state 2

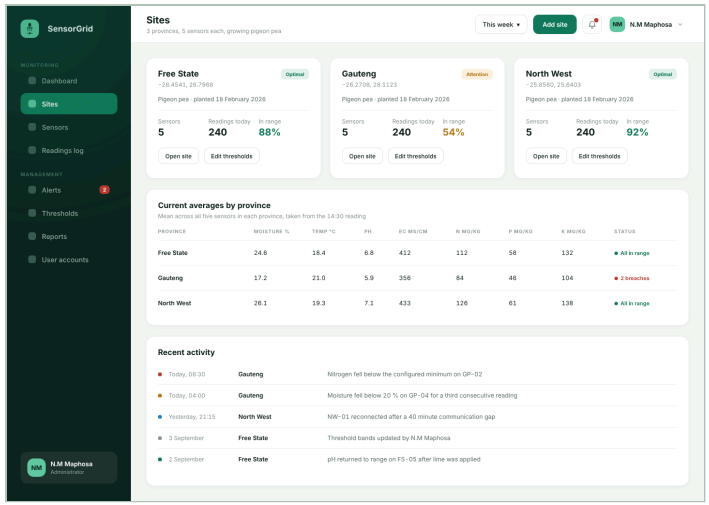


W2 Dashboard

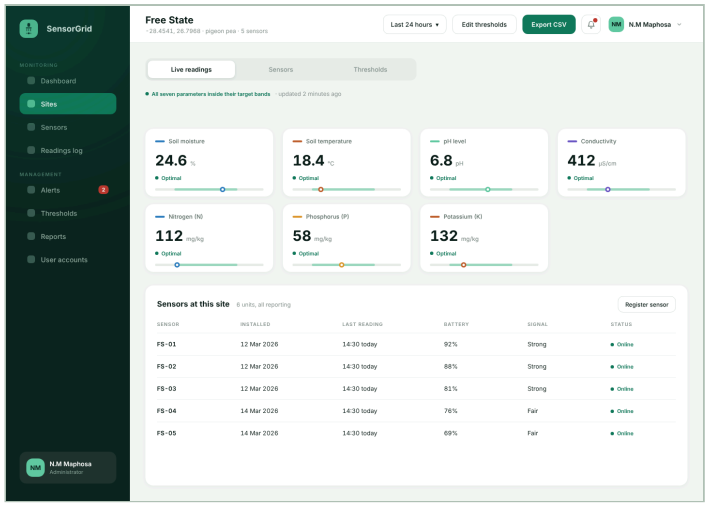


W2 Dashboard with the account menu open

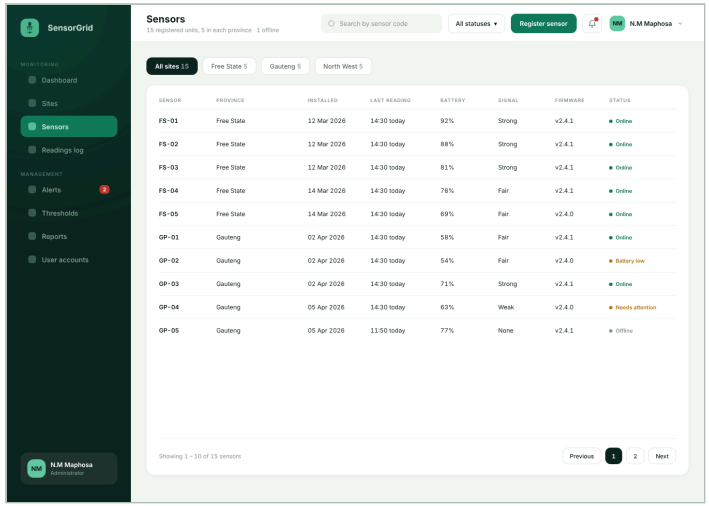
Figure 22.15 — ASP.NET web pages (2 of 6)



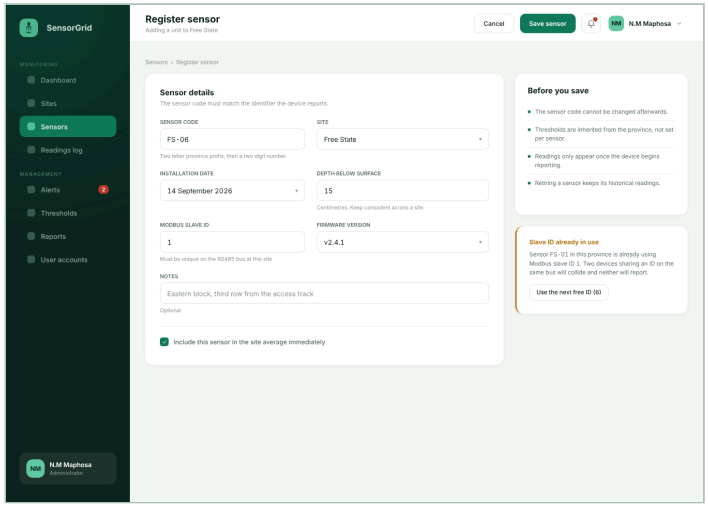
W3 Sites



W4 Site detail

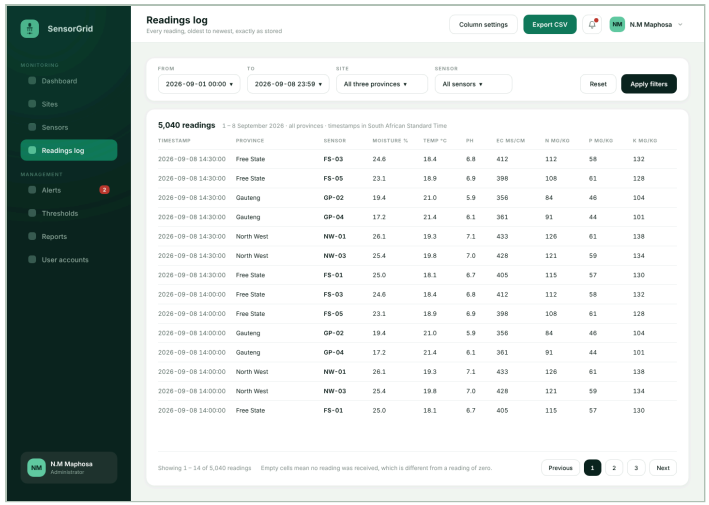


W5 Sensors

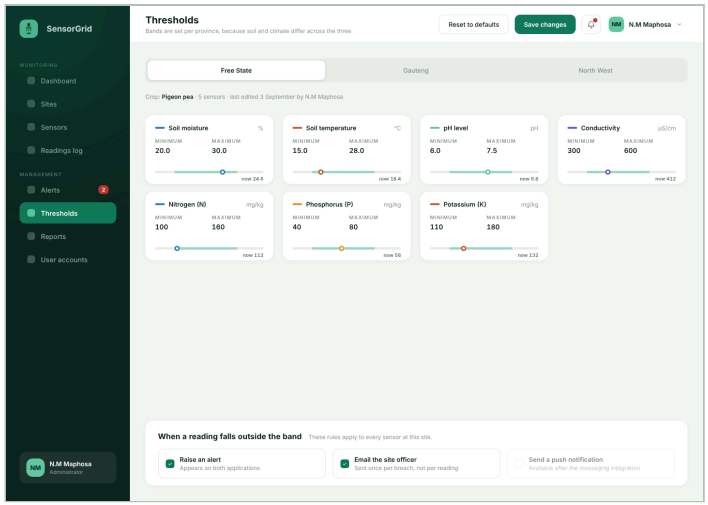


W6 Register sensor

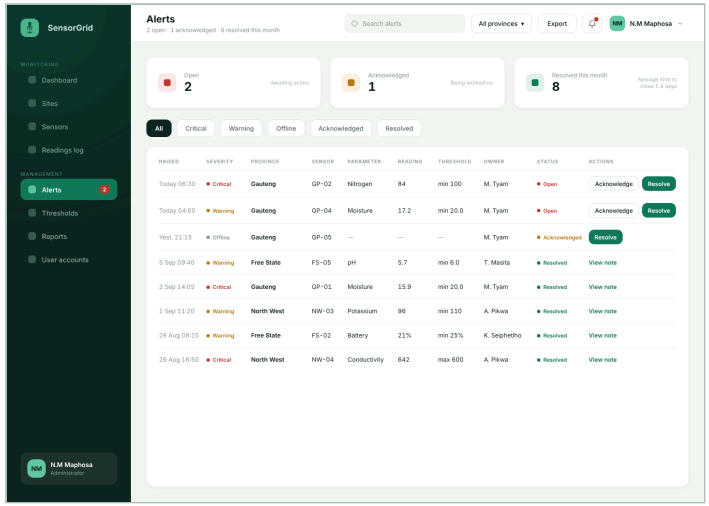
Figure 22.16 — ASP.NET web pages (3 of 6)



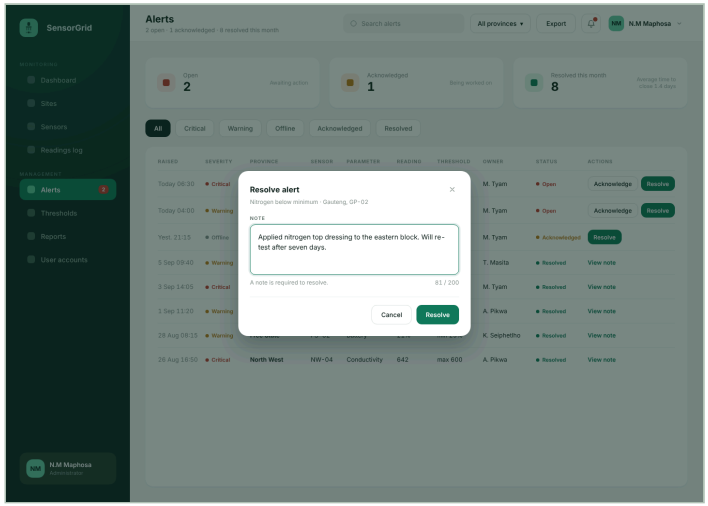
W7 Readings log



W8 Thresholds

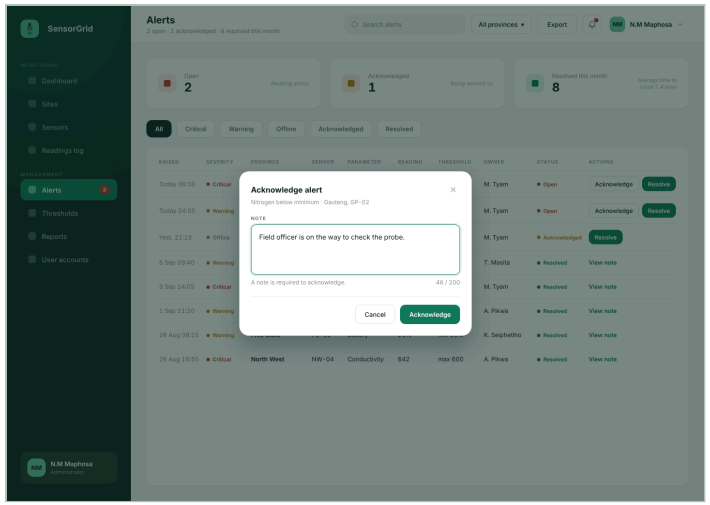


W9 Alerts

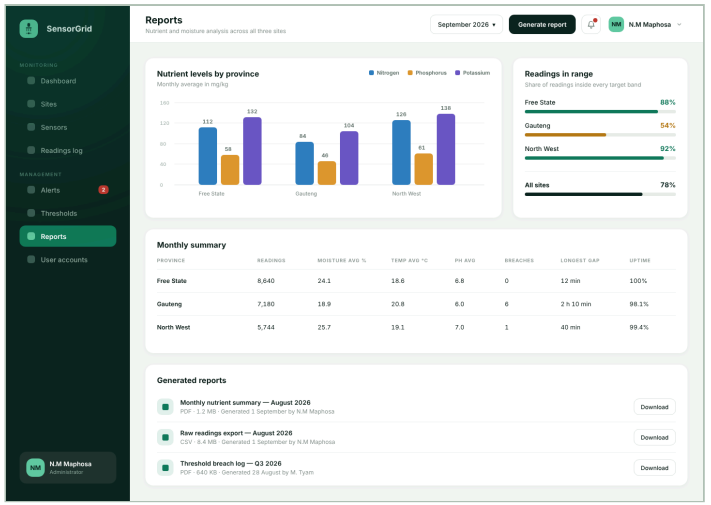


W9b Resolve alert dialog

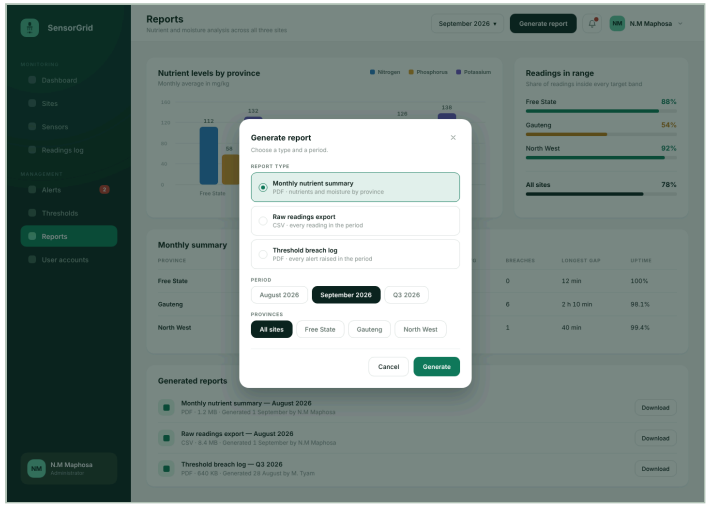
Figure 22.17 — ASP.NET web pages (4 of 6)



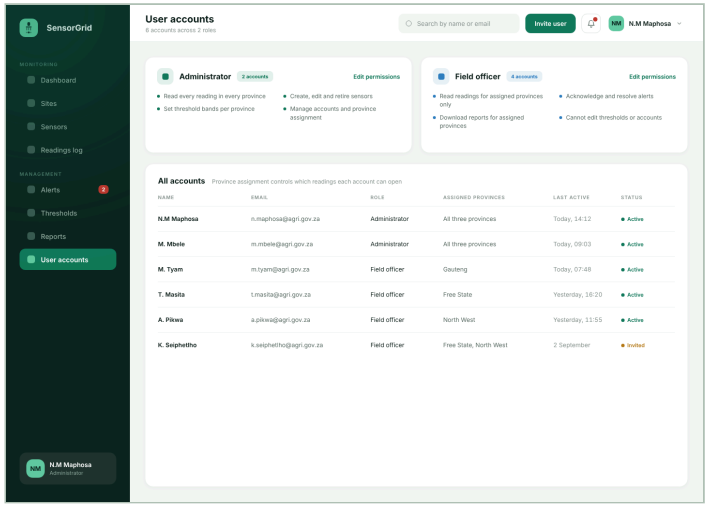
W9c Acknowledge alert dialog



W10 Reports

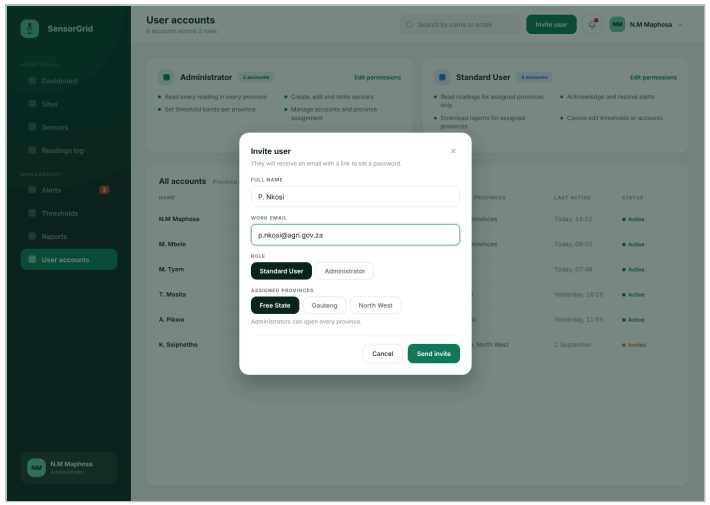


W10b Generate report dialog

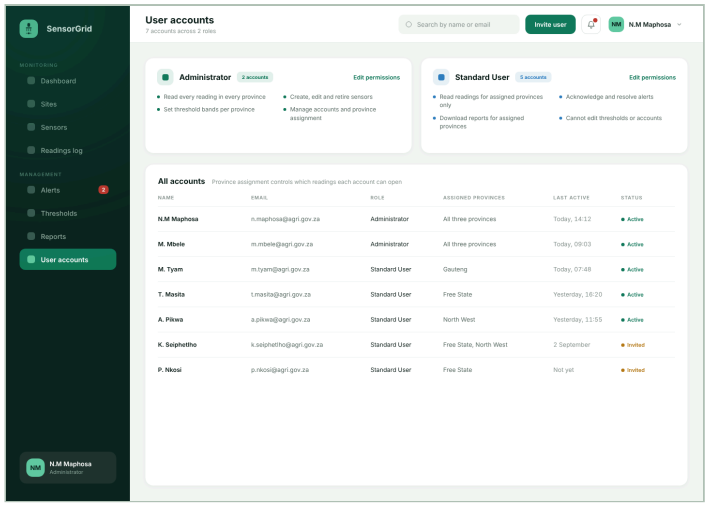


W11 User accounts

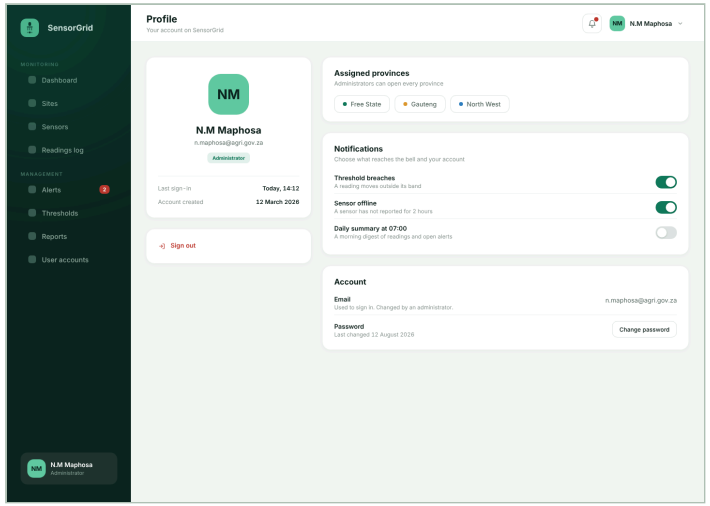
Figure 22.18 — ASP.NET web pages (5 of 6)



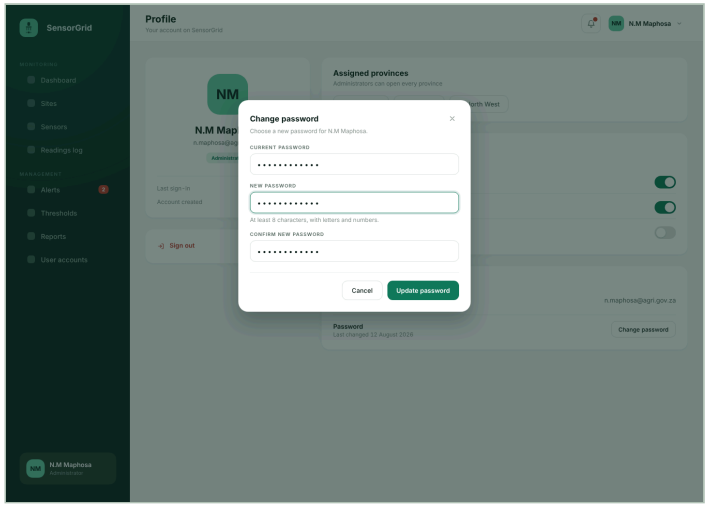
W11b Invite user dialog



W11c User accounts, invite sent

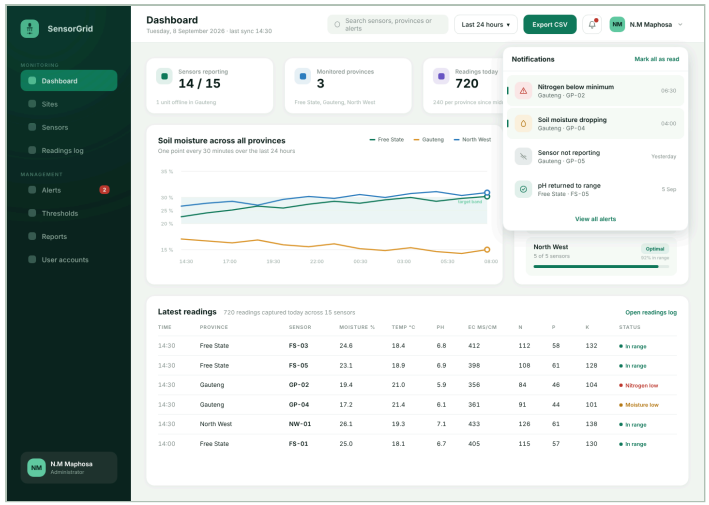


W12 Profile

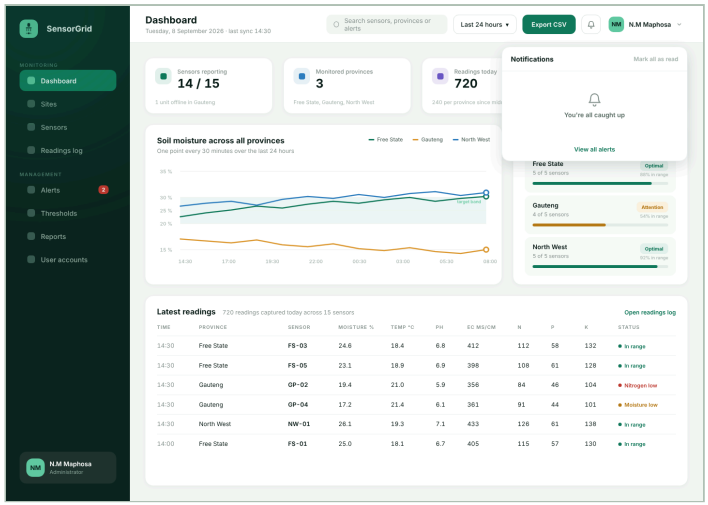


W12b Change password dialog

Figure 22.19 — ASP.NET web pages (6 of 6)



Bell Notification panel



Bell Notification panel, empty state

22.5.4 Navigation

The maps show how a user moves between screens. Every arrow is a row in the navigation tables that follow.

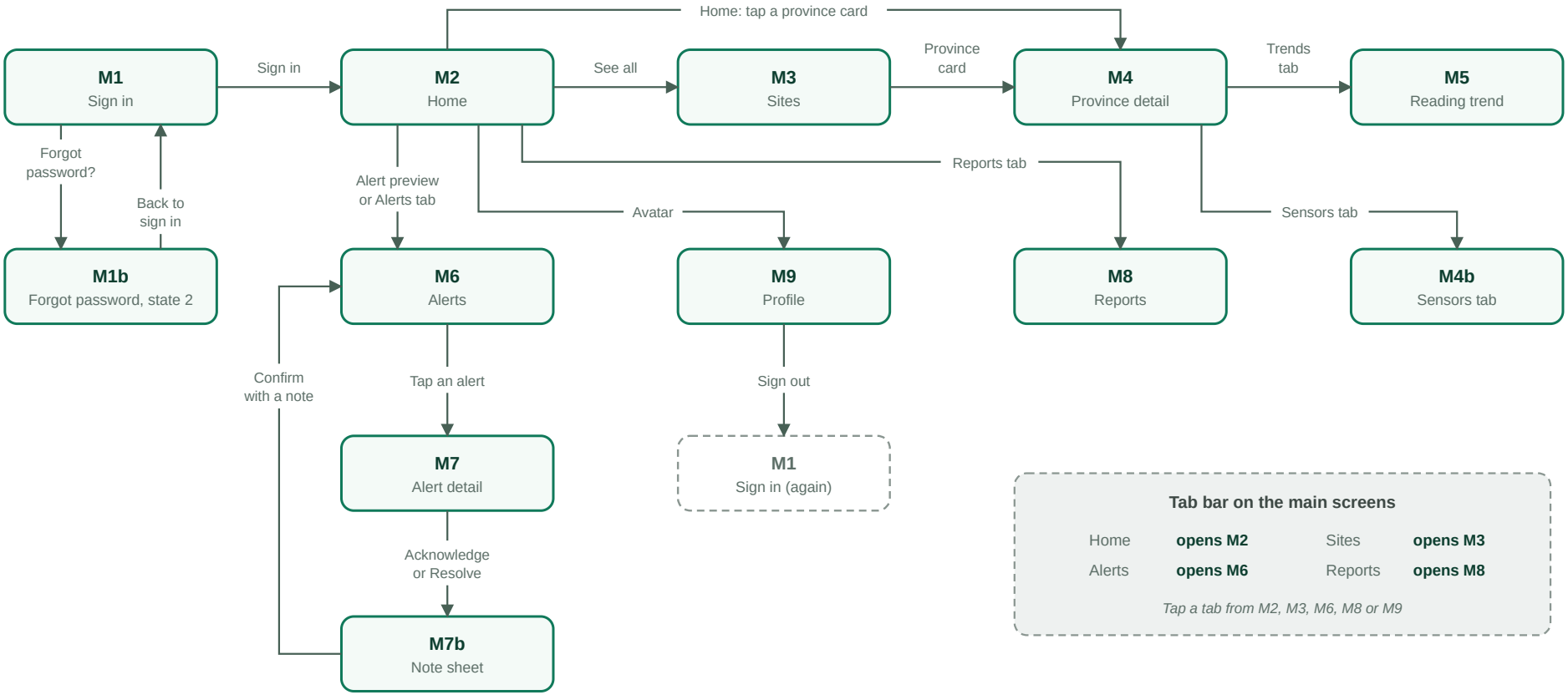


Figure 22.20 — Flutter navigation map.

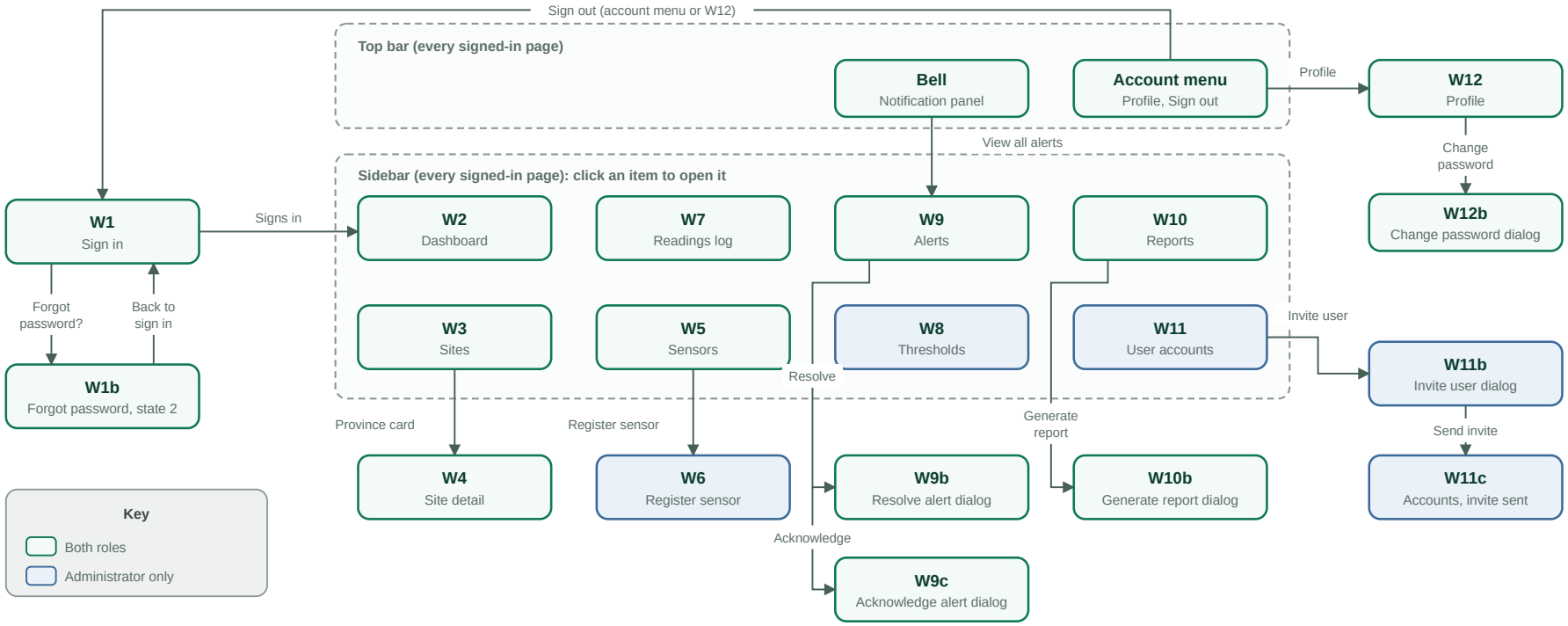


Figure 22.21 — ASP.NET navigation map.

22.5.5 Navigation Tables

Flutter. Each row is one arrow of the Flutter navigation map.

Starting screen	User action	Next screen
M1	Taps Sign in	M2 Home
M1	Taps Forgot password?	M1b
M1b	Taps Back to sign in	M1
M2	Taps a location card	M4 Province detail
M2	Taps See all, or the Sites tab	M3 Sites
M2	Taps the alert preview, or the Alerts tab	M6 Alerts
M2	Taps the avatar	M9 Profile
M2, M3, M6, M9	Taps the Reports tab	M8 Reports
M3	Taps a location card	M4 Province detail
M4	Opens the Trends tab	M5 Reading trend
M4	Opens the Sensors tab	M4b Sensors tab
M6	Taps an alert	M7 Alert detail
M7	Taps Acknowledge or Resolve	M7b Note sheet
M7b	Confirms with a note	M6, alert updated
M8	Taps Download report	Confirmation on M8
M9	Taps Sign out	M1 Sign in

ASP.NET. Each row is one arrow of the ASP.NET navigation map.

Starting page	User action	Next page
W1	Signs in	W2 Dashboard
W1	Clicks Forgot password?	W1b, then Back to sign in returns to W1
Any page	Clicks a sidebar item	Dashboard W2, Sites W3, Sensors W5, Readings log W7, Alerts W9, Thresholds W8, Reports W10, User accounts W11
W3	Clicks a location card	W4 Site detail
W5	Clicks Register sensor	W6 Register sensor
W9	Clicks Resolve	W9b Resolve alert dialog. Confirm or Cancel returns to W9.
W9	Clicks Acknowledge	W9c Acknowledge alert dialog. Confirm or Cancel returns to W9.
W10	Clicks Generate report	W10b Generate report dialog. Generate or Cancel returns to W10.

Starting page	User action	Next page
W11	Clicks Invite user	W11b Invite user dialog. Cancel returns to W11.
W11b	Clicks Send invite	W11c User accounts, with the new account shown as Invited
Any page	Clicks the bell, then View all alerts	W9 Alerts
Any page	Clicks the avatar, then Profile	W12 Profile
W12	Clicks Change password	W12b dialog, Cancel returns to W12
W12 or avatar menu	Clicks Sign out	W1 Sign in

22.5.6 Mobile and Web Integration Check

Each row shows an action in one application, the Firebase data it changes, and what the other application then shows.

Action or display	Firebase data	Effect in the other application
Standard User resolves an alert with a note (M7b)	Alert_action (Resolved) and Alert: state	The Administrator sees the alert as Resolved and opens the note on W9.
Administrator acknowledges an alert on W9c	Alert_action (Acknowledged) and Alert: state	M6 and M7 show the alert as Acknowledged with the note.
Administrator changes a band on W8	Threshold: min_value, max_value, basis	M4 shows the new band. The next reading is checked against it (IR06).
A reading arrives from the gateway	Reading and Sensor_status	W2, W4, W5 and W7 show it, and so do M2, M4 and M5.
A new alert is raised	Alert and Notification	The badge on M6 and the bell and W9 show it with the same state (IR03).
Administrator assigns locations on W11b or W11	User_location	M3 and M2 list only those locations for that user.
Administrator invites a user on W11b	Firebase Authentication, User and User_location	The new user sets a password from the email and signs in on M1 (IR05).
Standard User changes a toggle on M9	User: notify_breach, notify_offline, notify_daily	W12 shows the same toggle states.
A report is generated on W10b	Report, Report_location and the file in Cloud Storage	M8 downloads the same file.

22.5.7 Interface Information Supported by the ERD

Interface information	Shown on	Entity	Fields
Location name, coordinates, crop, planting date	M3, W3, W4	Location	name, latitude, longitude, crop, planting_date
Sensor code, depth, firmware, installed date	W4, W5, W6	Sensor	sensor_id, depth_cm, firmware_version, installation_date, modbus_slave_id, is_active
Battery, power, signal, state, last communication	M4b, W4, W5	Sensor_status	battery_level, power_status, signal, status, last_communication

Interface information	Shown on	Entity	Fields
Seven parameter values and capture time	M2 to M5, W2, W4, W7	Reading	captured_at, soil_moisture, soil_temperature, conductivity, ph, nitrogen, phosphorus, potassium
Band, basis and last edit	W8, M4 range bars	Threshold	min_value, max_value, basis, updated_by, updated_at
Alert severity, state, value, threshold, time	M6, M7, W9	Alert	severity, state, measured_value, threshold_value, raised_at
Offline and low-battery alerts	M6, W9	Alert	alert_type, parameter (empty for Offline)
Notes, owners, acknowledge and resolve times	M7b, W9, W9b, W9c	Alert_action	action, user_id, note, acted_at
Unread badge and notification list	M6 tab, bell panel	Notification	is_read, created_at, alert_id
Name, email, role, status	W11, W11b, W11c, W12, M9	User	name, email, role, account_status, created_at
Assigned locations	M3, M9, W11, W11b, W12	User_location	user_id, location_id
Notification toggles	M9, W12	User	notify_breach, notify_offline, notify_daily
Generated reports list and the dialog choices	M8, W10, W10b	Report, Report_location	name, format, report_type, period_start, period_end, generated_at, generated_by, file_path, location_id
Last data load and failed loads	W2, M2 (last updated time)	Data_load	started_at, finished_at, source, outcome
Password last changed	W12	Firebase Authentication	Held by Firebase Authentication, not in the ERD

22.5.8 Interface Updates Required

The drawn screens differ from the approved requirements in the places below. These updates are made to the designs and prototypes before the build.

Screen	Update needed	Reason
M7	Rename the Recommended action card to a plain comparison of the reading with its threshold. Remove the treatment advice.	NFR17, C12
W11	Change the role label Field officer to Standard User.	FR04, NFR12
W8	Add a Basis field for each band.	FR48, NFR19
W8	Offer in-application notifications only. Do not show email or push options.	C09
W9	Show the Owner column as empty while an alert is Open.	Alert_action holds the user once someone acts
W4, W7, W9	Show Export CSV and Export to the Administrator only.	FR67, FR68
W6	Remove the Notes field.	C24
W11	Remove the Last active column.	C24
W2, M2	Show the time of the last data load and a notice when a load failed.	FR15, FR78

Screen	Update needed	Reason
All screens	Use one word for a location in every label. The designs use Site, Sites and Province in different places.	NFR12

22.6 Design Alignment

Each requirement area traces to its use cases, sequence diagrams, ERD entities and screens, so the architecture, the UML diagrams, the ERD and the interfaces describe the same system. SD is a sequence diagram (22.3).

Requirement area	SRS reference	Use cases	SD	ERD	Flutter	ASP.NET
Sign in, password	FR01–FR03, FR07, FR79	UC01, UC02, UC15	SD5, SD6	User	M1, M1b, M9	W1, W1b, W12, W12b
Roles and location access	FR04–FR06, FR82, FR83	UC01, UC17	SD4, SD6	User, User_location	M3, M9	W11, W11b, W11c
Dashboard	FR08–FR15	UC03, UC04	SD7	Location, Sensor, Sensor_status, Reading, Alert	M2	W2
Locations	FR16–FR22	UC03, UC23		Location	M3, M4	W3, W4
Sensors and equipment condition	FR23–FR30, FR39–FR44	UC04, UC18	SD3, SD7	Sensor, Sensor_status	M4, M4b	W4, W5, W6
Readings	FR31–FR38	UC03	SD7	Reading	M2, M4	W2, W4
Thresholds	FR45–FR50	UC16	SD2	Threshold	M4 (read only)	W8
Alerts	FR51–FR56, FR59	UC20, UC21, UC24–UC26, UC08	SD7	Alert, Sensor	M6, M7	W9
Acknowledge and resolve with a note	FR57, FR80, FR81	UC09, UC10, UC11	SD1	Alert, Alert_action	M7, M7b	W9, W9b, W9c
Notifications	FR58, FR84, FR85	UC22, UC12	SD7	Notification, User	M6 badge, M9	Bell, W12
History and trends	FR60–FR66	UC05, UC06		Reading	M5	W2, W7
Reporting and export	FR67–FR69	UC07, UC13		Report, Report_location	M8	W7, W9, W10, W10b
User administration	FR70–FR74	UC17	SD4	User, User_location		W11, W11b, W11c
Data provisioning	FR75–FR78, IR04, IR07	UC19	SD7	Reading, Sensor_status, Data_load		W2 (last updated)
Integration	IR01–IR08	All	SD1, SD2, SD4, SD7	All entities	All	All

22.7 Changes to the Baseline Requirements

This design makes the changes below to the Phase 1 baseline (version 3.0). They continue the change log in Section 19, which ends at C14. Section 21.3 records the Supabase project used in Phase 1; it is superseded by C15 and stays in this document as the record of the baseline.

22.7.1 Change Log Entries

Ref	Original position	Change	Reason
C15	Shared backend provided by Supabase (SR03, SR05, SR06, IR01 and Sections 8 and 12).	Replaced by Firebase: Authentication, Realtime Database and Cloud Storage. The structure is unchanged: one shared backend, a Flutter client and an ASP.NET client.	Delivery team decision. The stakeholder also stated that sensor data is read into Firebase (Appendix E), so Firebase removes a transfer step.
C16	SR04: the database must be relational and enforce referential integrity.	The data model stays relational in design (the ERD with keys and cardinality). The Realtime Database is one JSON tree, so each entity is a path, a foreign key is a key in that path, and integrity is kept by Security Rules, the web server's checks and fixed keys.	Follows from C15. The ERD remains the logical model.
C17	C04: humidity removed because it was not a stakeholder requirement.	Confirmed. Humidity is not stored. The seven soil parameters are the only reading values.	Delivery team decision: the data model keeps only the fields the requirements need.
C18	FR44: an administrator can configure the period after which a sensor is inactive (C07: two hours, configurable).	The period is fixed at two hours for every sensor. An Offline alert is its own alert type with no parameter.	The Thresholds page holds bands for the seven soil parameters. A period setting would need its own control and rule.
C19	FR57: a user can acknowledge an alert.	Resolve is added as a separate step. Both Acknowledge and Resolve require a note, each stored with its own user and time. Severity and state are separate fields.	The designs show acknowledged and resolved alerts with notes (FR80, FR81).
C20	Section 17.2: the mobile application is read-only.	The mobile application also writes: alert acknowledge and resolve with a note, notification read flags and notification preferences.	FR57 is marked Both, and the designs show these actions on the phone.
C21	Access is by role only (FR05).	Access is also by location. A Standard User sees only assigned locations. An Administrator assigns them.	The designs (M3, M9, W11, W11b) show location assignment (FR82, FR83).
C22	Sections 8.1 and 8.3: data provisioning is a scheduled backend function that loads representative sample readings (C01, C02).	The system works with the actual readings of the 15 sensors. The sensor gateway is the data provisioning component: it writes the readings and the sensor status and is their only writer (IR07). Alert evaluation is a separate component. Each load is recorded in Data_load (FR78).	The sensors are operated by a third party who writes to Firebase (Appendix E). Keeps IR07: one writer, replaceable without changing any other component.
C23	Section 16: seven minimum entities (Users, Locations, Sensors, Readings, Sensor status, Thresholds, Alerts).	Extended to thirteen. Added: User_location, Alert_action, Notification, Report, Report_location and Data_load. Location gains latitude, longitude, crop and planting date.	Section 16 says the final structure is settled in design. The new entities and fields are what the screens and FR58, FR69, FR78, FR80–FR83 need.

Ref	Original position	Change	Reason
C24	Section 16 and the designs: Sensor.notes, User.last_active_at and Data_load.error_message.	The three fields are removed from the data model and from the W6 and W11 screens (22.5.8). A failed load is still shown from Data_load.outcome.	Delivery team decision: the data model keeps only the fields the requirements need.
C25	FR80 and FR81: who resolved an alert, when, and the note are stored with the alert.	Each acknowledge and each resolve is an Alert_action record under the alert, linked to the alert and to the signed-in user. Alert keeps one foreign key, sensor_id, and location_id is not repeated on Reading, Alert or Notification.	Keeps Alert free of empty fields and shows every link in the ERD.

22.7.2 New Requirements

ID	Requirement	Component	Priority
FR79	A user must be able to request a password-reset email when the password is forgotten.	Both	Should
FR80	A user must be able to resolve an alert, which records who resolved it and when.	Both	Should
FR81	The system must require a note when an alert is acknowledged or resolved and store the note with the alert.	Both	Must
FR82	The system must limit a Standard User to the locations assigned to that user.	Backend	Must
FR83	An administrator must be able to assign locations to a Standard User and change the assignment.	Web	Must
FR84	A user must be able to see their notifications and an unread count, and mark notifications as read.	Both	Should
FR85	A user must be able to choose which notifications they receive (breach, offline, daily summary).	Both	Could
IR08	A change to a user's role or assigned locations must apply in both applications at that user's next request.	Integration	Must

22.7.3 Amended Requirements

ID	Phase 1 wording	Amended wording
SR03	The shared backend must be provided by Supabase.	The shared backend must be provided by Firebase (Authentication, Realtime Database and Cloud Storage for Firebase).
SR04	The database must be relational and must enforce referential integrity between locations, sensors, readings, thresholds and alerts.	The data model must be relational in design, and referential integrity between locations, sensors, readings, thresholds and alerts must be kept by Firebase Security Rules, the web server's checks and fixed keys.
SR05	Authentication must be provided by the shared backend.	Authentication must be provided by Firebase Authentication rather than implemented separately in each application.
SR06	Role-based access must be enforced by database access rules.	Role-based access must be enforced by Firebase Security Rules, and by the web server before every write for operations that bypass those rules.

ID	Phase 1 wording	Amended wording
IR01	Both applications must connect to the same Supabase project and read from the same tables.	Both applications must connect to the same Firebase project and read from the same Realtime Database paths.
NFR05	Only representative sample data and test accounts may be used in the prototype and in the demonstration.	Only the actual readings of the project's sensors and test accounts may be used in the prototype and in the demonstration.
IR04	Readings written by the data provisioning component must be available to both applications.	Readings written by the sensor gateway must be available to both applications.
IR07	The data provisioning component must be the only writer of readings and sensor status, so that it can later be replaced by a live integration without changing any other component.	The sensor gateway must be the only writer of readings and sensor status, so that it can be replaced without changing any other component.
FR44	An administrator must be able to configure the period after which a sensor is treated as inactive.	A sensor must be treated as inactive after two hours without a reading. The period is the same for every sensor (C18).
FR57	A user must be able to acknowledge an alert so that it is removed from the open list.	A user must be able to acknowledge an alert, with a note, so that it is removed from the open list (FR81).
FR69	The system must produce a summary report of conditions across the three locations. (Web)	The web application generates the summary report. Both applications download it. (Both)

22.7.4 Open Decisions and Assumptions

Open decision or assumption	Current position	To settle
Blaze plan	Cloud Storage (report files) and Cloud Functions (alert evaluator) both need the paid Blaze plan. The Spark plan has neither. If the project stays on Spark, the evaluator runs as a background service in the web application and reports are generated on request instead of being stored.	Decision before alerts and reports are built
How ASP.NET reaches the database	REST API with a service-account token. The Realtime Database Admin documentation covers Java, Node.js, Python and Go, not .NET. One data access class hides the REST calls.	Confirm during the build
Live updates	The approved requirements refresh data when a screen is opened. The database can push changes to a listening phone with no change to the structure.	Decide whether the phone listens live
Reading layout the gateway writes	readings/{location_id}/{sensor_id}/{yyyyMMddHHmm} with all values in one record.	Confirm with the gateway operator
Critical or Warning	Severity is stored. The rule that picks between them is an evaluator setting.	Agree the rule with the stakeholder
Password last changed (W12)	Firebase Authentication records when the password was last updated. W12 reads it from there, so it is not an ERD field.	Confirm during the build

References: Firebase pricing, firebase.google.com/pricing (Spark plan allowances for the Realtime Database, Cloud Storage and Cloud Functions). Cloud Storage for Firebase plan requirement, firebase.google.com/docs/storage/faqs-storage-changes-announced-sept-2024.

22.8 Design Sources, Project Plan and Repository

22.8.1 Original Design Sources

The diagrams and the interface designs were drawn in Figma. The table lists where each design lives.

Design	Tool	Link
System architecture diagram	Figma	https://www.figma.com/design/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=0-1&t=7lJzgtAz3s2bOrsj-1
Use case diagram	Figma	https://www.figma.com/design/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=0-1&t=7lJzgtAz3s2bOrsj-1
Sequence diagrams 1 to 7	Figma	https://www.figma.com/design/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=0-1&t=7lJzgtAz3s2bOrsj-1
Entity relationship diagram	Figma	https://www.figma.com/design/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=0-1&t=7lJzgtAz3s2bOrsj-1
Navigation maps	Figma	https://www.figma.com/design/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=0-1&t=7lJzgtAz3s2bOrsj-1
Flutter interface designs	Figma prototype	https://www.figma.com/proto/H0fkapW22cNvwNadtpKjip/Mobile?node-id=72-2&p=f&t=IJKiFTTGSL0aGauq-1&scaling=scale-down&content-scaling=fixed&page-id=61%3A544&starting-point-node-id=72%3A2
ASP.NET interface designs	Figma prototype	https://www.figma.com/proto/KmYxvZZnIxJXbAuP3n39Hb/Web-Application?node-id=114-875&p=f&viewport=1067%2C46%2C0.47&t=DKfFo55Ux5OjYWjw-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=114%3A875&page-id=6%3A15

22.8.2 Source Repository

The source code and the project documentation, including this specification, are held in one GitHub repository that all members of the delivery team can write to.

Item	Detail
Source repository	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Repository name	soil-sensor-monitoring
Repository owner	Nido-Maphosa86
Repository contents	README, a documentation folder holding the specification and the design sources, the project schedule, and the source code for the Flutter application, the ASP.NET application and the backend
Shared backend	One Firebase project: Authentication, Realtime Database and Cloud Storage for Firebase
Not in the repository	Passwords, API keys, the Firebase service-account key, telephone numbers and identity numbers are never committed. The key is loaded from the server's environment.

22.8.3 Project Plan

The project schedule is kept in Microsoft Project. The screenshot below shows Phase 2, Architecture and Design: its tasks, durations, dates, predecessors, owners and progress.

28			**PHASE 2: ARCHITECTURE AND DESIGN**	**23 days**	**Tue 26/09/15**	**Thu 26/10/15**			**95%**
29	✓		System architecture design	3 days	Tue 26/09/15	Thu 26/09/17		Mbasa Tyam	100%
30	✓		UML diagrams: use case, class and sequence	6 days	Fri 26/09/18	Fri 26/09/25	27	Mamello Machedi, Thobani Mjiyakho	100%
31	✓		Entity relationship diagram	4 days	Fri 26/09/25	Wed 26/09/30	29	Thobani Mjiyakho	100%
32	✓		Opened firebase account	1 day	Mon 26/09/21	Mon 26/09/21		Nido Maphosa	100%
33	✓		Mobile interface prototype	7 days	Sat 26/09/26	Mon 26/10/05		Ayatolla Pikwa,Caitlin Freem	100%
34	✓		Web interface prototype	18 days	Sun 26/09/13	Tue 26/10/06		Caitlin Freeman,Mamello Ma	100%
35	✓		Stakeholder design feedback	0 days	Wed 26/10/07	Wed 26/10/07	29,30,31,33,34	Nido Maphosa	100%
36	✓		Mentor consultation	1 day	Fri 26/10/09	Fri 26/10/09		Ayatolla Pikwa,Caitlin Fre	100%
37	✓		Design portfolio compilation	0 days	Mon 26/10/12	Mon 26/10/12	35	Nido Maphosa	100%
38	✓		Phase 2 design portfolio submitted	0 days	Tue 26/10/13	Tue 26/10/13	37	Nido Maphosa	100%
39			Phase 2 presentation and design approval	1 day	Thu 26/10/15	Thu 26/10/15	38		0%
40			Design approved to proceed to Phase 3	1 day	Thu 26/10/15	Thu 26/10/15			0%

Figure 22.22 — Project plan in Microsoft Project: Phase 2, Architecture and Design.

23. As-Built System and Final Status

This section records the system as it was delivered. Sections 1 to 22 and Appendices A to E are kept as approved. Where the delivered system differs from them, this section gives the final position and the reason. The facts were checked against the final code in the project repository (commit `5513bf1`, 2 November 2026: web console and mobile app). Where something was planned in Sections 1 to 22 but built differently, or not built, this section says so.

Where the Phase 3 assessment items are answered

Phase 3 assessment item	Where it is in this SRS
Overview of the implemented Flutter, ASP.NET and shared backend components	23.1 The Delivered System at a Glance; 23.2 Architecture as Built; 23.3 Data as Built
Table linking the approved requirements to the implemented functions, with working or incomplete status	23.6 Final Status of Every Requirement; 23.7 Acceptance Criteria Results
Screenshots of the important mobile and web interfaces and the shared processes	23.8 Screens as Built; 23.11 Shared Processes Demonstrated; 23.14 Progress Tracker Evidence
Approved changes made during implementation and the remaining limitations	23.4 Changes to the Requirements; 23.5 New Requirements; 23.10 Open Decisions, Risks, Limitations and Test Evidence
Working GitHub link and where the mobile and web source code is	23.12 Repository and Source Code
Screenshot of the updated Microsoft Project schedule	23.13 Project Schedule, Phase 3

Shared backend. The Phase 3 assessment sheet names Supabase. The approved shared backend of this project is Firebase, recorded as change C15 in Section 22.7.1 (the same single shared backend, used by both applications). Both applications use the same Firebase project.

23.1 The Delivered System at a Glance

- **One shared backend:** one Firebase project, `sensor-grid-73366`. Firebase Authentication holds the accounts and passwords; the Firebase Realtime Database holds all data as one JSON tree.
- **Two applications:**
  - **Web console:** ASP.NET Core MVC on .NET 10. Used by Administrators, and by field officers as a fallback. It reaches Firebase only from the server (REST API with a service-account token) and checks the role and provinces before every read and write.
  - **Mobile app:** Flutter (Android, app name **SensorGrid**). For field officers (Standard Users) only. It talks to Firebase directly as the signed-in person, limited by the Security Rules.
- **Real sensor data, not sample data:** a third-party gateway writes each probe's readings under the probe's code, about **every 6 seconds**. The system has **no sample data**.
- **Alerts:** raised by an **alert evaluator inside the web console**, which checks every **10 seconds**:
  - **Breach:** a value outside its band. **Critical** when 15 % or more past the limit, otherwise **Warning**.
  - **Offline:** no reading for **1 minute**. Always Critical.
- **Notifications:** one per new alert for each Active user who wants that kind: field officers assigned to that province, and all administrators. Shown on the web bell and, on the phone, as a bell count, an inbox and a banner.

- **Live behaviour:**
  - The phone is **pushed** changes: alerts and notifications arrive in about 1–2 s, readings are redrawn every 10 s.
  - The web **re-asks** on a schedule: every 10 s on Sensors and Site detail, 60 s on the Dashboard, 15 s for the bell.
- **Average or one sensor:** where a province has two or more sensors, the live readings (web Site detail gauges, phone province screen) show the **average of the reporting sensors** by default, with a switch **Average | GP-01 | GP-02** to show one sensor instead. A line always says which ("Average of 2 sensors" or "GP-01 · Online · last reading 14:32").
- **Deployment at the time of writing:** The sites and sensors in use are those registered in the web console. The system supports any number of sites and sensors, added from the web console.

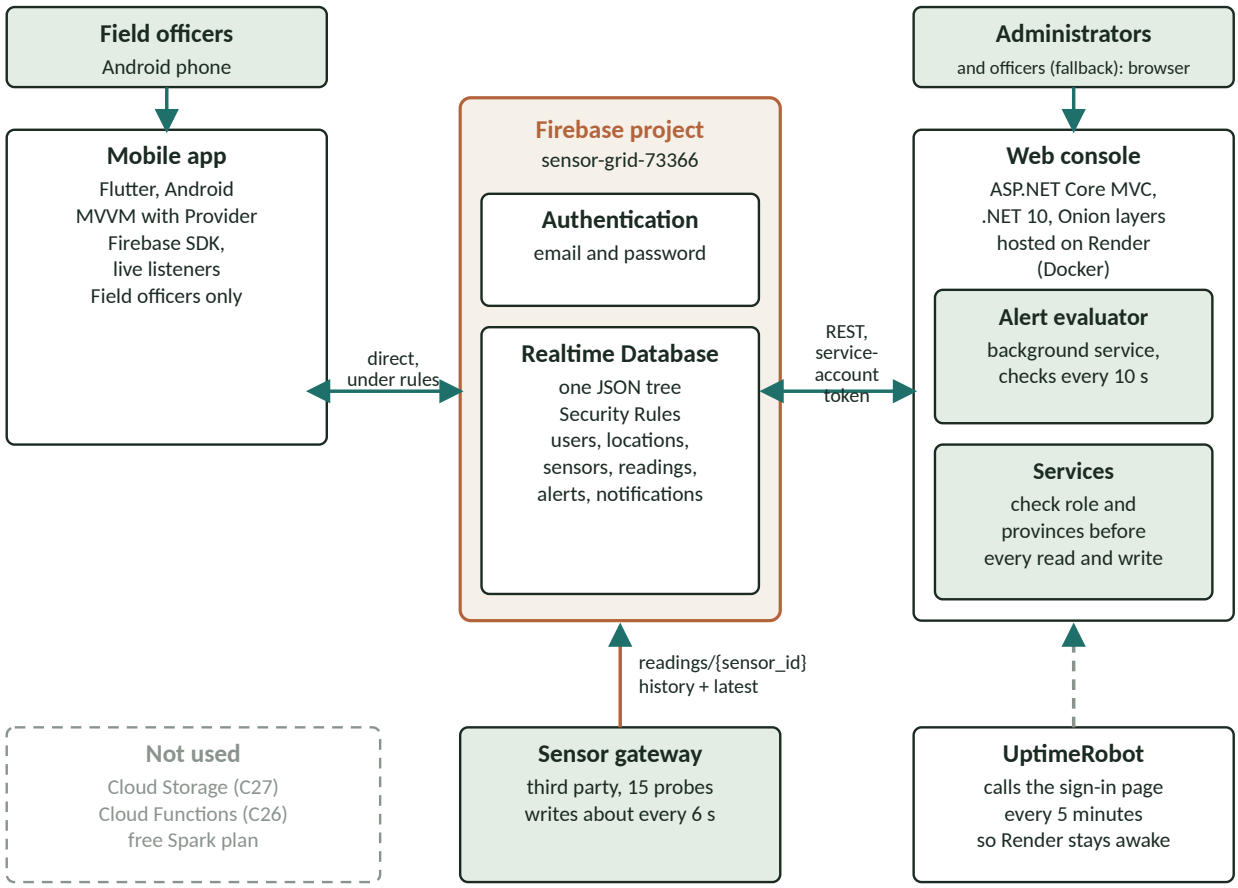
23.2 Architecture as Built

23.2.1 Components

Component	As built	Difference from v4.0 (22.1)
Web console	ASP.NET Core MVC, .NET 10, one project in Onion layers (Domain, Repository, Service, Web). Hosted on Render in Docker.	v4.0 said Admin SDK; built with the REST APIs (no .NET Admin SDK for the Realtime Database).
Mobile app	Flutter, MVVM with Provider, Android. Firebase SDK, live listeners.	Live listening added (v4.0 left it open).
Firebase Authentication	Email and password, reset and invitation emails, disabling deactivated accounts.	Same.
Firebase Realtime Database	All data. Security Rules in `web/firebase/database.rules.json`.	Tree differs in places (23.3).
Cloud Storage	**Not used.** Report files are built when downloaded.	Changed (C27).
Cloud Functions	**Not used.** The alert evaluator runs inside the web console every 10 s.	Changed (C26).
Sensor gateway	Third party. Writes `readings/{sensor_id}/history/{push_id}` and `readings/{sensor_id}/latest` about every 6 s.	Path, interval and fields differ (C28, C29).
Render + UptimeRobot	The web console runs on Render's free plan, which sleeps after 15 minutes without visitors. A free UptimeRobot monitor calls the sign-in page every 5 minutes to keep it awake, so alerts keep being checked.	New (C30).

23.2.2 Architecture Diagram

Figure 23.1 replaces Figure 22.1. The alert evaluator is part of the web console, Cloud Storage and Cloud Functions are not used, the gateway writes about every 6 seconds under the probe code, and the web console is hosted on Render.



Solid arrows: data in use. Dashed box and arrow: not part of the delivered system.

Figure 23.1 — Architecture as built (replaces Figure 22.1).

23.2.3 Figures of Section 22 That the As-Built System Replaces

Figure	What changed	Where the as-built version is
22.1 Architecture	Cloud Storage and Cloud Functions removed; the alert evaluator is inside the web console; the gateway writes about every 6 s under `readings/{sensor_id}`; Render and UptimeRobot added	Figure 23.1
22.2 Use cases	UC26 Raise Low-Battery Alert removed (not built: the gateway sends no battery data). Added for the Administrator: Edit Sensor, Retire Sensor, Delete Sensor, Edit Location, Delete Location. UC12 View Notifications now also covers the phone	This table
22.9 Sequence Diagram 7	The evaluator is the web console's background check every 10 s, not a Cloud Function; Offline is 1 minute; the phone receives the alert through its live listener	Figure 23.4 (in 23.9)
22.10 ERD	Sensor_status and Data_load removed; Sensor_location and Daily_stats added; Reading, Sensor, User, Report, Alert and Notification fields adjusted	Figure 23.2 and Table 23.3.3
Figures 22.11 to 22.19 (screens)	Replaced by the list of screens as built, where they differ	23.8

23.3 Data as Built

23.3.1 The Realtime Database Tree

```
users/{uid}                                 name, email, role ("Administrator" | "Standard User"),
                                            account_status (Active | Invited | Deactivated),
                                            created_at, last_active_at, notify_breach, notify_offline
user_locations/{uid}/{location_id}          true       always written together, in one update
location_users/{location_id}/{uid}          true
locations/{location_id}                     name, description, crop, latitude, longitude, planting_date, created_at
sensors/{location_id}/{sensor_id}           name, model, is_active, include_in_average, installation_date,
                                            depth_cm, modbus_slave_id
sensor_locations/{sensor_id}                "{location_id}"   (lets the rules find a sensor's province)
readings/{sensor_id}/history/{push_id}      written by the gateway, one row about every 6 s
readings/{sensor_id}/latest                 the newest row, written by the gateway
thresholds/{location_id}/{parameter}        min_value, max_value, basis, updated_by, updated_at
alerts/{location_id}/{alert_id}             sensor_id, alert_type (Breach | Offline), parameter, measured_value,
                                            threshold_value, severity (Critical | Warning),
                                            state (Open | Acknowledged | Resolved), raised_at
alert_actions/{location_id}/{alert_id}/{acknowledged | resolved}    user_id, note, acted_at
notifications/{uid}/{alert_id}              location_id, created_at, is_read, read_at
daily_stats/{sensor_id}/{yyyy-MM-dd}        readings, in_range, first_at, last_at, longest_gap_ms, offline_ms,
                                            {parameter}_min, {parameter}_sum, {parameter}_max
reports/{report_id}                         generated_by, name, format (PDF | CSV), report_type,
                                            period_start, period_end, generated_at
report_locations/{report_id}/{location_id}   true
```

Formats:

- Times are UTC milliseconds, shown in South African time (UTC+2). Dates are `yyyy-MM-dd`. Empty values are not stored.
- Reading keys are Firebase **push IDs**, which start with their time, so a period is a key range. No screen ever reads a whole readings branch.

23.3.2 The Gateway's Reading Row (Field Mapping)

Gateway field	Meaning	Unit	Note
`timestamp`	Capture time	UTC ms	
`moisture`	Soil moisture	%	**Divided by 10** (281 = 28.1 %), setting `GatewayMoistureDivisor`
`temperature`	Soil temperature	°C	
`ec`	Electrical conductivity	µS/cm	v4.0 Section 15 used dS/m. 1 dS/m = 1000 µS/cm
`ph`	pH		
`nitrogen`, `phosphorus`, `potassium`	N, P, K	mg/kg	
`device_id`, `humidity`	Also in the row		**Not used** (humidity stays out, C04 and C17)

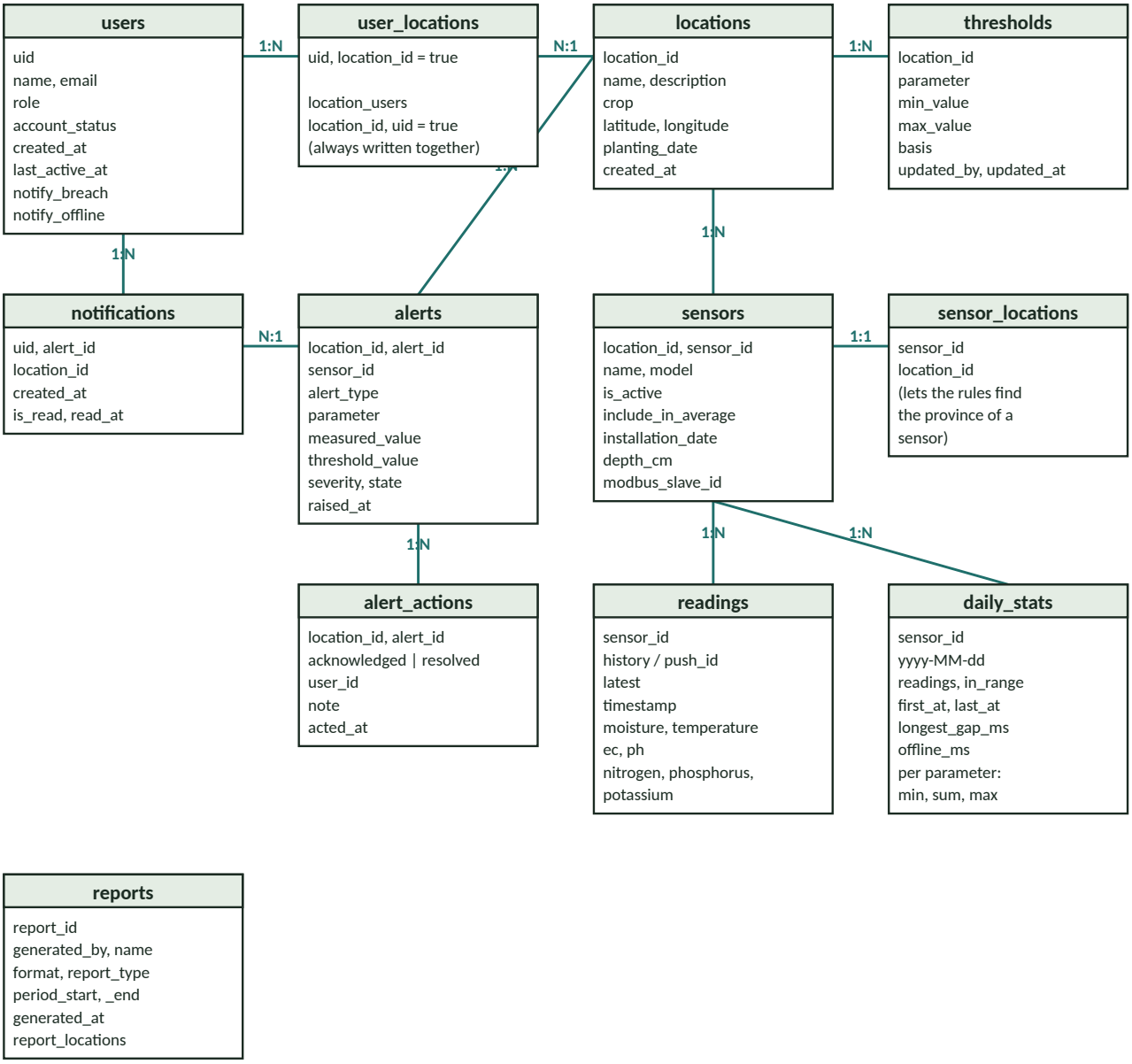
- A row where moisture, conductivity, N, P and K are all 0 means the probe is not in the soil. It is shown, but it never raises a breach and is left out of figures and charts.
- The gateway sends **no battery, power or signal values**.

One row as the gateway wrote it on 29 October 2026 (the time `1793280937532` is 15:35 in South African time):

```
readings/FS-01/history/-P56rssw_D_QqsDu1_oP
{ "device_id": "FS-01", "ec": 105, "humidity": 5.7, "moisture": 356, "nitrogen": 0, "ph": 6.5,
  "phosphorus": 7, "potassium": 0, "temperature": 30.9, "timestamp": 1793280937532 }
```

23.3.3 ERD Changes from Version 4.0 (Figure 22.10)

Entity	Change
Sensor_status	**Removed.** "Last communication" is the `timestamp` of `readings/{sensor_id}/latest`; Online or Offline is worked out from it (C29).
Data_load	**Removed.** The gateway writes no load records. A failing gateway shows as Offline sensors and Offline alerts (C33).
Sensor_location	**Added.** `sensor_locations/{sensor_id} = location_id`, written together with the sensor. The gateway's rows carry no location, so the rules use it.
Daily_stats	**Added.** One record per sensor per day with the day's figures (count, in range, lowest, sum, highest per parameter, longest gap, offline time), so a monthly report does not read hundreds of thousands of raw rows.
Reading	Key is a push ID under the sensor, not `yyyyMMddHHmm` under location and sensor (C28).
Sensor	`firmware_version` removed; `include_in_average` added.
User	`notify_daily` removed (C35); `last_active_at` kept (C24 reversed, C36).
Report	`file_path` removed (C27).
Alert	`alert_type` is Breach or Offline only (no Battery, C32).
Notification	`read_at` added.



Each box is a path of the Realtime Database; a line is a key that points to another path. Key fields only.

Figure 23.2 — Data as built (replaces Figure 22.10). The ERD is the logical model; the Realtime Database stores it as one JSON tree.

23.3.4 Security Rules as Built

- **Closed by default.** Only field officers (Standard User, Active or Invited) get any access through the SDK. Administrators use the web console.
- A field officer **reads**:
  - their own `users`, `user_locations` and `notifications`;
  - `locations`, `sensors`, `thresholds`, `alerts` and `alert_actions` of their assigned provinces;
  - `readings/{sensor}` only when `sensor_locations/{sensor}` points to one of their provinces.
- A field officer **writes** only:
  - their own `account_status` (Invited to Active, once), `last_active_at`, `notify_breach`, `notify_offline`;
  - an alert `state`, forward only;
  - a new `alert_actions` entry with their own user ID and a non-empty note of at most 500 characters;
  - their own `is_read` and `read_at`.
- `alerts/{location_id}` is indexed on `state` and `raised_at`.

- The web console's service-account token is not limited by the rules, so **every service checks the role and the assigned provinces itself** before every read and write.

> **Evidence without secrets**
> The service-account key, the database settings and the first administrator's details are held in user-secrets or environment variables and are never part of the repository or of this document (NFR03, NFR04).

23.4 Changes to the Requirements (C26 to C47)

These entries continue the change log of Section 19 and Section 22.7.1. A changed or dropped requirement keeps its ID; the change is explained here.

Ref	Original position	Change	Reason
C26	22.1.3: the alert evaluator is a Cloud Function	The evaluator runs inside the web console as a background service, every 10 seconds (setting `RunAlertEvaluatorInWebApp`, on by default)	22.7.4: Cloud Functions need the paid Blaze plan; the project stays on the free Spark plan
C27	22.1.3: report files stored in Cloud Storage; the phone downloads them	Reports are listed in `reports` and **built when downloaded**, from saved daily figures and alerts. No file storage. The phone shows a 24-hour summary and no download.	Cloud Storage needs the Blaze plan. Building on download also always gives current figures
C28	22.4.5: `readings/{location_id}/{sensor_id}/{yyyyMMddHHmm}`	`readings/{sensor_id}/history/{push_id}` and `readings/{sensor_id}/latest`, exactly as the gateway writes them	Confirmed with the gateway operator (open decision in 22.7.4)
C29	30-minute readings, a separate sensor status record (Sensor_status)	A row about every 6 s. No status record: last communication is the newest row's timestamp	The real gateway's behaviour
C30	Hosting not specified	Web console hosted on Render (Docker) with UptimeRobot calling it every 5 minutes	Free hosting; Render's free plan sleeps after 15 minutes, which would stop the alert check
C31	FR44 amended in v4.0: Offline after a fixed 2 hours	Offline after a fixed **1 minute**	With a reading every 6 s, 1 minute is 10 missed readings
C32	FR39, FR40, 22.1.7: battery, power and signal shown; low-battery alert below 25 %	Not built. No Battery alert type	The gateway sends no battery, power or signal values
C33	FR78, Data_load: every load recorded	Not built	The gateway writes no load records; a stopped gateway shows as Offline sensors and alerts
C34	C09, 20.2: push notification simplified to in-app alerts	In-app notifications on both apps: web bell; phone bell count, inbox (M10) and a banner on any screen. Still no push while the app is closed	Push (Firebase Cloud Messaging) was not added in the time available
C35	FR85: breach, offline, daily summary	Daily summary removed (switch and field)	Nothing produced a daily summary; a switch that does nothing was removed

Ref	Original position	Change	Reason
C36	C24: `User.last_active_at` removed	Kept. W11 shows "Last active"	Useful to administrators; written at sign-in on the phone
C37	FR30: move a sensor to another location	Not built	A sensor code starts with its province code and the gateway writes under that code
C38	22.5.8: Export only for the Administrator	Readings and alert exports are available to field officers too, for their own provinces only. Readings export is limited to 7 days	The data is already theirs to see; 7 days keeps an export under about 100,000 rows per sensor
C39	20.2: live dashboard updates simplified to refresh on view	Live: the phone listens to Firebase; the web refreshes key pages on a schedule (23.9)	Readings arrive every 6 s; refresh-on-view would look broken
C40	22.5.8: role label "Standard User" everywhere	The web still shows "Field officer"; the phone shows "Standard user"	Not corrected before submission. Recorded under NFR12
C41	Severity rule open (22.7.4)	Critical when 15 % or more past the limit; otherwise Warning. Offline is Critical	Agreed rule, written once in the web console's Domain rules
C42	22.4: Sensor_location and Daily_stats not in the ERD	Added (23.3.3)	Rules need a sensor's province; monthly reports need daily figures
C43	Location and sensor removal (FR22, FR29)	Retire and reactivate a sensor; delete only a retired sensor (its readings are kept); delete only a site with no sensors; both after typing the code	Safe removal with no lost readings
C44	22.1.2: an Administrator on the phone gets Standard User functions	Administrators are refused on the phone ("Use the web console")	The rules give the SDK access to field officers only; administrators work on the web
C45	v4.0 sample screens (M7 "Recommended action", M8 report download, M9 daily summary)	Removed from the built app	NFR17 (no advice), C27, C35
C46	22.5: the live readings of a location (W4 gauges, M4) show one set of values, the average of its sensors	A switch **Average | sensor | sensor** on W4 and M4 when a location has two or more active sensors. Average (of the reporting sensors that count towards it) is the default; choosing a sensor shows only its newest reading. On the phone, a value tapped with a sensor chosen opens that sensor's trend	With several sensors in a location, an average alone hides which probe is dry or failing
C47	W4 with no sensor reporting showed every value as 0 ("Low", in red)	The gauges show the last values received, in grey, marked "Last known", as the phone already did	A value of 0 looked like a real, alarming reading

23.5 New Requirements (FR86 to FR92)

ID	Requirement	Component	Priority
FR86	An administrator must be able to delete a retired sensor's registration and alerts, without deleting its readings, after confirming the sensor code.	Web	Should

ID	Requirement	Component	Priority
FR87	The applications must show new readings, sensor state changes and alerts without the user refreshing the screen.	Both	Should
FR88	The mobile application must show a banner on any screen when a new notification arrives while the app is open.	Mobile	Could
FR89	A province given to or taken from a field officer must reach the mobile application without signing out.	Mobile	Should
FR90	When the database has no accounts, the web console must create the first Administrator from its settings and send a password-setup email.	Web	Must
FR91	The web console must show a "Waiting for the Firebase connection" page naming any missing setting, instead of failing.	Web	Should
FR92	Where a location has two or more sensors, a user must be able to choose whether its live readings show the average of its reporting sensors or one chosen sensor, and the screen must say which.	Both	Should

23.6 Final Status of Every Requirement

Status values: Met; **Met (changed)**, where the requirement is met in a changed form (see the change ref); **Partly met**; **Not met**, with the reason. Screen codes: W = web, M = mobile.

23.6.1 Functional Requirements

ID	Status	Evidence / note
FR01	Met	W1, M1
FR02	Met	One message that never says which part was wrong
FR03	Met	Web account menu and W12; M9
FR04	Met	Administrator, Standard User
FR05	Met	Admin pages refused on the server (not only hidden); the phone refuses administrators ("Use the web console")
FR06	Met	Cookie authentication on every web page; rules closed by default
FR07	Partly met	W12b on the web. The phone has no change-password screen; phone users use Forgot password (FR79) or the web
FR08	Met	W2, M2
FR09	Met	W2 "Monitored provinces"
FR10	Met	W2 "Sensors reporting x / y"; W5 "N registered units"
FR11	Met	W2 reporting count; M2 network card
FR12	Met	W2 "n units offline"; W5 subtitle
FR13	Met	W2 "Open alerts"; M2; Alerts tab badge
FR14	Met	W2 site status; M2 cards (Optimal or Attention)
FR15	Met	Latest-reading times on W2 and M2; W4 "updated … ago"
FR16	Met	W3, M3
FR17	Met	W4, M4

ID	Status	Evidence / note
FR18	Met	W4 gauges and summary; M4
FR19	Met	W9 filtered by province; W4 summary of parameters outside bands
FR20	Met	W3 Add site
FR21	Met	Edit site page (new in v5.0)
FR22	Met	Delete site, only when it has no sensors, after typing its code
FR23	Met	W4, W5, M4 Sensors
FR24	Met	Sensor code, for example FS-01
FR25	Met	Sensor rows (code, installed, last reading, state); the Average / sensor switch shows one sensor's live values on W4 and M4; administrators open Edit sensor; M5 trend per sensor
FR26	Met	W7 Sensor column; M5
FR27	Met	W6 Register sensor
FR28	Met	Edit sensor page (new in v5.0)
FR29	Met	Retire / Reactivate; readings are kept
FR30	Not met (C37)	A sensor code must start with its province code, and the gateway writes under that code, so a sensor cannot move. Retire it and register the probe under the new province's code
FR31–FR37	Met	W4 and M4 for a chosen sensor (switch), W7, M5
FR38	Met	Traffic-light gauges and range bars on W4 and M4
FR39	Not met (C32)	The gateway sends no battery level
FR40	Not met (C32)	The gateway sends no power status
FR41	Met (changed)	Online, Offline or Retired
FR42	Met	Last reading time on W4, W5 and M4
FR43	Met (changed, C31)	Offline after 1 minute without a reading
FR44	Met (changed, C31)	Fixed at 1 minute for every sensor (v4.0 amended it to a fixed 2 hours)
FR45–FR48	Met	W8: minimum below maximum, basis required, updated_by and updated_at saved
FR49	Met	Field officers see the bands on W4 and M4 but cannot open W8
FR50	Not met (Could)	Not built. W8 offers "Reset to defaults" instead
FR51	Met	Breach alerts, checked every 10 s
FR52	Met	Offline alerts
FR53	Met	Sensor, location (path), parameter, measured value, threshold value, raised at
FR54–FR56	Met	W9, M6, M7; the same alert record in both applications
FR57	Met (C19)	Acknowledge with a note
FR58	Met (changed, C34)	In-app: web bell; phone bell, inbox and banner. No push when the app is closed

ID	Status	Evidence / note
FR59	Met	W9 chips (All, Critical, Warning, Offline, Acknowledged, Resolved), province filter and search
FR60	Met	Readings are never deleted, even when a sensor is deleted
FR61	Met	W7, M5
FR62–FR64	Met	W2 chart (value and range selectable); M5 (15 min, 24 h, 7 days, 30 days)
FR65	Met	W3 "Current averages by province"; W2 chart with one line per province
FR66	Met	W7 province, sensor and date-range filters; all seven parameters shown as columns
FR67	Met (changed, C38)	Export CSV of the filtered set, at most 7 days. Available to field officers too, for their own provinces
FR68	Met	W9 Export (alert history CSV); W10 Threshold breach log for a month
FR69	Met (changed, C27)	W10 Monthly summary (PDF or CSV), built when downloaded. The phone shows a 24-hour summary but downloads no files
FR70–FR74	Met	W11: invite, role, change role, deactivate (also disables sign-in), list
FR75	Met (changed, C29)	Readings come from the gateway about every 6 s, not every 30 minutes
FR76	Not met (C29)	No separate status record; the state is worked out from the newest reading
FR77	Met (changed)	Each gateway row has its own push ID, so a row is never written twice. This is owned by the gateway
FR78	Not met (C33)	The gateway writes no load records. A failure shows as Offline sensors and alerts
FR79	Met	W1b, M1 forgot password; the same answer whether or not the email exists
FR80, FR81	Met	Note required on the web, the phone and in the rules
FR82, FR83	Met	Province assignment on W11; enforced in the services and the rules
FR84	Met	Web bell (unread dot, Mark all as read); phone bell count, inbox and Mark all read
FR85	Met (changed, C35)	Threshold breaches and Sensor offline. Daily summary removed
FR86–FR92	Met	New, see 23.5

23.6.2 Non-Functional Requirements

ID	Status	Note
NFR01	Met	Passwords held and hashed by Firebase Authentication
NFR02	Met	Security Rules, plus server checks before every web read and write
NFR03	Met	Key file and settings in user-secrets (local) and environment variables plus a secret file (Render); `.gitignore` blocks key files. See R03 in 23.10
NFR04	Met	The app holds only public project identifiers, passed in at build time
NFR05	Met (amended in v4.0)	Actual readings and test accounts
NFR06	Met	Thresholds store updated_by and updated_at; alert actions store the user
NFR07	Met	Every page is in the sidebar or tab bar, one or two taps from the dashboard

ID	Status	Note
NFR08	Met	Traffic-light colours, red values out of range
NFR09	Met	Form errors appear under their field, inside the dialog
NFR10	Met	Android release build tested on a phone
NFR11	Met	Green and red messages on the web; snack bars and states on the phone
NFR12	Partly met	The web shows the Standard User role as **"Field officer"**; the phone shows **"Standard user"**. v4.0 22.5.8 asked for "Standard User" everywhere. "Province" is used for a location in both
NFR13	Met	Server-side short-lived cache; pages read only what they show
NFR14	Met	30-day trends read one row per 6 hours (about 120 reads)
NFR15	Met	A stopped gateway leaves earlier readings in place; both apps keep showing them
NFR16	Met, at risk	Readings are never deleted, but at one row every 6 s the free plan's 1 GB fills quickly. See R11 in 23.10
NFR17	Met	Values against bands only; no advice text (v4.0's "Recommended action" card was removed)
NFR18	Met	Every reading shows its time
NFR19	Met	Basis stored for every band
NFR20	Met	Not-built functions are listed in 23.10.3

23.6.3 System and Integration Requirements

ID	Status	Note
SR01	Met	Flutter
SR02	Met	ASP.NET Core MVC (.NET 10)
SR03	Met (changed, C27)	Firebase Authentication and Realtime Database. Cloud Storage not used
SR04–SR06	Met (as amended in v4.0)	
SR07	Met	GitHub repository `Nido-Maphosa86/soil-sensor-monitoring`
SR08	Met	Android
IR01	Met	One Firebase project, the same paths
IR02	Met	Sites, sensors and thresholds set on the web appear on the phone within 10–30 s
IR03	Met	One alert record; the phone shows changes in about 2 s
IR04	Met	Gateway readings shown in both
IR05	Met	An account invited on W11 signs in on M1
IR06	Met	The next 10-second check uses a changed band
IR07	Met	Only the gateway writes readings

ID	Status	Note
IR08	Met	Web: checked on every request. Phone: a province change applies within seconds

23.6.4 Updated Section 14.1 Counts

Section 14.1 gives the same counts by priority. The totals were counted again from the lists of requirements in Sections 10 to 13 and 22.7.2, plus FR86 to FR92.

Type	IDs	Total	Met	Met (changed)	Partly met	Not met
Functional (FR)	FR01 to FR92	92	75	10	1	6
Non-functional (NFR)	NFR01 to NFR20	20	19	0	1	0
System (SR)	SR01 to SR08	8	7	1	0	0
Integration (IR)	IR01 to IR08	8	8	0	0	0
Total	—	**128**	**109**	**11**	**2**	**6**

Type	Prefix	Total	Must	Should	Could
Functional	FR	92	67	21	4
Non-functional	NFR	20	16	4	0
System	SR	8	8	0	0
Integration	IR	8	8	0	0
Total	—	**128**	**99**	**25**	**4**

A requirement met in the form approved in version 4.0 (for example SR04 to SR06 and NFR05) counts as Met. A requirement met in a form changed in this section (a reference such as C27 is given) counts as Met (changed). The six requirements not met are FR30, FR39, FR40, FR50, FR76 and FR78; each has its reason in the table above.

23.7 Acceptance Criteria Results

Ref	Result	Note
AC1	Met	Wrong details and deactivated accounts are refused
AC2	Met	W2: provinces, sensors reporting x of y, offline, open alerts
AC3	Met	
AC4	Met for the registered sensors	Fifteen were planned; any number can be registered, and the sensors in use are those registered in the web console
AC5	Met	All seven values and the reading time
AC6	Partly met	Last communication and state are shown; battery and power are not (C32)
AC7	Met	
AC8	Met	1 minute (C31)
AC9	Met	

Ref	Result	Note
AC10	Met	Locations, sensors, thresholds and users can be created, edited and removed
AC11	Met	Refused on the server
AC12	Met	
AC13	Met	Changes appear in the other app within seconds (23.9)
AC14	Met	

23.8 Screens as Built

23.8.1 Web Console

Page	As built
W1, W1b	Sign in, forgot password. Known issue: the left panel still shows the design's sample text ("Fifteen sensors", "every thirty minutes", fixed sample values). It does not affect signing in or any reading
W2 Dashboard	Tiles: sensors reporting, monitored provinces, readings today, open alerts. Chart per province: last 15 minutes (live), 24 h, 7 days, 30 days, with a choice of value. Site status, latest readings.
W3 Sites	Province cards (Open site, Edit thresholds, Edit site), current averages by province, recent activity. **Add site** dialog.
W4 Site detail	Live gauges for seven parameters (every 10 s) with target bands; **Average / sensor switch** when the site has two or more sensors, with a line saying what is shown; last known values in grey when nothing reports; sensors at the site; Export CSV (24 h or 7 days); Edit site.
Edit site (new)	Name, crop, coordinates, planting date, description. Delete site when empty, after typing the code.
W5 Sensors	Province chips, search, Online / Offline / Retired filter, 10 per page, Edit button (admin).
W6 Register sensor	Proposes the next code and the next free Modbus ID; "Use the next free ID" fix.
Edit sensor (new)	Installation date, depth, Modbus ID, include in average. Retire / Reactivate. Delete (retired only, type the code).
W7 Readings log	Province, sensor and date filters; Newest / Newer / Older paging (14 rows); at most 31 days shown; Export CSV of at most 7 days.
W8 Thresholds (admin)	Minimum, maximum and basis per parameter per province; Reset to defaults.
W9 Alerts	Chips (All, Critical, Warning, Offline, Acknowledged, Resolved), province filter, search; Acknowledge (W9c) and Resolve (W9b) with a required note; View note; Export CSV.
W10 Reports	Monthly summary (PDF or CSV), raw readings (CSV, up to 7 days), threshold breach log; for the current month and the two before; built when downloaded.
W11 User accounts (admin)	Invite (status Invited, setup email), resend invitation, edit role, status and provinces, deactivate (also disables sign-in); Last active column.
W12, W12b Profile	Details, provinces, the two notification switches, change password, sign out.
Bell	Latest notifications, unread dot, Mark all as read; refreshes every 15 s.
Setup	"Waiting for the Firebase connection", naming the missing setting.

23.8.2 Mobile App

Screen	As built
M1, M1b	Sign in, forgot password; "Use the web console" for administrators; "Access denied" for deactivated accounts.
M2 Home	Greeting, **bell with unread count** (new), sensor network (Live or Offline), newest open alert, province cards (last hour).
M3 Sites	Province cards with sensors reporting, readings in the last hour, share in range, averages.
M4, M4b Province detail	Headline moisture, Readings (traffic light), Trends, Sensors (Online / Offline / Retired, last reading). **Average / sensor switch** under the header when the province has two or more sensors; tapping a value opens the trend on the chosen sensor. Offline: last values in grey with a banner naming the sensor.
M5 Trend	15 min (live, point per 10 s), 24 h, 7 days, 30 days; lowest, average, highest; sensor switcher; "No readings" with a longer period offered.
M6 Alerts	Counters, the six filters, grouped by day; open-alert badge on the tab.
M7, M7b	Alert detail as a plain comparison (no advice); Acknowledge and Resolve with a required note.
M8 Reports	Nutrient levels and share in range over the last 24 hours. No download (C27).
M9 Profile	Account, provinces, the two notification switches with an explanation, sign out.
M10 Notifications (new)	Newest first, grouped by day, unread dots, Mark all read; tapping opens the alert.
Banner (new)	Drops in on any screen when a new alert arrives, with View and Dismiss; disappears after 6 s.
Waiting screen	Shown when the app was built without its Firebase settings.

23.9 Timing and Live Behaviour

What happens	Web console	Mobile app
New reading (about every 6 s)	Sensors and Site detail: within 10 s. Dashboard: within 60 s. Sites and Readings log: on reload	Province screen and 15-minute trend: within 10 s. Home and Sites: within 30 s
Sensor stops sending	Offline 1 min to 1 min 10 s after the last reading (Dashboard 1–2 min)	Offline 1 min to about 1 min 20 s after the last reading
New alert (raised 0–10 s after the cause)	Bell within 15 s; Alerts page on reload; Dashboard within 60 s	Banner and bell about 1 s; Alerts tab about 2 s; Home within 30 s
Alert acknowledged or resolved on the other app	Alerts page on reload (up to 10 s late)	About 2 s
Threshold changed on the web	At once for whoever saved it; others within 30 s	Province screen within 10 s; Home within 30 s
Province given or taken away	The officer's next page, within 30 s	Within a few seconds, no sign-out
Account deactivated	Signed out at the next click	Refused by the rules within 10–30 s; signed out at the next start

Why they differ: the phone keeps a live Firebase connection and is pushed changes. The browser never talks to Firebase (the key stays on the server), so it re-asks the server on a schedule. The server remembers answers for

3 s (newest reading), 10 s (alerts, notifications) and 30 s (sites, sensors, thresholds, accounts); changes made through the web console itself show at once.

23.9.1 Alert Sequence as Built

Figure 23.4 replaces Figure 22.9 (Sequence Diagram 7). The numbers follow the order of events.

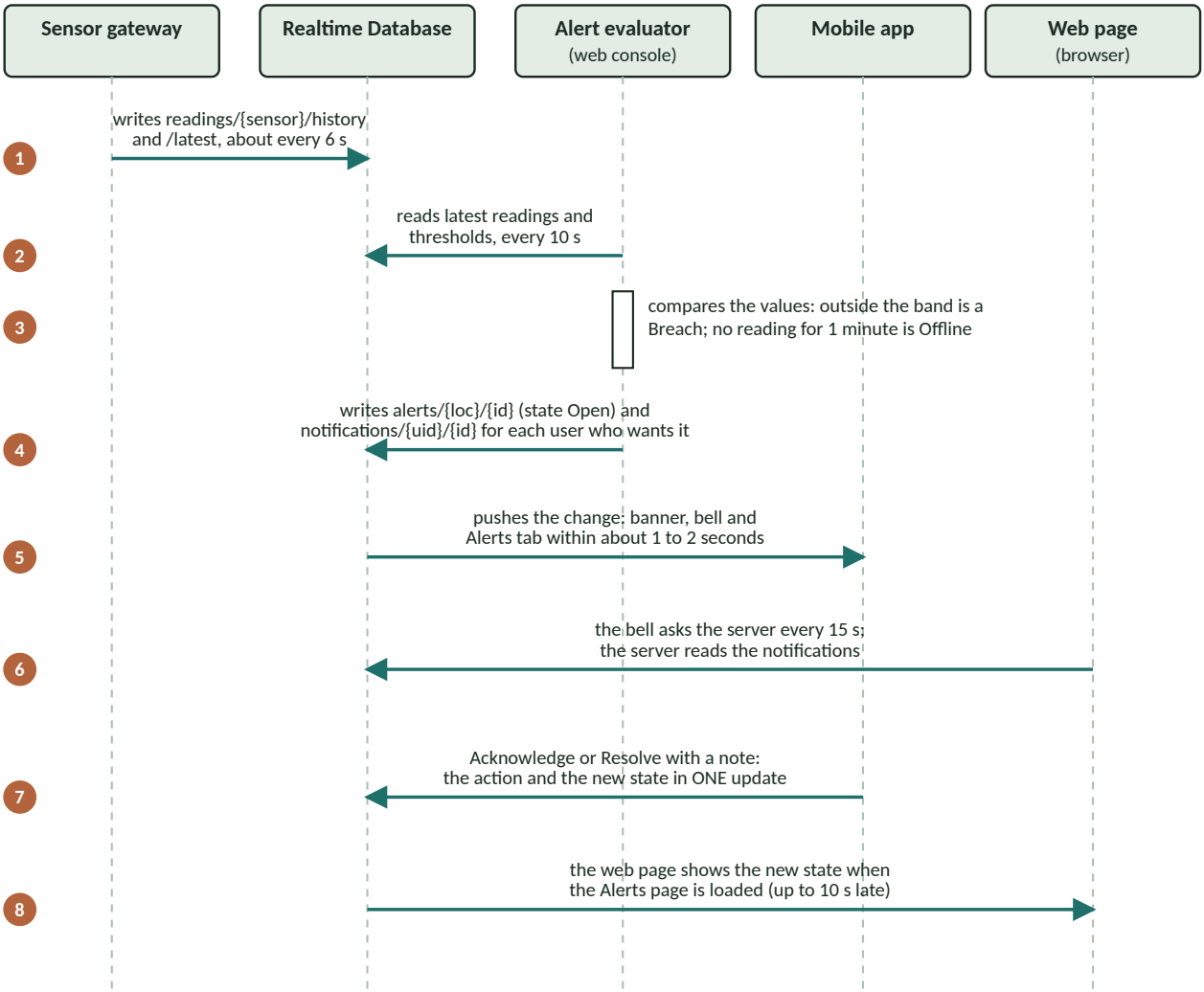


Figure 23.4 — A reading arrives and an alert is raised, as built (replaces Figure 22.9).

23.10 Open Decisions Resolved, Risks, Limitations and Test Evidence

23.10.1 Open Decisions from 22.7.4, Resolved

Decision	Outcome
Blaze plan	Stayed on the free Spark plan: evaluator in the web console (C26), reports built on download (C27)
How ASP.NET reaches the database	REST API with a service-account token, signed in the app (no NuGet package), behind one data-access interface
Live updates	Yes: the phone listens live; the web refreshes key pages on a schedule (C39)
Reading layout	`readings/{sensor_id}/history/{push_id}` plus `latest` (C28)
Critical or Warning	15 % or more past the limit is Critical; Offline is Critical (C41)

Decision	Outcome
Password last changed (W12)	Not shown on W12

23.10.2 Risk Register Updates (Section 18)

Risk	Update
R01 Time	Closed. The system was delivered; the functions not built are listed in 12.3.
R02 Integration	Closed. Both apps share one database; integration checked (23.9).
R03 Credentials	No key or password was ever committed to the repository.
R04 Thresholds	Open. The bands must be reviewed with real readings (Section 15.2).
R11 Free limits	**Raised to High.** At one row every 6 s per sensor (about 14,400 a day), each sensor stores roughly 3–4 MB a day. The Spark plan's 1 GB fills in under a year with one sensor, and in weeks with fifteen. Needs an archiving plan, a longer gateway interval, or the Blaze plan.
R13 (new) Hosting sleep	Render's free plan sleeps after 15 minutes without visitors, which stops the alert check. Mitigated with UptimeRobot; contingency: keep the web console open during demonstrations.
R14 (new) Third-party gateway format	The system depends on the gateway's paths and field names (23.3.2). A change on their side needs a change in the Repository layer only.

23.10.3 Limitations: Functions Not Built (NFR20)

1. Battery, power and signal values, and the low-battery alert. The gateway does not send them.
2. Push notifications while the app is closed. Only in-app notifications.
3. Moving a sensor to another province (FR30).
4. Copying one province's bands to another (FR50).
5. A configurable offline period (fixed at 1 minute).
6. Data-load records (FR78).
7. Report download on the phone, and a change-password screen on the phone.
8. Stage-aware thresholds and climate temperature (Section 15.1 and Appendix E.4).
9. Report files are not stored: a report is rebuilt from current data each time it is downloaded, so it changes if the data or the provinces change (for example, after a province is deleted).

23.10.4 Test Evidence

Check	Result
Web console build (`dotnet build`, fresh clone from GitHub)	Build succeeded, 0 warnings, 0 errors
Mobile static analysis (`flutter analyze`)	No issues found
Android release build	`app-release.apk` built (about 49 MB)
Edit, retire and delete for sensors and sites	Tested against a local stand-in for Firebase: field errors, retire closes open alerts with a note, delete removes the registration and alerts and **keeps the readings**, a site with sensors cannot be deleted (even by a forged request), an empty site is deleted and taken off every user

Check	Result
In-app notifications	Tested in a browser preview with made-up data: bell count, banner arriving and disappearing after 6 s, inbox, opening marks read, Mark all read
Average / sensor switch	Web, against the stand-in database with one province holding two sensors: the average was correct (25.0 % from 22 % and 28 %); each sensor showed only its own values; a silent sensor turned grey by itself within the 10-second refresh; the average then counted only the reporting sensor; with no sensor reporting, last values were shown instead of zeros. Phone, in the preview: the switch, an offline sensor in grey, the trend opening on the chosen sensor, and no switch for a one-sensor province
Final repository check	Commit `5513bf1` on GitHub identical to the tested packages; fresh clone builds (web) and analyses clean (mobile); hosted console responding
End-to-end on a phone with live data	Shared Processes 1 and 2 checked across the integrated system on 29 October 2026 (Section 23.11): passed
Hosted console	Responds at `https://sensorgrid.onrender.com` (sign-in page)

23.11 Shared Processes Demonstrated

Both applications use the same Firebase project. Two shared processes are demonstrated in the final presentation. In each, an action in one place changes the shared information and the other application shows the same result. Both processes were completed and checked across the integrated system on 29 October 2026 (tasks P3-07 and P3-08, Section 23.13.1).

23.11.1 Process 1: Data Submission and Retrieval

Step	What happens	Where
1	The sensor gateway writes a reading row about every 6 seconds	`readings/{sensor_id}/history/{push_id}` and `readings/{sensor_id}/latest`
2	The Realtime Database keeps the row; the Security Rules allow the gateway to write and field officers to read only their provinces	Firebase project
3	The web console reads the newest row from the server (Sensors and Site detail every 10 s); the phone is pushed the change (province screen within 10 s)	Web W4, W5; mobile M4
4	Both applications show the same values and the same reading time. Where a province has two or more sensors, both can show the average or one chosen sensor	Web W4; mobile M4

Requirements: FR31 to FR37, FR75, FR87, FR92, IR04 and IR07.

23.11.2 Process 2: Shared Update and Display

Step	What happens	Where
1	The alert evaluator raises an alert and a notification for each user who wants that kind of alert	`alerts/{location_id}/{alert_id}`, `notifications/{uid}/{alert_id}`
2	A field officer opens the alert on the phone and chooses Acknowledge or Resolve. A note is required, and it is checked in the ViewModel and by the Security Rules	Mobile M7, M7b

Step	What happens	Where
3	The phone writes the action and the new state in ONE update. The state can only move forward: Open, Acknowledged, Resolved	`alert_actions/{location_id}/{alert_id}/{acknowledged | resolved}` and `alerts/.../state`
4	The web console shows the new state and the note when the Alerts page is loaded (up to 10 s late). The same alert record is shown in both applications	Web W9
5	The same works in the other direction: an alert acknowledged on the web appears on the phone in about 2 seconds	Web W9; mobile M6

Requirements: FR54 to FR58, FR80, FR81, FR84, IR03 and acceptance criterion AC13.

23.11.3 Other Shared Processes Available for the Demonstration

Process	What the panel sees	Requirements
Invitation and sign-in	An account invited on the web (W11) signs in on the phone (M1) with the same email and password	FR70, FR90, IR05
Threshold change	An administrator saves a band on W8; the next 10-second check uses it, and the phone shows the new band within 10 s	FR45 to FR48, IR02, IR06
Province assignment	An administrator gives or takes a province on W11; the phone changes within seconds, with no sign-out	FR82, FR83, FR89, IR08
Sensor goes silent	No reading for 1 minute: the sensor turns Offline on both and an Offline alert is raised	FR43, FR52

23.12 Repository and Source Code

Item	Where
GitHub repository	`https://github.com/Nido-Maphosa86/soil-sensor-monitoring`
Web console source (ASP.NET Core MVC, .NET 10)	`web/` folder, with `web/README.md` (set-up and run instructions)
Mobile app source (Flutter, Android)	`mobile/` folder, with `mobile/README.md` (set-up and run instructions)
Database Security Rules and set-up	`web/firebase/database.rules.json`
Documents (Phase 1, Phase 2 and Phase 3)	`documentation/` folder
Version demonstrated	Commit `5513bf1` (2 November 2026). The submitted ZIP and the repository hold the same version
Hosted web console	`https://sensorgrid.onrender.com` (sign-in page)

Secrets

No password, key or confidential information is in the repository, the ZIP or this document. The service-account key is held outside the repository, and the build settings of the mobile app are public project identifiers only (NFR03, NFR04). The web console is configured from user-secrets (local) or environment variables (Render).

23.13 Project Schedule, Phase 3

The project schedule is kept in Microsoft Project. The screenshots below show Phase 3, Development, Integration and Evaluation: its tasks, durations, dates, predecessors, resources and progress. The original editable file is submitted with this SRS, and the Excel Progress Tracker is submitted separately.

23.13.1 Phase 3 Task Record

The Excel Progress Tracker records thirteen Phase 3 tasks for the reporting period 15 October to 4 November 2026. All thirteen are marked completed at 100%. The responsible students for each task are named in the tracker. All dates are in 2026.

Task	Area	What was done	Planned	Done
P3-01	Firebase backend	Required tables, fields, keys and relationships implemented	15 Oct	15 Oct
P3-02	Firebase backend	Authentication and access configured where required	19 Oct	20 Oct
P3-03	Flutter	Approved mobile screens and navigation implemented	21 Oct	22 Oct
P3-04	Flutter	Main mobile functions connected to Firebase	23 Oct	27 Oct
P3-05	ASP.NET	Approved web pages and navigation implemented	21 Oct	22 Oct
P3-06	ASP.NET	Main web functions connected to Firebase	23 Oct	27 Oct
P3-07	Integration	Shared Process 1 working across the integrated system	28 Oct	29 Oct
P3-08	Integration	Shared Process 2 working across the integrated system	28 Oct	29 Oct
P3-09	Integration	Data fields and status values consistent across the applications	30 Oct	30 Oct
P3-10	SRS	Implementation status, approved changes and known limitations updated	19 Oct	1 Nov
P3-11	Microsoft Project	Schedule updated to reflect actual development progress	15 Oct	1 Nov
P3-12	GitHub	Current source code, README and relevant documentation organised	26 Oct	29 Oct
P3-13	Final demonstration	Prototype and demonstration flow prepared for the panel	—	—

23.13.2 Microsoft Project Schedule

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
40		Design approved to proceed to Phase 3	1 day	Wed 26/10/14	Wed 26/10/14		
41		**PHASE 3: DEVELOPMENT, INTEGRATION AND EVALUATION**	**14 days**	**Thu 26/10/15**	**Tue 26/11/03**		
42		**Sprint 1: Foundation**	**6 days**	**Thu 26/10/15**	**Thu 26/10/22**		
43		Firebase schema implementation	0 days	Thu 26/10/15	Thu 26/10/15		Mbasa Tyam,Nido Maphosa
44		Authentication and role based access	2 days	Mon 26/10/19	Tue 26/10/20		Thobani Mjiyakho
45		Representative sample data generation	2 days	Fri 26/10/16	Mon 26/10/19		Mbasa Tyam,Nido Maphosa
46		Flutter: authentication, dashboard and navigation	2 days	Wed 26/10/21	Thu 26/10/22	44	Ayatolla Pikwa,Katlontle Seiphetlho,Thobani Mjiyakho,Tiisetso Masita
47		ASP.NET: authentication, dashboard and navigation	2 days	Wed 26/10/21	Thu 26/10/22	44	Caitlin Freeman,Mbasa Tyam,Nido Maphosa,Mamello Machedi
48		**Sprint 2: functionality and integration**	**5 days**	**Fri 26/10/23**	**Thu 26/10/29**		
49		Flutter: location, sensors and readings	3 days	Fri 26/10/23	Tue 26/10/27	46	Ayatolla Pikwa,Katlontle Seiphetlho,Thobani Mjiyakho,Tiisetso Masita
50		ASP .NET: management and configuration screen	3 days	Fri 26/10/23	Tue 26/10/27	47	Caitlin Freeman,Mamello Machedi,Mbasa Tyam,Nido Maphosa
51		Threshold configuration and alert generation	3 days	Fri 26/10/23	Tue 26/10/27	47	Caitlin Freeman,Mamello Machedi,Mbasa Tyam,Nido Maphosa
52		Historical readings and graphs	3 days	Mon 26/10/26	Wed 26/10/28	45,46	Ayatolla Pikwa,Katlontle Seiphetlho,Tiisetso Masita
53		Integration testing across both applications	2 days	Wed 26/10/28	Thu 26/10/29	49,50	Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,Nido Maphosa,Thobani Mjiyakho,Tiisetso Masita

Figure 23.5 — Project plan in Microsoft Project: Phase 3, Sprint 1 and Sprint 2 (tasks 40 to 53).

GANTT CHART

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
52		Historical readings and graphs	3 days	Mon 26/10/26	Wed 26/10/28	45,46	Ayatolla Pikwa,Katlontle Seiphetlho,Tiisetso Masita
53		Integration testing across both applications	2 days	Wed 26/10/28	Thu 26/10/29	49,50	Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,Nido Maphosa,Thobani Mjiyakho,Tiisetso Masita
54		**Testing and evaluation**	**5 days**	**Wed 26/10/28**	**Tue 26/11/03**		
55		Manual functional testing	2 days	Wed 26/10/28	Thu 26/10/29	49,50	Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,
56		Prototype problem log and retesting	1 day	Wed 26/10/28	Wed 26/10/28	55SS,51SS,52SS	Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,
57		Corrections and defect resolution	2 days	Fri 26/10/30	Mon 26/11/02	56,53	
58		Stakeholder evaluation of the prototype	0 days	Fri 26/10/30	Fri 26/10/30	57SS	Nido Maphosa
59		Known limitations document	1 day	Fri 26/10/30	Fri 26/10/30		Nido Maphosa
60		Final GitHub release and Microsoft Project report	1 day	Tue 26/11/03	Tue 26/11/03	57,59	Caitlin Freeman,Katlontle Seiphetlho,Nido Maphosa
61		Final presentation preparation and rehearsal	3 days	Sun 26/11/01	Tue 26/11/03		Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,Nido Maphosa,Thobani Mjiyakho,Tiisetso Masita
62		Project Submitted	0 days	Mon 26/11/02	Mon 26/11/02		Nido Maphosa
63		Final prototype presentation and project defence	0 days	Tue 26/11/03	Tue 26/11/03		Ayatolla Pikwa,Caitlin Freeman,Katlontle Seiphetlho,Mamello Machedi,Mbasa Tyam,Nido Maphosa,Thobani Mjiyakho,Tiisetso Masita
64		Final group reflection and project closure	0 days	Tue 26/11/03	Tue 26/11/03		

Figure 23.6 — Project plan in Microsoft Project: Phase 3, Sprint 2 and Testing and evaluation (tasks 52 to 64).

23.14 Progress Tracker Evidence

The screenshots below are the sheets of the Excel Progress Tracker for Phase 3, as completed by the group. The tracker is submitted as a separate file.

Phase 3 Implementation Evidence

dence ID	Application / Area	Function / Feature	Evidence Type	Evidence / Link / File	Related SRS Requirement	Responsible Student	Notes
	ASP.NET	Admin Login	Screenshot		User Authentication Requirement	Nido Maphosa	
	ASP.NET	User Registration	Screenshot		User Management Requirement	Caitlin Freeman	
	ASP.NET	User Approval and management	GitHub Commit/PR		User Approval requirement	Tyam	
	Flutter	Farmer Login	Screenshot		User Authentication Requirement	GK Seiphetlho	
	Flutter	Soil Monitoring Dashboard	GitHub Commit/PR		Soil Data Monitoring Requirement	Mjiyakho Thobani	
	ASP.NET	Soil data synchronisation	Screenshot		Data Synchronisation requirement	Mamello Machedi	
	Integration	API integration	GitHub Commit/PR		System Integration Requirement	Ayatolla Pikwa	
	Supabase Backend	Database tables and soil readings	GitHub Commit/PR		Soil Readings requirement	Tiisetso Masita	

Figure 23.7 — Progress Tracker: Implementation Evidence.

Phase 3 Shared Process and Integration Progress

Flutter Action	ASP.NET Action	Shared Supabase Information	Current Status	Responsible Student(s)	Evidence / Link	Notes / Limitation
Provides login and registration interfaces and submits user credentials for authentication	Handles authentication requests, manages user registration and approval, validates login credentials and enforces user access privileges	Stores user account information, authentication details, unique user IDs, account approval status and assigned user roles	Completed	Mjiyakho,Nido, Tyam		
Requests and displays soil monitoring data on the mobile application	Handles data requests and retrieves the relevant soil monitoring information.	Soil sensor readings, moisture, temperature, humidity, pH, nutrient levels and timestamps	Completed	Pikwa, Tyam		
Retrieves and displays updated soil monitoring information on the mobile application	Processes data requests and ensures the applications retrieve the latest soil monitoring records	Shared soil sensor readings, parameter values, sensor IDs and timestamps.	Completed	Caitlin,Masita		
Retrieves and displays information from the shared database through the application's backened/API	Handles database requests, processes information and communicates with Firebase	User records, user roles, soil monitoring readings, sensor information and other stored system data	Completed	Nido Maphosa,Mjiyakho		
Sends requests to the ASP.NET API to retrieve and submit relevant system information	Receives API request, processes them and returns the appropriate responses to the applications	Soil monitoring records,user information and other data accessed through the API	Completed	Mjiyakho, Nido		
Displays soil monitoring readings and parameter information through the mobile dashboard	Retrieves and provides soil monitoring data required by the dashboard	Soil moisture, temperature, humidity, pH, nitrogen, sulfur and other sensor readings	Completed	Seiphetlho,Machedi		
			Not Started			
			Not Started			
			Not Started			
			Not Started			
			Not Started			

Figure 23.8 — Progress Tracker: Shared Process and Integration Progress.

Phase 3 Individual Contributions

Student Number	Development Area	Assigned Responsibility	Work Personally Completed	Date Completed	Hours Spent	Evidence / GitHub Link	Verified by Group	Verification Date	Quality / Review Notes	Contribution Status
223039784	ASP.NET	ASP.NET backend development, system integration and ensuring communication between the web and mobile applications	Developed and worked on the ASP.NET backend, handled the integration of the applications,ensured that the web and mobile applications communicated correctly and tested the overall system functionality.	2026/10/30	20hrs		Yes	2026/11/01		Completed
224097774	ASP.NET	ASP.NET development and Microsoft Project implementation	Worked on the Microsoft project and ASP.NET coding. Assisted with backend functionality, tested the application and verified that implemented features aligned with the SRS	2026/10/30	15hrs		Yes	2026/10/31		Completed
224135694	Flutter	Flutter mobile application development	Developed the flutter mobile application, implemented mobile application features and assisted with testing to ensure that the application functioned correctly and met the specified requirements	2026/10/28	14hrs		Yes	2026/10/31		Completed
222056708	ASP.NET	ASP.NET backend development and system integration	Assisted with ASP.NET development and supported with integrating the web and mobile applications	2026/10/29	16hrs		Yes	2026/10/31		Completed
222088478	Flutter	Flutter development, system integration and database support	Contributed to the flutter mobile application, assisted with the system integration and helped with the firebase database setup and functionality	2026/10/30	18hrs		Yes	2026/10/31		Completed
222001607	ASP.NET	SRS documentation updates and ASP.NET development	Updated and finalised the SRS documentation, contributed to ASP.NET development and assisted with testing and checking system functionality against the requirement	2026/10/31	14hrs		Yes	2026/11/01		Completed
223043636	Flutter	Flutter mobile application development	Worked on developing the flutter mobile application, implementing features and testing the application to ensure that it worked correctly	2026/10/28	14hrs		Yes	2026/10/31		Completed
224130045	Flutter	Progress report, flutter mobile application development and database support	Prepared and maintained the project progress report, contributed to flutter mobile application development and assisted with database-related tasks	2026/10/31	15hrs		Yes	2026/11/01		Completed

Figure 23.9 — Progress Tracker: Individual Contributions.

Phase 3 Development Progress

Task ID	Development Area	Task / Function	Responsible Student(s)	Planned Date	Actual Completion Date	Status	% Complete	Evidence / GitHub Link	Notes / Action Required
P3-01	Firebase Backend	Required tables, fields, keys and relationships implemented	Nido Maphosa, Tyam	2026/10/15	2026/10/15	Completed	100%		
P3-02	Firebase Backend	Authentication and access configured where required	Mjiyakho Thobani	2026/10/19	2026/10/20	Completed	100%		
P3-03	Flutter	Approved mobile screens and navigation implemented	Mjiyakho, Seiphetlho, Pikwa, Masita	2026/10/21	2026/10/22	Completed	100%		
P3-04	Flutter	Main mobile functions connected to Firebase	Mjiyakho, Seiphetlho, Pikwa, Masita	2026/10/23	2026/10/27	Completed	100%		
P3-05	ASP.NET	Approved web pages and navigation implemented	Nido Maphosa, Tyam, Freeman, Machedi	2026/10/21	2026/10/22	Completed	100%		
P3-06	ASP.NET	Main web functions connected to Firebase	Nido Maphosa, Tyam, Freeman, Machedi	2026/10/23	2026/10/27	Completed	100%		
P3-07	Integration	Shared Process 1 working across the integrated system	Nido Maphosa, Tyam, Freeman, Machedi	2026/10/28	2026/10/29	Completed	100%		
P3-08	Integration	Shared Process 2 working across the integrated system	Thobani Mjiyakho, Seiphetlho, Pikwa, Masita	2026/10/28	2026/10/29	Completed	100%		
P3-09	Integration	Data fields and status values consistent across applications	Nido Maphosa, Thobani Mjiyakho, Seiphetlho	2026/10/30	2026/10/30	Completed	100%		
P3-10	SRS	Implementation status, approved changes and known limitations updated	Nido Maphosa	2026/10/19	2026/11/01	Completed	100%		
P3-11	Microsoft Project	Schedule updated to reflect actual development progress	Caitlin Freeman, GK Seiphetlho	2026/10/15	2026/11/01	Completed	100%		
P3-12	GitHub	Current source code, README and relevant documentation organised	Nido Maphosa, Tyam, Thobani Mjiyakho, Freeman, Pikwa, Machedi, Masita, Seiphetlho	2026/10/26	2026/10/29	Completed	100%		
P3-13	Final Demonstration	Prototype and demonstration flow prepared for panel	Nido Maphosa	2026/11/30	2026/11/31	Completed	100%		

Figure 23.10 — Progress Tracker: Development Progress.

Group	Group O
Project Title	Multi-Site and Environmental Monitoring System
Stakeholder / Organisation	Goitsemang Diane(Department of Agriculture, Plant and Science)
Group Leader	Nido Maphosa
Number of Group Members	8
GitHub Repository Link	https://github.com/Nido-Maphosa86/soil-sensor-monitoring
Reporting Period	(2026/10/15-2026/11/04)
Overall Phase Status	Completed

Phase 3 Development Summary

Total tracked items	**13**
Completed	**13**
In Progress	**0**
Blocked	**0**
Average completion	**100%**

Phase 3 Reminder

Phase 3 focuses on implementing and demonstrating the integrated prototype. Track the shared Supabase backend, Flutter mobile application, ASP.NET web application, important shared processes, GitHub evidence, Microsoft Project and the final demonstration.

Figure 23.11 — Progress Tracker: Development Summary.

Integration Progress	Track important shared processes across the system	Flutter action, ASP.NET action, shared Supabase information and current working status
Implementation Evidence	Record evidence of the working prototype	Screenshots, recordings, GitHub commits/links and relevant SRS sections
Challenges & Actions	Record development/integration challenges and corrective action	Problem, effect, action, owner, target date and status
Phase Summary	Summarise Phase 3 progress	Completed work, limitations, participation and final readiness
Group Reflection	One shared final group reflection	Development experience, collaboration, challenges and lessons learned

Important

Phase 3 does not require a formal test plan, test cases or defect log. Record normal development checks, working functions, integration progress, challenges and corrections as part of implementation. Do not include passwords, API keys or confidential stakeholder/user information.

Figure 23.12 — Progress Tracker: Tracker guidance sheet.

Phase 3 Group Reflection

1. What did the group complete successfully during development and implementation?

During phase 3, the group made progress in developing and integrating the mobile and web applications. The group worked on connecting the applications to the database to ensure that information could be stored, retrieved and displayed correctly. We also worked on improving the functionality of the system and ensuring that the diffent componenets worked together to support the soil-monitoring process.

2. Which development or integration challenges were the most difficult, and how were they addressed?

One of the main challenges was ensuring that the applications communicated correctly with the database. Connecting the different components and making sure that data was transferred, stored and retrieved correctly required testing and troubleshooting.We also had to ensure that the mobile and web application used the correct data and displayed information consistently

3. How did the group ensure that Flutter and ASP.NET worked with the same Supabase backend?

The group ensured integration by working towards a shared database structure and making sure that both applications followed the same system requirements. Flutter was used for the mobile application, while ASP.NET was used for the web application. Both applications were designed to communicate with the database so that information could be accessed and managed within one system.

Figure 23.13 — Progress Tracker: Group Reflection, questions 1 to 3.

4. How was development work divided among group members, and was participation balanced?

Tasks were divided among group members according to different components and responsibilities of the project. Members worked on different areas, including application development, database management, integration, testing, and documentation. Although participation was not always equal at every stage, group members contributed towards completing the project and supporting one another when challenges arose.

5. Which parts of the prototype are strongest, and which limitations remain?

The strongest part of the prototype are the integration of the mobile and web applications with the database. The system allows different users to access the platform according to their roles and privileges. The web application provides administrative functions, while the mobile application allows farmers to access the system. Although the prototype meets its main objectives, further improvements could be made to enhance the user experience, system performance and overall functionality

6. What did the group learn from developing and integrating the prototype?

Developing the two applications taught us the importance of system integration and teamwork. We learned that developing a mobile application and web application requires proper planning, communication, and coordination. We also gained practical experience in connecting applications to databases, identifying technical problems and testing whether different components work together as expected

Figure 23.14 — Progress Tracker: Group Reflection, questions 4 to 6.

6. What did the group learn from developing and integrating the prototype?
Developing the two applications taught us the importance of system integration and teamwork. We learned that developing a mobile application and web application requires proper planning, communication, and coordination. We also gained practical experience in connecting applications to databases, identifying technical problems and testing whether different components work together as expected
7. What would the group improve if the project continued beyond the final assessment?
If the project continued beyond the final assessment, the group would focus on improving the existing features, enhancing the user interface, and strengthening the overall system performance. We would also consider adding more functionality based on user feedback and conducting further testing to ensure that the system remains reliable and easy to use.

Figure 23.15 — Progress Tracker: Group Reflection, questions 6 and 7.

Phase 3 Progress Summary			
Reporting Period	15 October-04 November 2026	**Phase 3 Completion Checklist**	**Complete?**
Overall Phase Status	Completed	Shared Supabase backend implemented	Yes
What was completed during Phase 3?	The group completed the development and integration of the flutter mobile application and ASP.NET web application. Database connectivity and user access according to assigned roles and privileges were implemented. The group also worked on testing the system and ensuring that the main functions operated as expected.	Main Flutter functions working	Yes
Which functions or processes remain incomplete?	No major functions or processes remain incomplete. The main project requirements have been addressed, and the prototype is ready for the final assessment.	Main ASP.NET functions working	Yes
What known limitations remain?	There are no major known limitations that affect the main functionality of the prototype. However, like any developing system, there may be opportunities for future improvements and additional features.	At least two shared processes working	Yes
Which contributions were particularly important?	The contributions of members working on application development, database management, system integration, testing and documentation were particularly important. Teamwork and communication also played an important role in bringing the different components together.	Integration progress recorded	Yes
Were there participation concerns?	Although members had different responsibilities and levels of involvement at different stages, the group worked together towards completing the project. Members contributed to their assigned tasks and supported the overall development process.	Implementation evidence recorded	Yes
Is the prototype ready for the final demonstration?	Yes	SRS updated	Yes

Figure 23.16 — Progress Tracker: Progress Summary.

Main priority before final presentation	The main priorities are to conduct final system testing, verify that all major functionalities work correctly, check the user interfaces, prepare the presentation and ensure that all group members understand the system and can explain their contributions	Microsoft Project updated	Yes
		GitHub current	Yes
		Individual contributions recorded	Yes
		Challenges/actions updated	Yes
		Known limitations recorded	Yes
		Group reflection completed	Yes
		Final demonstration prepared	Yes

Figure 23.17 — Progress Tracker: Progress Summary, checklist continued.

Phase 3 Challenges and Corrective Actions								
Date Identified	**Development / Integration Challenge**	**Effect on Progress or Function**	**Action Taken / Required**	**Responsible Student**	**Target Date**	**Status**	**Resolved Date**	**Notes / Known Limitation**
2026/10/16	Uneven distribution of tasks among group members, with some members having more responsibilities than	Created an imbalance in workload and participation	The group reviewed and adjusted task allocation to ensure a fairer distribution of responsibilities among members	Nido Maphosa	2026/10/17	Completed	2026/10/18	
2026/10/28	Difficulty connecting the applications to the database and ensuring proper communication between the systems	Delayed some development activities due to troubleshooting and additional testing	The group investigated the connection issues, made the necessary adjusments and tested the database integration to ensure that data could be transferred correctly	Mjiyakho,Tyam	2026/10/29	Completed	2026/10/29	

Figure 23.18 — Progress Tracker: Challenges and Corrective Actions.

SUPPORTING EVIDENCE

Appendices A to E

Stakeholder engagement evidence for the Multi-Site Smart Soil and Environmental Monitoring System

Appendix	Evidence	Date
A	Follow-up questions issued to the stakeholder	26 August 2026
B	Stakeholder written responses to the follow-up questions	27 August 2026
C	Project requirements document supplied by the stakeholder	20 August 2026
D	Stakeholder written confirmation of the findings	3 September 2026
E	Engagement record — meetings, correspondence and attendance	19 Aug – 3 Sep 2026

Traceability

Every requirement in this specification traces to one of these sources. Section 2 draws on the meeting of 19 August and on Appendix B; sections 5 and 10 to 13 draw on Appendix C; section 14.2 draws on Appendix B; section 15 draws on the meeting of 19 August and on Appendix D.

Appendix A — Follow-Up Questions Issued

Issued	26 August 2026
From	Nido Maphosa, on behalf of Group O
To	Ms Mpho Mbele
Reason	A meeting could not be arranged during that week, so the questions were put in writing

The covering note read: "Thank you again for the time you have given to this project so far, and for the requirements document you shared with us. As we move into the planning and design we have a short list of follow-up questions that will help us complete our documentation accurately and design the system correctly."

#	Question
1	Before this monitoring system was proposed, how was soil condition on these three sites being assessed — was there a process in place at all, or is this a completely new capability for your operation?
2	What specifically prompted you to commission this system — was there a particular moment or realisation that made you decide you needed it?
3	Could you confirm, for our documentation, your organisation's full name, what your organisation does, and your role and title?
4	What crop or crops are grown at each of the three locations, and what soil type is typical at each?
5	Who specifically is affected day to day by not having this system — which roles or team members?
6	What are the practical consequences when this information isn't available in time — for example, crop stress, wasted water, or a delayed response?
7	Of the difficulties you've described, which one causes the most damage if left unaddressed?
8	Roughly how many people will use the system, and are there any user types beyond Administrator and Standard User that we should plan for?
9	Could you share a sample of how a single sensor reading looks once it is stored in Firebase — for example a screenshot or the field names — so that we can design our data integration correctly?
10	Does the data in Firebase include battery level, power status, and a last-communication timestamp for each sensor node?
11	What access will we be given to read the data in Firebase, and who is the right technical contact if the format ever changes?
12	If development time runs short, which functions matter most to you, and which could wait for a later version?
13	Beyond the technical requirements already defined, how would you personally know that this system has solved the problem for your operation?
14	Was anyone else involved in preparing the original requirements document, or should we consider you the sole author for our records?
15	Once we send you a written summary of what we've discussed, would you be able to confirm it is accurate by email?

Appendix B — Stakeholder Written Responses

Received	27 August 2026
From	Ms Mpho Mbele
Form	The same document returned with answers written beneath each question

Preamble supplied with the responses

"For the current prototype, you may use realistic dummy sensor data. Firebase integration is not required at this stage and can be considered later if needed."

Question 1

Before this monitoring system was proposed, how was soil condition on these three sites being assessed — was there a process in place at all, or is this a completely new capability for your operation?

Response	This is a new centralised monitoring capability for the project. At present, there is no single mobile and web system that continuously brings together soil and environmental readings from all three locations. The proposed system will make it easier to view and compare information from the different sites in one place. At the moment they only take soil samples and send them to the lab only once when they start planting just to see which fertilizer will be suitable.

Question 2

What specifically prompted you to commission this system — was there a particular moment or realisation that made you decide you needed it?

Response	The main reason for the system is the need to monitor conditions across three different locations without relying only on physical checks. The project will use 15 sensors, with five sensors at each location, so a central system is needed to organise the readings, show the condition of each site and help users identify changes that may need attention.

Question 3

Could you confirm, for our documentation, your organisation's full name, what your organisation does, and your role and title?

Response	This is for research purpose, you can call it environmental researchers, responsible for monitoring environment for agricultural benefit.

Question 4

What crop or crops are grown at each of the three locations, and what soil type is typical at each?

Response	The crop is pigeon pea, soil type is not a must to know. This does not prevent you from

	continuing with the system design and prototype development.

Question 5

Who specifically is affected day to day by not having this system — which roles or team members?

Response	Site or farm manager, agricultural staff, field workers, researchers.

Question 6

What are the practical consequences when this information isn't available in time — for example, crop stress, wasted water, or a delayed response?

Response	When the information is not available in time, changes in soil conditions may be noticed late. This can lead to delayed responses to problems such as low soil moisture, unusual nutrient levels or other environmental changes. The system should help users identify changes earlier and respond more quickly.

Question 7

Of the difficulties you've described, which one causes the most damage if left unaddressed?

Response	The most important problem to address is the delay in identifying significant changes in soil or environmental conditions across the three locations. A central monitoring system should make it easier to see which location or sensor needs attention.

Question 8

Roughly how many people will use the system, and are there any user types beyond Administrator and Standard User that we should plan for?

Response	The exact number of users is not fixed at this stage. For the first version, the system should support at least two user types: Administrator and Standard User. The Administrator can manage users, locations and sensor information, while the Standard User can view dashboards, readings, history and alerts. Additional roles can be added later if the stakeholder requires them.

Question 9

Could you share a sample of how a single sensor reading looks once it is stored in Firebase — for example a screenshot or the field names — so that we can design our data integration correctly?

Response	A dummy reading can include fields such as sensor ID, location ID, nitrogen (N), phosphorus (P), potassium (K), soil moisture, soil temperature, pH, electrical conductivity, battery level and date/time. The exact live-data structure can be integrated at a later stage if required.

Question 10

Does the data in Firebase include battery level, power status, and a last-communication timestamp for each sensor node?

Response	For the prototype, the dummy data should include battery level, sensor status and a last-communication date/time so that the system can demonstrate sensor-health monitoring. This is important because there will be 15 sensors across three locations. You do not need to obtain these values from Firebase at this stage.

Question 11

What access will we be given to read the data in Firebase, and who is the right technical contact if the format ever changes?

Response	No Firebase access is required for the current prototype. You can continue using dummy data and focus on the required system functionality, user interface, dashboards and monitoring features. Real sensor or database integration can be treated as a later integration step when access and the final data format are available.

Question 12

If development time runs short, which functions matter most to you, and which could wait for a later version?

Response	If development time becomes limited, priority should be given to user login and access control, the three monitoring locations, display of all 15 sensors, current sensor readings, battery and sensor status, historical readings, alerts for abnormal conditions or inactive sensors, and a clear dashboard for each location. More advanced analytics, prediction and complex reporting can be added in a later version.

Question 13

Beyond the technical requirements already defined, how would you personally know that this system has solved the problem for your operation?

Response	The system will be considered successful if authorised users can easily view the condition of all three locations from the mobile and web applications, identify which sensor produced a reading, notice unusual conditions, see whether a sensor is active or inactive, view previous readings and receive useful alerts. The aim is to make monitoring information easier to access, more organised and available sooner.

Question 14

Was anyone else involved in preparing the original requirements document, or should we consider you the sole author for our records?

Response	Consider me the sole author.

Question 15

Once we send you a written summary of what we've discussed, would you be able to confirm it is accurate by email?

Response	Yes.

Appendix C — Project Requirements Document

Received	20 August 2026
From	Ms Mpho Mbele, on behalf of the stakeholder organisation
Title	Project Requirements — Multi-Site Smart Soil Monitoring System
Form	PDF, thirty-one numbered sections

The document as received is reproduced after this summary. Its sections map to this specification as follows.

Section of the requirements document	Where it is addressed
1 Project overview	§4 Aim and objectives
2 Physical deployment — three locations, fifteen sensors	§2.1, §5.2 AC3 and AC4
3 Soil parameters — seven per sensor	§8.3.1, §10.5
4 User roles — Administrator and Standard User	§7.2
5 Authentication and access control	§10.1, §12
6 Location management	§10.3
7 Sensor management	§10.4
8 Sensor data collection	§8.3, §10.12
9 Sensor reading storage	§10.9, §16
10 Battery monitoring	§10.6
11 Device health monitoring	§10.6
12 Monitoring thresholds	§10.7, §15
13 Alerts	§10.8
14 Mobile application requirements	§9
15 Web application requirements	§9
16 Location comparison	FR65
17 Filtering	FR66
18 Reporting	§10.10
19 Data visualisation	§10.9
20 Supabase requirements	§8.1, §12
21 Minimum database entities	§16
22 Data validation	§10.12, NFR
23 System security	§11.1
24 Reliability	NFR15
25 Usability	§11.2, NFR8

Section of the requirements document	Where it is addressed
26 Performance	§11.3
27 Audit and timestamps	NFR18, §16
28 Testing requirements	Phase 3 planning
29 Required final demonstration	§5.2 Acceptance criteria
30 Integration requirement	§13, change log C01
31 Required technical components	§12, §5.3

Appendix D — Stakeholder Confirmation of the Findings

Received	3 September 2026
From	Ms Goitsemang Dikane, Plant Sciences (nGAP), Department of Agriculture
To	Ms Mpho Mbele and Nido Maphosa
Subject	Confirmation of findings — Multi-Site Smart Soil and Environmental Monitoring System
Outcome	Findings confirmed, requirements approved, no blocking corrections

The reply in full

> "Dear Ms Mbele
>
> Kindly note that I have gone through the document and have realized that some of the minor corrections can be amended while the process unfolds. The calibration and setting of range measurements can also be amended later on once the platform has been established. We shall revisit the pipeline and amend where necessary, for now I believe the document can be used as it meets the entry requirements as the end user.
>
> Warm regards
>
> Goitsemang Dikane, Plant Sciences: nGAP, Work Integrated Learning, Department of Agriculture, Faculty of Health and Environmental Sciences"

How this was applied

What she said	Where it is recorded
The document meets the entry requirements of the end user and can be used as it stands	§21.1 Confirmation
Minor corrections can be amended while the process unfolds	§21.1, recorded as non-blocking
Calibration and range settings can be amended once the platform is established	§15.2 and change log C14
The pipeline will be revisited and amended where necessary	§15.2

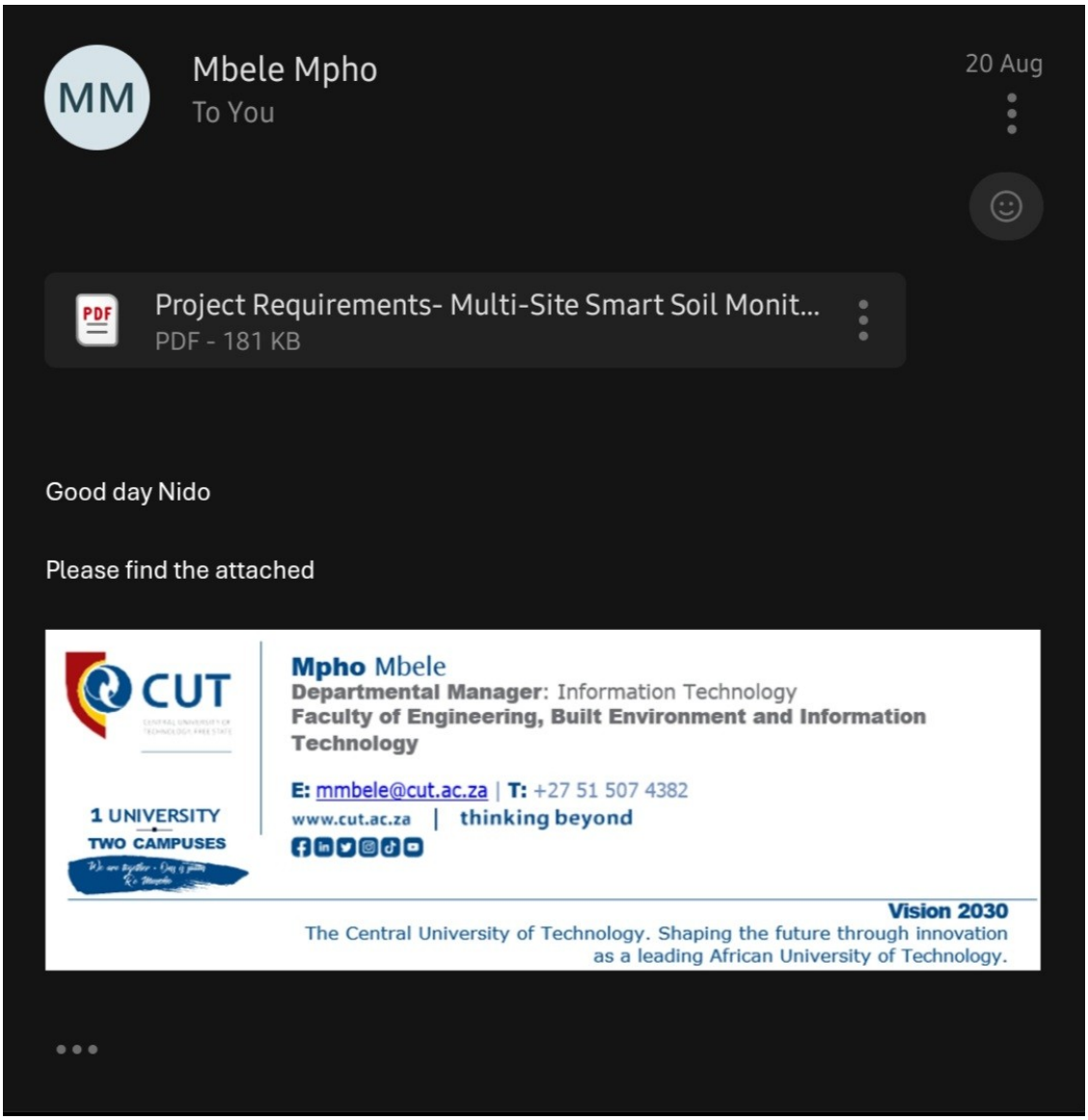
Appendix E — Engagement Record

E.1 Engagement Log

#	Date	Type	With	Subject	Evidence
1	19 Aug 2026	Consultation meeting	Stakeholder	Scope and architecture briefing	Recording and transcript
2	20 Aug 2026	Email received	Ms Mpho Mbele	Project requirements document supplied	Email, screenshot
3	26 Aug 2026	Email sent	Ms Mpho Mbele	Fifteen follow-up questions issued	Email, screenshot
4	27 Aug 2026	Email received	Ms Mpho Mbele	Written responses returned	Email, screenshot
5	30 Aug 2026	Email sent	Ms Mpho Mbele	Findings summary and specification issued	Email, screenshot
6	3 Sep 2026	Email received	Ms Goitsemang Dikane	Findings confirmed and approved	Email, screenshot

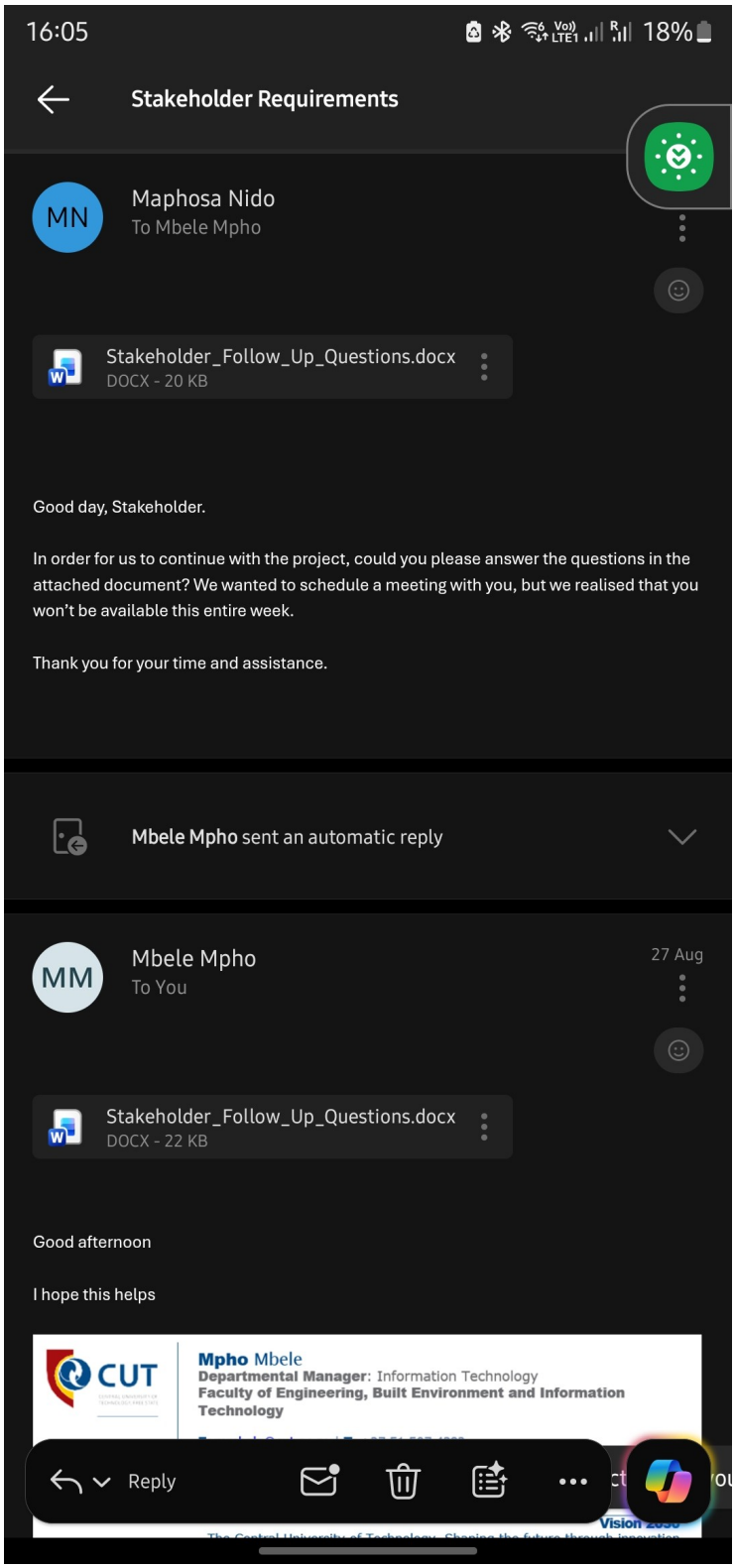
E.2 Correspondence Record

Screen captures of the email exchange, in date order. Each confirms the sender, the recipient, the date and the attachment. The consultation meeting that preceded this exchange is recorded in E.3.



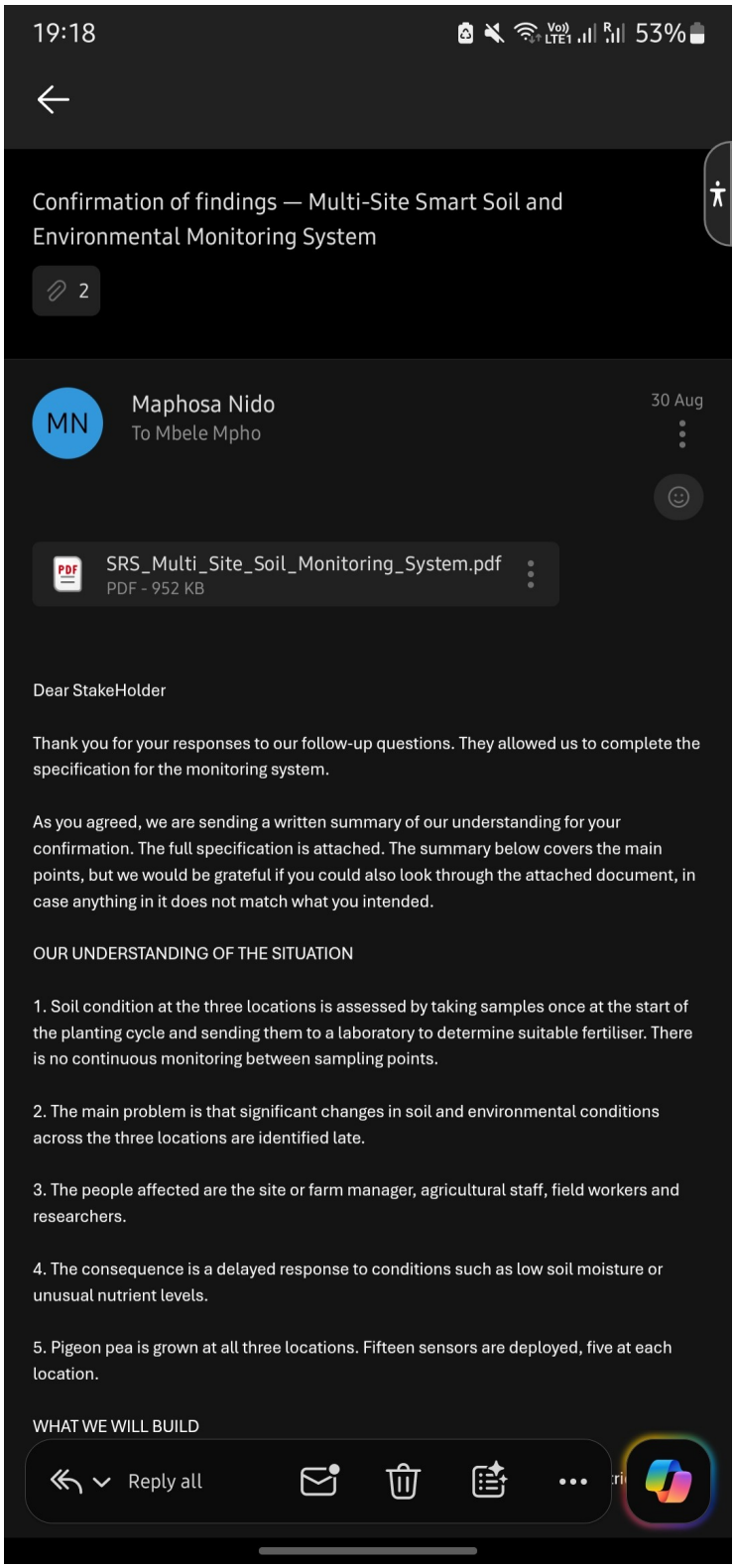
20 August 2026 — Ms Mpho Mbele to Nido Maphosa

Project requirements document supplied. Evidences Appendix C.



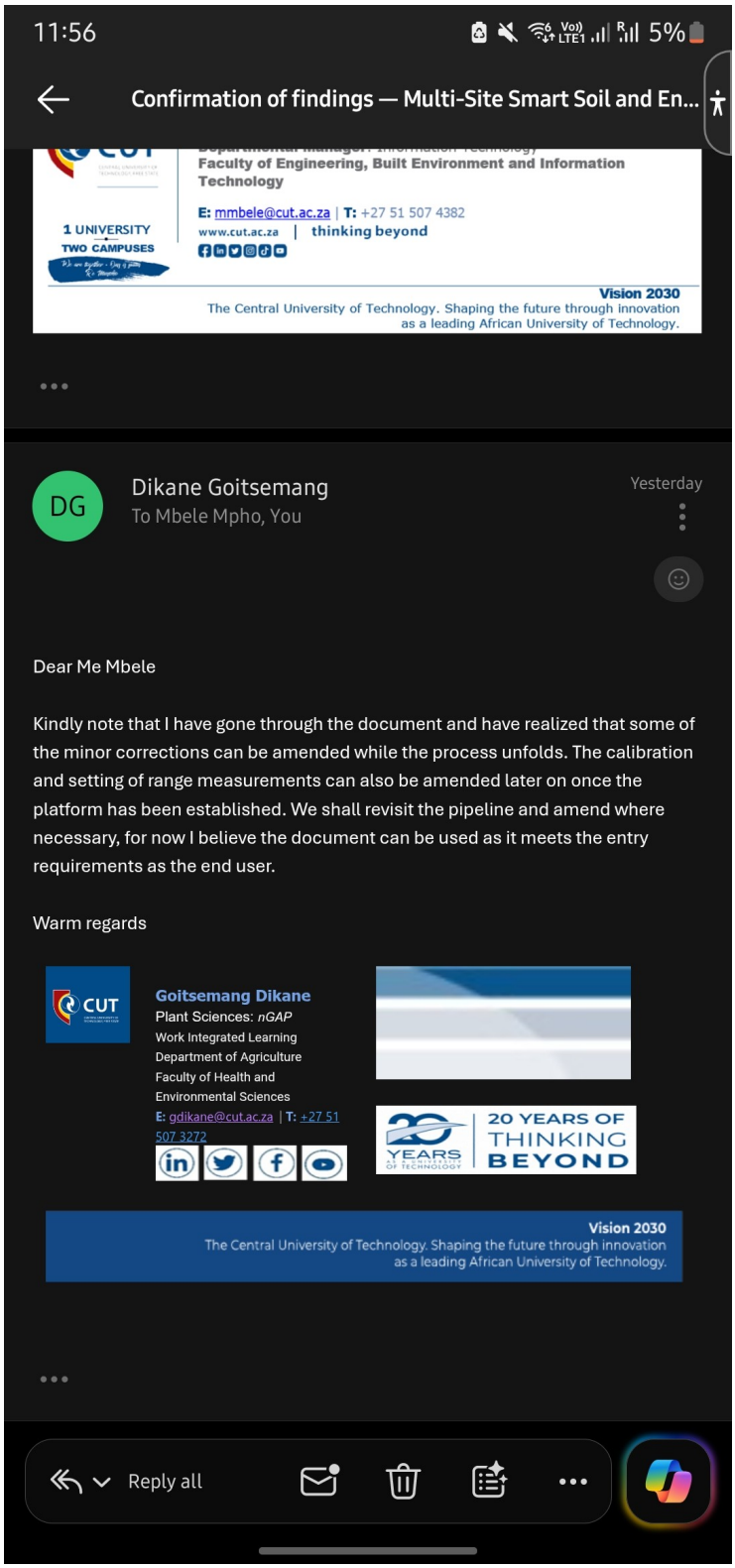
26 and 27 August 2026 — questions issued and answered

Sent by Nido Maphosa; responses returned by Ms Mbele. Evidences Appendices A and B.



30 August 2026 — Nido Maphosa to Ms Mpho Mbele

Findings summary issued with the specification attached, for confirmation.



3 September 2026 — Ms Goitsemang Dikane to Ms Mbele and Nido Maphosa

Findings confirmed and requirements approved. Evidences Appendix D.

E.3 Consultation Meeting, 19 August 2026

Subject	IoT soil-monitoring platform — scope, architecture and timeline
Present	The stakeholder, and all eight members of the Group O delivery team. One member spoke on behalf of the group during the recording.
Record	Two audio recordings, transcribed. Full transcript held with the project records.
Note	Speaker attribution in the transcript is inferred from turn-taking; no speaker names were captured in the audio.

What the meeting established

#	Established at the meeting	Where it appears
1	Sensor hardware is handled by a separate party. The delivery team does not work on the hardware.	§5.3
2	Readings are transmitted by Wi-Fi and staged in Firebase; the team's role is to bring that data into Supabase.	§8.3
3	Where sensor data is not yet available, the team is to work with representative data in the database.	§8.3, C01
4	Three locations, five sensors at each, fifteen in total.	§2.1, AC3, AC4
5	The database must link locations to sensors, and readings must trace to a named sensor at a named location.	§16
6	Device health is tracked separately. Solar panels and batteries are in use; a failing battery must be flagged.	§10.6
7	Web is the primary home for visualisation; mobile focuses on device and battery health.	§9
8	Readings must be interpreted, not merely displayed.	§15, NFR17
9	Thresholds are to be researched from South African sources; the team may set its own with a stated basis.	§15, §15.1
10	Thresholds are crop-specific; no single national threshold applies to all crops.	§15
11	A threshold breach must raise an alert. Push notification preferred, email secondary.	§10.8, C09
12	Data is fetched at intervals by a scheduled job, at approximately thirty-minute intervals.	§8.3, FR75
13	Climate temperature per location, beyond soil temperature, to be incorporated for context.	See E.4
14	Indicative timeline: Phase 1 early September, design end September, delivery end October.	§17.4

E.3.1 Verbatim extract — scope and data

Stakeholder

> “The data is transmitted by Wi-Fi from the sensor to the database, so there's a middleman — you guys don't really have to worry about the hardware parts. At the moment we're going to read data into Firebase, so your role will just be to take that data from Firebase into Supabase.”

Stakeholder

> “Even if the data isn't ready, we'll just check the type of data that the sensors are supposed to read. So you just have dummy data on your database for those sensors.”

Stakeholder

> “We'll be using three locations. We have fifteen sensors, so for each location we'll put five sensors. In your database you must have locations, and then you need to link locations with the sensors.”

E.3.2 Verbatim extract — interpretation and thresholds

Stakeholder

> “We're not going to leave you with just the raw data — you have to tell us the information and analyse what the data set is saying. So say, for example, pH is 6 — you guys must go and find out what pH 6 means.”

Delivery team

> “The thresholds — will we have to determine them ourselves, or will you provide them?”

Stakeholder

> “You have to go and find out if they exist. If there are no thresholds, then you can come up with your own, just to have something — I can't have you tweaking something with no basis.”

Stakeholder

> “Critical threshold tests for NPK depend on the specific crop type. Other than that, they don't have a general threshold here — it depends on the crop type.”

E.3.3 Verbatim extract — alerting and scheduled collection

Stakeholder

> “Your system must also give alerts. If a threshold is exceeded, then you give out an alert, or maybe send a notification.”

Delivery team

> “Should the notification be from the app, or an SMS?”

Stakeholder

> “From the app. Push notifications are fine, or you can send an email — there's an option for emails. But I think a push notification will do.”

Stakeholder

> “Even though we have solar panels, we're also going to use batteries — so if those batteries are going to die at some point, you need to flag it.”

Stakeholder

> “There's something called a cron job — how to fetch data at intervals, so it fills up your database automatically, instead of only reading once when something happens on the application.”

E.4 Items Raised and Not Yet Closed

Two matters raised in the consultation are recorded here because they are not fully reflected in the specification.

Raised	Position	Status
Climate temperature per location, in addition to soil temperature	The stakeholder undertook to obtain temperature data for each location. This would be an additional external data source beyond the sensors.	Not in the current specification. To be raised before Phase 2 design begins and recorded in the change log with the outcome.
Whether the sensor reports humidity as a measured parameter	The consultation described the sensor as reading temperature, humidity, pH and conductivity alongside nitrogen, phosphorus and potassium. The requirements document, section 3, defines seven parameters and does not list humidity, and the sample reading fields in Appendix B list the same seven.	The specification records seven parameters, following the written requirements document. Recorded as change log entry C04 and to be confirmed with the stakeholder.

E.5 Attendance

Group members present at or involved in each engagement. Also recorded in the Stakeholder Engagement sheet of the project progress tracker.

Member	19 Aug	20 Aug	26 Aug	27 Aug	30 Aug	3 Sep
Nido Maphosa	P	P	P	P	P	P
Caitlin Freeman	P	P	P	P	P	P
Mbasa Tyam	P	P	P	P	P	P
Tiisetso Masita	P	P	P	P	P	P
Machedi Mamello	P	P	P	P	P	P
Ayatolla Pikwa	P	P	P	P	P	P
Katlontle Seiphetlho	P	P	P	P	P	P
Thobani Mjiyakho	P	P	P	P	P	P

P indicates present or involved. All eight members took part in every engagement: the consultation was attended by the group with one member speaking on its behalf, and each item of correspondence was drafted and reviewed by the group before it was sent or acted upon.

E.6 Evidence Held but Not Reproduced

Item	Reason	Where held
Audio recordings of the consultation meeting	Two recordings, 19 August 2026. Not reproduced in a written submission.	Project records
Full meeting transcript	Contains discussion unrelated to the project. The technical content is extracted in E.2.	Project records

Confidentiality

Telephone numbers, identity numbers, credentials and confidential organisational records are excluded from this document and from the source repository. Where a source item contains such information it is redacted before inclusion.